

Bayesian Additive Voronoi Tessellations for Variable Selection in High-Dimensional Settings

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Abstract

This thesis develops and evaluates variable selection methodologies for Bayesian Additive Regression Trees (BART) and Bayesian Additive Voronoi Tessellations (AddiVortes) in continuous regression. BART is a Bayesian tree-based ensemble model capable of capturing complex non-linear relationships and interactions through recursive axis-aligned partitions. AddiVortes provides an alternative partitioning framework based on flexible Voronoi tessellations. Existing approaches to BART variable selection either summarise posterior information from an unconstrained BART model or incorporate variable selection directly into the prior specifications. In this thesis, these approaches are reviewed and a framework for variable selection with AddiVortes is developed. Through a series of simulation studies, we investigate the ability of AddiVortes to identify important predictors under a range of regression settings. The results show that the additional flexibility of Voronoi-based partitions can improve variable selection performance, particularly in settings where axis-aligned partitions provide a less effective representation of the response surface.

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Chapter 1

Introduction

Modern statistical modelling often aims to describe relationships between a response and a large number of predictors, where the underlying relationships may be non-linear and involve complex interactions between variables. Classical parametric models provide a simple and interpretable framework for this task, but they require the form of the relationship to be specified in advance. As the structure of the underlying response becomes more complex, these assumptions can become restrictive. Non-parametric methods instead allow the relationship to be determined more flexibly from the data. In particular, tree-based methods have become widely used because they can capture non-linear effects and interactions without requiring a prespecified functional form.

Chipman et al. (2010)’s Bayesian Additive Regression Trees (BART) is a Bayesian ensemble method for regression which models a response surface as a sum of decision trees. By combining many relatively simple trees, BART can capture complex non-linear relationships and interactions without imposing a predefined functional form. Each tree recursively partitions the predictor space using axis-aligned splits. Bayesian Additive Voronoi Tessellations (AddiVortes), developed in Stone and Gosling (2025a), provide an alternative to this approach by replacing the tree-based partitioning structure with Voronoi tessellations. The resulting partitions are not restricted to axis-orthogonal partitions of the predictor space and can therefore represent more complex response surfaces.

Chapter 2 develops the BART model, beginning with the historical development of decision trees. This leads through the development of tree-based ensemble methods to BART. The chapter then gives a full specification of the BART model, including its prior distributions and posterior sampling strategy. Particular attention is given to the Metropolis-Hastings (MH) updates within the Gibbs sampler used to explore the space of tree structures.

Chapter 3 develops the AddiVortes model in a similar manner. The chapter begins by introducing the use of tessellations in statistical modelling and follows their development to ensemble methods. This leads to the AddiVortes framework, followed by its prior specification and MH sampling procedure. The different partitioning mechanisms of BART and AddiVortes are then considered, highlighting the distinction between axis-aligned tree partitions and the more flexible partitions produced by Voronoi tessellations.

Recent work by Stone and Gosling (2025a) showed that AddiVortes can provide improved regression and predictive performance compared with BART and several established machine learning methods. This raises the question of whether the greater flexibility of the AddiVortes partitioning mechanism can also improve variable selection. BART has a well established literature in this area, with existing methods broadly falling into two approaches. The first uses posterior output from an otherwise unconstrained BART model to construct measures of variable importance, such as variable inclusion frequencies. The second incorporates variable selection directly into the model through prior specifications that restrict or regularise the predictors available during model construction. These approaches include Variable Inclusion Proportions (VIP) (Bleich et al., 2014a), Metropolis Information (MI) (Luo et al., 2021), Dirichlet priors (Linero, 2018), and Approximate Bayesian Computation (ABC) Bayesian forests (Liu et al., 2021). Chapter 4 develops a variable selection framework for AddiVortes by adapting these existing BART approaches to the AddiVortes setting. The resulting methods are evaluated through a simulation study based on the framework of Luo et al. (2021).

Chapter 2

Bayesian Additive Regression Trees (BART)

2.1 Decision Trees and a Historical Overview of Tree-based Methodologies

A central objective in statistical modelling is the estimation of the relationship between a response variable and a collection of explanatory variables. Classical approaches, such as linear and generalised linear models, provide an interpretable framework for this task. However, these methods typically require the practitioner to specify the functional form of the relationship in advance. In many practical applications, the true relationship may be highly non-linear and involve complex interaction effects between predictors that are difficult to identify *a priori*. This challenge motivated the development of non-parametric learning methods capable of discovering such structures directly from the data. Among the most influential of these approaches are decision trees, which estimate unknown response functions by recursively partitioning the predictor space into increasingly homogeneous regions.

This chapter introduces the fundamental concepts underlying tree-based learning and traces their historical development. Beginning with the earliest decision tree methodologies and subsequent recursive partitioning frameworks, we examine how researchers sought to model increasingly complex data structures through hierarchical partitioning of the predictor space. We then discuss the emergence of Classification and Regression Trees (CART), ensemble methods such as Random Forests and Gradient Boosting, and the Bayesian reformulation of tree-based models.

This progression culminates in Bayesian Additive Regression Trees (BART), one

of the primary methodologies considered throughout this thesis. By combining the predictive strength of additive tree ensembles with a fully Bayesian framework for regularisation and uncertainty quantification, BART connects the results of several decades of methodological development in statistical learning and Markov Chain Monte-Carlo (MCMC) methods. Understanding this historical progression provides the necessary context for the theoretical and methodological developments presented in the subsequent chapters. We begin our discussion with an introduction to the central tool of our study: the decision tree.

2.1.1 Decision trees: Illustrative example

Decision trees are a widely used modelling method, commonly applied to both classification and regression tasks, with origins dating back to the 1960s (Morgan and Sonquist, 1963). They are particularly popular due to their interpretability, ability to handle both numerical and categorical data, and minimal requirement for data preprocessing. First consider the following illustrative example. Suppose that we want to predict whether a student will pass or fail an exam based on two features:

Table 2.1: Description of features for decision tree illustration

Feature	Description
$\mathbf{x}_1$	Number of hours studied
$\mathbf{x}_2$	Number of hours slept on the night before the exam

A simple decision tree might first split students based on study hours, then further split based on sleep hours, and assign a prediction (Pass or Fail). Suppose that, from previous experience, the teacher organising this examination has observed the following pattern:

Teacher's Hypothesis

- Students who study less than 5 hours and sleep less than 6 hours are predicted to **Fail**.
- Students who study less than 5 hours but sleep at least 6 hours are predicted to **Pass**.
- Students who study 5 hours or more are predicted to **Pass** regardless of sleep.

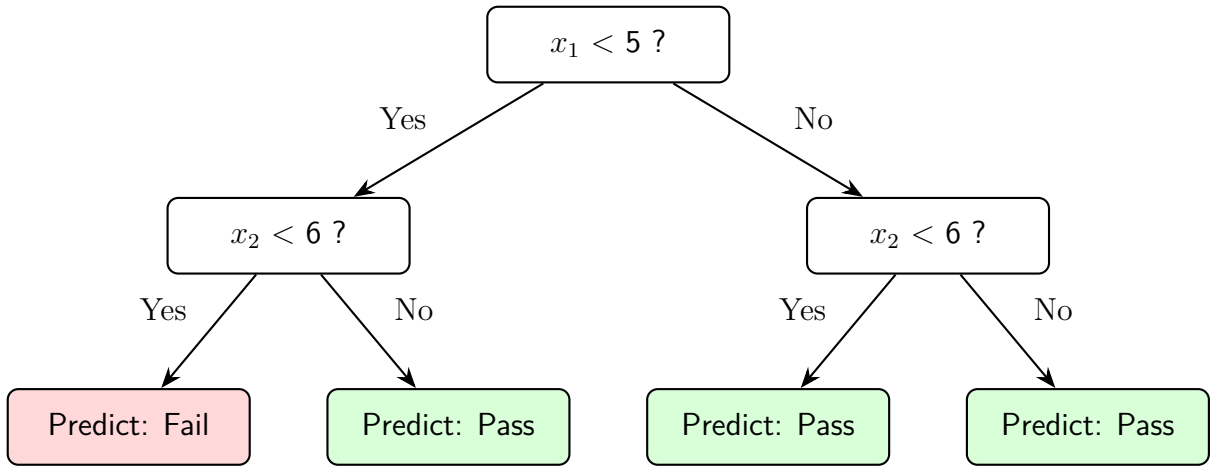


Figure 2.1: A decision tree illustrating a teacher's predictions on the result (Pass or Fail) for a student given the number of hours they studied and the number of hours slept on the night before the exam.

Figure 2.1 illustrates the decision tree constructed from the teacher's hypotheses. Generating a prediction requires only traversing the branches from the root node to a terminal leaf based on the observed values. For instance, a student who studied for 4 hours and slept for 5 hours follows the left branch at each split, resulting in a prediction of Fail. This simple walkthrough highlights the interpretability of decision trees: individual predictions can be explicitly audited through a deterministic sequence of rules. This transparency stands in contrast to competing predictive frameworks such as neural networks (Rumelhart et al., 1986) and support vector machines (Cortes et al., 1995), which obscure feature effects within deep, layered transformations or implicit high-dimensional feature spaces.

2.1.2 Decision trees: Formal definition

We now formalise the definition of a regression tree. Let $\mathcal{X} \subseteq \mathbb{R}^p$ denote the predictor space, let $x = (x_1, \dots, x_p)^\top \in \mathcal{X}$ denote a p -dimensional vector of covariates and let $y \in \mathbb{R}$ denote a continuous response variable. Decision trees represent a general class of non-parametric models that estimate the relationship between x and y by recursively partitioning $\mathcal{X}$ into disjoint regions via a sequence of binary decisions. When applied to categorical response variables, these models are termed **classification trees**; when estimating the conditional expectation of a continuous response, $\mathbb{E}[y \mid x]$, they are termed **regression trees**.

A tree consists of a connected directed graph comprising **internal nodes** and **terminal nodes** (or **leaves**). Beginning at the **root node**, a deterministic splitting rule is applied to the input space. For continuous covariates, internal splits typically take the form:

$$x_j \leq c, \quad (2.1)$$

where x_j is the chosen splitting variable, $j \in \{1, \dots, p\}$, and $c \in \mathbb{R}$ is the corresponding splitting threshold. Observations satisfying the condition are directed to the left child node, while those failing it proceed to the right child node. This recursive binary partitioning continues through successive internal nodes until a stopping criterion is met, terminating at leaf nodes where no further splits are performed.

Let T denote the tree structure, which encompasses the graph topology, the interior splitting variables, and their corresponding split values. Each unique path from the root node to a terminal leaf $b \in \{1, \dots, B\}$ defines a specific sequence of splitting rules. The intersection of these rules forms a distinct sub-region of the predictor space, which we denote as $\mathcal{A}_b$. Together, the structure T partitions the entire predictor space $\mathcal{X}$ into B mutually disjoint hyper-rectangles:

$$\mathcal{X} = \bigcup_{b=1}^B \mathcal{A}_b, \quad \text{where } \mathcal{A}_j \cap \mathcal{A}_k = \emptyset \text{ for } j \neq k. \quad (2.2)$$

Associated with these terminal regions is a set of terminal parameters:

$$M = (\mu_1, \mu_2, \dots, \mu_B), \quad (2.3)$$

where each $\mu_b \in \mathbb{R}$ represents the scalar mean prediction assigned to all points falling into partition $\mathcal{A}_b$. Given the tree structure T and leaf parameters M , a single regression tree maps an input to its corresponding region's mean, defining a

piecewise constant function $g(x; T, M)$:

$$g(x; T, M) = \sum_{b=1}^B \mu_b \mathbb{I}(x \in \mathcal{A}_b), \quad (2.4)$$

where $\mathbb{I}(\cdot)$ is the indicator function satisfying $\mathbb{I}(x \in \mathcal{A}_b) = 1$ if $x \in \mathcal{A}_b$ and 0 otherwise. Consequently, the tree models the conditional expectation as:

$$\mathbb{E}[y \mid x] \approx g(x; T, M). \quad (2.5)$$

To illustrate how the splitting rules define the region $\mathcal{A}_b$, consider an observation with $x_1 = 20$ and $x_2 = 30$ evaluated against the tree in Figure 2.2. Tracing the path through the root split ($x_1 < 100$) and a subsequent left split ($x_2 < 50$) at Internal Node A directs the observation to Terminal Node 1. The sequence of these conditions mathematically defines the region as $\mathcal{A}_1 = \{x \in \mathcal{X} \mid x_1 < 100 \text{ and } x_2 < 50\}$. Because our observation $(20, 30)^\top$ falls within the boundaries of $\mathcal{A}_1$, the indicator function $\mathbb{I}(x \in \mathcal{A}_1)$ equals 1, yielding the prediction:

$$g((20, 30)^\top; T, M) = \mu_1. \quad (2.6)$$

This traversal corresponds to the green highlighted path in Figure 2.2, while Figure 2.3 displays the resulting geometric regions $\mathcal{A}_1, \dots, \mathcal{A}_4$ dividing the predictor space. To aid later comparison, we emphasise the viewpoint of decision trees partitioning the input space into hyper-rectangles as we will later consider alternative partitioning mechanisms which extend to non-rectangular regions.

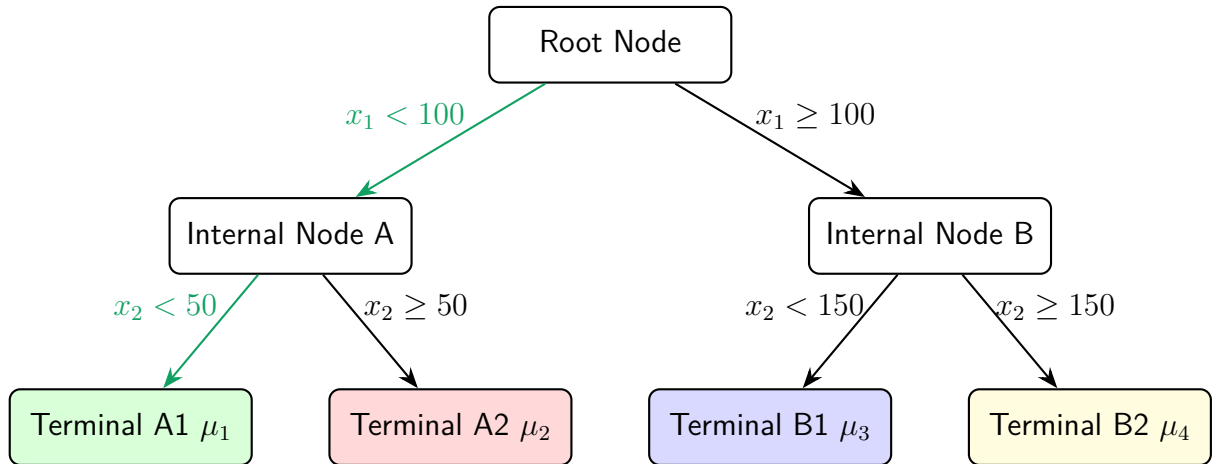


Figure 2.2: Example Decision Tree illustrating the definition of Internal, Terminal Nodes and Root nodes.

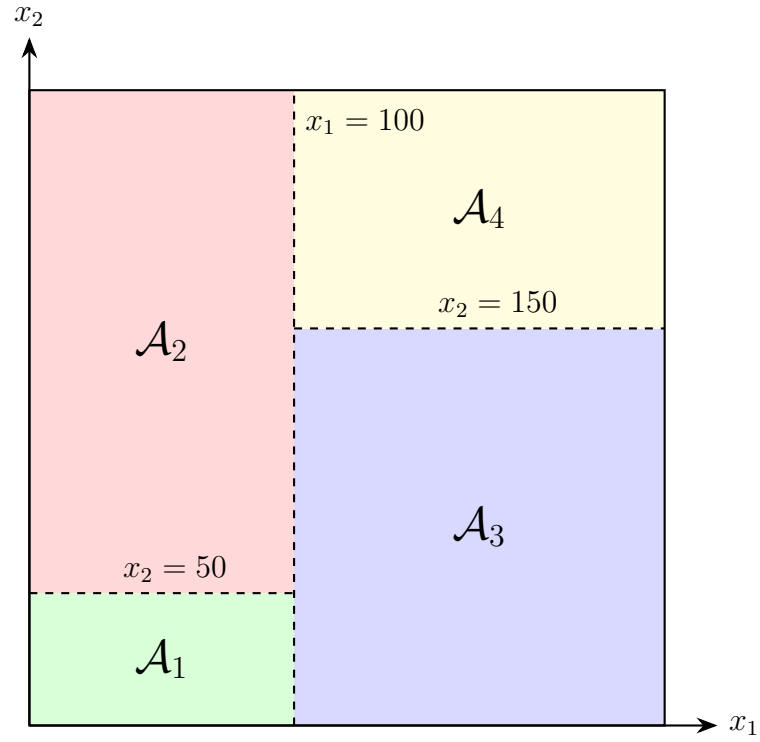


Figure 2.3: Partition of the (x_1, x_2) input space induced by the decision tree given in Figure 2.2.

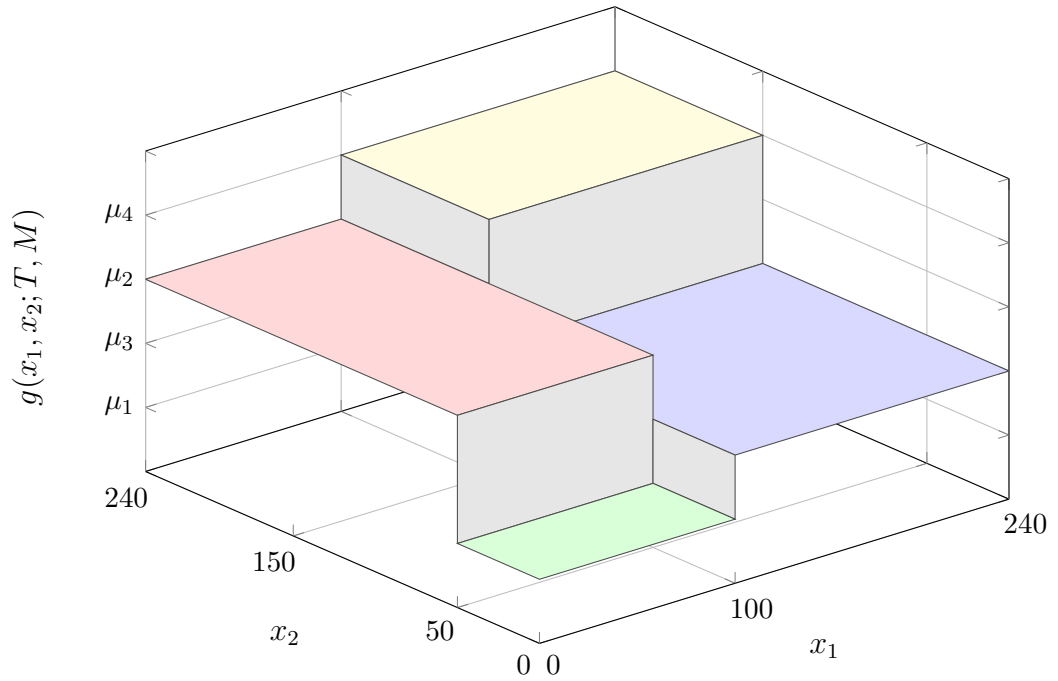


Figure 2.4: 3D surface plot representing the piecewise constant function $g(x_1, x_2)$ defined by the decision tree, assuming arbitrary $\mu_1 < \mu_2 < \mu_3 < \mu_4$.

2.1.3 Step Function Representation and Interaction Modelling

A principal strength of decision trees, and by extension tree-based ensemble methods, is their ability to capture nonlinearities and complex interaction effects without explicit model specification. This property becomes evident when representing a regression tree as an explicit piecewise constant step function via products of indicator functions. For the tree structure presented in Figure 2.2, the mapping function $g(x_1, x_2; T, M)$ can be expressed analytically as:

$$\begin{aligned} g(x_1, x_2; T, M) = & \mu_1 \mathbb{1}(x_1 < 100) \mathbb{1}(x_2 < 50) \\ & + \mu_2 \mathbb{1}(x_1 < 100) \mathbb{1}(x_2 \geq 50) \\ & + \mu_3 \mathbb{1}(x_1 \geq 100) \mathbb{1}(x_2 < 150) \\ & + \mu_4 \mathbb{1}(x_1 \geq 100) \mathbb{1}(x_2 \geq 150), \end{aligned} \tag{2.7}$$

where $\mathbb{1}(\cdot)$ denotes the indicator function. Formulating the tree in this manner illustrates how recursive partitioning models predictor interactions. Each path from the root node to a terminal leaf represents an intersection of split conditions. For example, the terminal region $\mathcal{A}_1$ is defined as

$$\mathcal{A}_1 = \{x \in \mathcal{X} \mid x_1 < 100\} \cap \{x \in \mathcal{X} \mid x_2 < 50\}, \tag{2.8}$$

yielding the joint indicator $\mathbb{1}(x \in \mathcal{A}_1) = \mathbb{1}(x_1 < 100) \mathbb{1}(x_2 < 50)$. The model therefore estimates a localised interaction between x_1 and x_2 without manual specification. In contrast, parametric approaches such as Generalized Linear Models (GLMs) (Dobson et al., 2018) can represent interaction effects by including terms such as $x_1 x_2$ in the linear predictor, but these interactions must be specified explicitly as part of the model structure. Regression trees can discover interaction structures through the recursive partitioning process, allowing the relevant combinations of predictor values to be learned directly from the data without requiring their functional form to be specified in advance. Figure 2.4 illustrates the resulting step function response surface corresponding to Figure 2.2 under the arbitrary ordering $\mu_1 < \mu_2 < \mu_3 < \mu_4$.

2.1.4 Historical Development of Tree-Based Methods

The development of tree-based statistical learning methods spans more than half a century, driven by the need for flexible models capable of capturing nonlinear relationships and interaction effects without requiring strong parametric assumptions. The origins of this literature can be traced to the Automatic Interaction Detector (AID) introduced by Morgan and Sonquist (1963), which is widely regarded as the first implementation of a regression tree algorithm. AID operated through a greedy, forward-selection mechanism. Starting with the entire dataset contained within a root node, the algorithm evaluated every possible binary split across all available explanatory variables. The objective function chooses the split that produces the largest reduction in within-group variation. This represented a substantial departure from traditional parametric modelling approaches, as interaction effects and non-linear relationships could be discovered automatically rather than specified by the researcher in advance. While revolutionary for exploratory data analysis, AID suffered from being highly susceptible to extreme overfitting, generating models that failed to generalise to out-of-sample data. The algorithm lacked a principled stopping mechanism, halting only when the variance reduction fell below an arbitrary threshold or when sample sizes became too small.

Subsequent developments sought to address these limitations through alternative splitting criteria and tree construction procedures. Notable examples include the Theta Automatic Interaction Detector (THAID) of Morgan and Messenger (1973), which extended AID to accommodate a broader range of response variables, and the Chi-squared Automatic Interaction Detector (CHAID) introduced by Kass (1980), which selected predictor splits using chi-squared significance tests and merged statistically indistinguishable categories. While these methods represented important advances in recursive partitioning, they remained specialised algorithms and lacked a unified framework for tree construction, pruning, and statistical inference.

However, it was the work of Breiman et al. (1984) that established decision trees as a general statistical learning methodology through the introduction of Classification and Regression Trees (CART). CART formalised recursive partitioning through binary decision rules and demonstrated that a decision tree can be viewed as a non-parametric estimator of the form

$$f(x) = \sum_{b=1}^B \mu_b \mathbb{1}(x \in \mathcal{A}_b)$$

where the regions $(\mathcal{A}_m)$ are defined through a sequence of recursive splits. Beyond providing a unified framework for both regression and classification, CART intro-

duced cost-complexity pruning and cross-validation procedures to control overfitting, transforming decision trees from largely heuristic tools into statistically principled predictive models. They additionally provided the rigorous asymptotic theory and large sample proofs that the methodology had previously lacked, re-framing tree-based models as a rigorously defined statistical tool.

Despite these advances, a fundamental limitation of individual decision trees remained their instability. Small perturbations in the training data can produce substantially different tree structures and predictions, resulting in high variance and poor predictive robustness. This observation motivated a major shift in the literature towards ensemble methods, in which multiple trees are combined to form a single predictive model.

One of the earliest and most influential ensemble approaches was Bootstrap Aggregation, or Bagging, introduced by Breiman (1996). Bagging constructs multiple bootstrap samples from the training data, fits a separate CART tree to each sample, and aggregates the resulting predictions through averaging for regression or majority voting for classification. By averaging across many trees, Bagging substantially reduces variance while preserving the flexibility of individual decision trees.

Building upon Bagging, Breiman (2001) introduced Random Forests. In addition to bootstrap resampling, Random Forests randomly restrict the set of candidate predictors available at each split. This additional source of randomness diversifies the trees within the ensemble, producing more flexible learners and further reducing variance. Random Forests rapidly became one of the most successful machine learning algorithms due to their strong predictive performance, robustness, and minimal tuning requirements (Biau, 2012). Random forests have also seen popularity in the task of variable selection (identifying the set of most important predictors), making it a more complete inferential framework.

A second major ensemble paradigm emerged through boosting. Unlike Bagging and Random Forests, which construct trees independently, boosting builds trees sequentially, with each new tree attempting to improve upon the errors of the existing ensemble. Early boosting algorithms such as AdaBoost in Freund et al. (1997) demonstrated that combining many weak learners could yield highly accurate predictive models. This framework was subsequently generalised by Friedman et al. (2000), culminating in Gradient Boosting (Friedman, 2001). Gradient Boosting interprets model fitting as an optimisation problem, where each successive tree is fit to the negative gradient of a specified loss function. This produces an additive model consisting of many weak decision trees, capable of approximating highly complex nonlinear functions.

The development of Bagging, Random Forests and Gradient Boosting established that ensembles of trees consistently outperform individual decision trees. However, these methods remained fundamentally frequentist and relied on greedy optimisation procedures to determine tree structures. Consequently, they provide only a single estimate of the underlying predictive function and offer limited mechanisms for quantifying uncertainty. These limitations motivated a Bayesian reformulation of tree-based learning, beginning with Bayesian CART models in the late 1990s and culminating in the development of Bayesian Additive Regression Trees (BART), which forms a primary focus of this thesis.

2.2 The Bayesian Additive Regression Trees (BART) model

The tree based methods discussed so far, including CART, Random Forests and Gradient Boosting, differ substantially in their algorithmic construction yet share a common statistical objective: the production of a single point estimate for the predictive function. CART selects a single tree structure from the space of possible trees, Random Forests construct a single ensemble predictor by averaging a collection of independently fitted trees, and Gradient Boosting sequentially builds a single additive model through the accumulation of weak learners. In each case, these algorithms yield a single fixed estimator $\hat{f}(x)$. However, in many applications, there may exist numerous tree structures or ensembles that provide similarly plausible explanations of the observed data. By selecting only one model, the uncertainty arising from this structural multiplicity is entirely discarded, and all subsequent predictions and inferences are conditionally restricted to this single choice.

A Bayesian approach adopts a fundamentally different perspective. Rather than selecting a single model and treating it as fixed, the model structure itself is regarded as uncertain. Tree structures and their associated parameters are therefore assigned prior distributions, and inference is based on the posterior distribution over the space of plausible models supported by the observed data. Consequently, the objective is no longer to identify a single “best” tree, but rather to characterise the collection of plausible trees supported by the data and their associated posterior probabilities.

2.2.1 The Sum of Trees Model

In the Bayesian paradigm, predictions are therefore typically obtained by averaging over a posterior distribution of models rather than relying on a single fitted model; naturally accounting for uncertainty in the tree topology and parameter values. As we will see, a key advantage here is that it provides a principled mechanism for regularisation through prior formulations. This Bayesian perspective forms the foundation of Bayesian Additive Regression Trees (BART), where posterior inference is performed over an ensemble of trees rather than a single tree.

The BART model, introduced by Chipman et al. (2010), is a flexible, non-parametric ensemble method capable of modelling complex, non-linear relationships and interactions while mitigating overfitting through carefully designed regularisation priors. BART’s Bayesian framework allows for quantification of predictive uncertainty, which has contributed to its widespread popularity across sectors such

as finance and environmental sciences (Zhang et al., 2008; Li, 2020). Posterior inference in BART relies on Markov Chain Monte Carlo (MCMC) methods that extend techniques originally developed for single tree Bayesian models. In particular, BART employs a blocked Metropolis-within-Gibbs strategy, as developed for Bayesian CART (Chipman et al., 1998; David G. T. Denison et al., 1998; Wu et al., 2007). Chipman et al. (2010) extends this sampler to the ensemble of trees setting and demonstrates its efficiency in exploring the high dimensional space of additive trees.

We begin our introduction to BART by first outlining the objective of this model. Our objective here is the classical regression goal of constructing a model which can capture the relationship between a vector of covariates x_i and a response variable y_i for observations $i = 1, \dots, n$. We assume that the data is generated by an unknown response generating function $f(x_i)$ subject to additive noise. The aim is to estimate $f(x_i)$ by a “Sum of trees” model, as originally outlined in Chipman et al. (2010), with the aim of using it for prediction. Continuing notation established previously, a single regression tree models the response by

$$y_i = g(x_i; T, M) + \varepsilon_i, \quad \varepsilon_i \sim \mathcal{N}(0, \sigma^2) \quad (2.9)$$

Instead, BART sums the output of an ensemble of regression trees to produce the following additive model:

$$y_i = \sum_{j=1}^m g(x_i; T_j, M_j) + \varepsilon_i, \quad \varepsilon_i \sim \mathcal{N}(0, \sigma^2), \quad (2.10)$$

where T_j and M_j represent the j^{th} tree structure and terminal node values respectively. As defined in Section 2.1.2, $g(x_i; T_j, M_j)$ represents the individual prediction from the j^{th} tree for observation x_i . The number of trees, m , controls the size of the ensemble. In their investigation into the model’s sensitivity to m , Chipman et al. (2010) found that increasing the number of trees beyond a certain point provided little additional improvement in predictive performance, leading them to use $m = 200$ as a default. Subsequent work by Kapelner et al. (2014a) found that as few as $m = 50$ trees could be sufficient in many settings.

It is important to note that BART combines trees through addition rather than averaging. In Random Forests, the final prediction is obtained by averaging the predictions of many independently grown trees, with each tree acting as a relatively strong learner. In contrast, BART represents the regression function as a sum of many weak trees, where each tree contributes only a small component of the overall

signal through its terminal node parameters (μ_{jb}). The objective is therefore not to construct a collection of competing predictions and average them, but rather to build the target function as an additive decomposition across many simple trees.

Aside from how the individual predictions from trees are aggregated, another important distinction between BART and other tree based ensemble methods is that in BART, the individual learners (trees) are inherently kept weak through regularisation. In practice, this means that individual trees are kept shallow and will be heavily penalised for growing deeper, unless the data demands it. In short, BART’s design as a “basis of weak learners” enables it to capture complex relationships, whilst maintaining flexibility. Section 2.2.2 provides further details on this property. Additionally, simple example of how a BART ensemble can be used for prediction can be found in Appendix A.1

2.2.2 Regularisation Priors

Consider the task of modelling a complex response surface using a single decision tree. To capture intricate non-linear relationships and interactions, the tree must introduce increasingly fine partitions of the predictor space. Such a tree can become heavily invested in particular local patterns present in the training data, making its predictions sensitive to small changes in the data and potentially leading to poor generalisation. BART mitigates this overfitting by ensuring that the modelling capability is spread across many weak learners. By restricting each tree to remain shallow, the model distributes the representation of the underlying signal across the full ensemble. The resulting model is highly flexible as each tree therefore makes only a small contribution to the overall fit and can be readily altered without substantially changing the fitted function. Model complexity is consequently achieved through the shared contribution of many simple partitions rather than through the intricate partition produced by a single complex tree.

The remaining question then becomes how BART can prevent individual trees from becoming too complex. Classical tree-based methods typically control tree complexity through algorithmic procedures such as cost-complexity pruning (Breiman et al., 1984), or through mechanisms such as early stopping and shrinkage (Friedman, 2001). BART adopts a different approach by incorporating regularisation directly into the probabilistic formulation through prior distributions. These priors place greater probability on shallow tree structures and constrain the magnitude of terminal node parameters, encouraging each tree to function as a weak learner within the ensemble. The Bayesian formulation therefore provides direct control

over the complexity of the individual components that make up the ensemble. The following sections describe the priors governing tree structure and terminal node parameters, and show how they work together to produce this regularised ensemble.

The prior distributions used in BART build directly upon the Bayesian CART framework introduced by Chipman et al. (1998), where prior distributions were used to control tree complexity in single-tree models. Chipman et al. (2010) subsequently adapted these priors to the additive ensemble setting, where many weak trees collectively represent the regression function. The resulting prior specification is governed by a small number of interpretable hyperparameters that control the complexity of individual trees and the magnitude of their contributions to the ensemble. In the following, we describe the joint prior specification and the conditional independence assumptions underlying the BART model.

2.2.3 Assumptions and Prior Simplification

The primary objective of BART is to conduct posterior inference over the collection of tree structures, terminal node parameters, and the residual variance:

$$\pi((T_1, M_1), \dots, (T_m, M_m), \sigma^2 \mid x, y), \quad (2.11)$$

where T_j denotes the topology of the j^{th} tree and $M_j = (\mu_{j1}, \dots, \mu_{jB_j})$ denotes its associated set of B_j terminal node parameters. Prior to observing the data, uncertainty over these quantities is characterised by the joint prior distribution

$$\pi((T_1, M_1), \dots, (T_m, M_m), \sigma^2). \quad (2.12)$$

Directly specifying a joint prior over this high-dimensional space is challenging due to the size and complexity of the parameter space. For example, one challenge is that the cardinality of M_j is conditionally dependent on the given tree structure T_j . Instead, Chipman et al. (2010) impose three conditional independence assumptions that allow the prior to be decomposed into simpler components.

BART Prior Independence Assumptions

-
- (A1)** $\{(T_1, M_1), \dots, (T_m, M_m)\} \perp\!\!\!\perp \sigma^2$
 - (A2)** $(T_j, M_j) \perp\!\!\!\perp (T_k, M_k) \quad \text{for all } j \neq k$
 - (A3)** $\mu_{jb} \perp\!\!\!\perp \mu_{jb'} \mid T_j \quad \text{for all } b \neq b'$
-

(**A1**) specifies prior independence between the residual variance and the ensemble of regression trees. (**A2**) specifies prior independence between the individual tree components of the ensemble. This supports BART’s dispersed representation of complex functions by preventing any individual tree from being given a privileged role or dominating the ensemble a priori. Finally, (**A3**) specifies that, conditional on the tree structure T_j , the terminal node parameters within a tree are independent. Applying these assumptions allows the joint prior in Equation (2.12) to be factorised as follows:

$$\begin{aligned}\pi((T_1, M_1), \dots, (T_m, M_m), \sigma^2) &= \pi(\sigma^2) \prod_{j=1}^m \pi(T_j, M_j) \\ &= \pi(\sigma^2) \prod_{j=1}^m \pi(M_j | T_j) \pi(T_j) \\ &= \pi(\sigma^2) \prod_{j=1}^m \left[\pi(T_j) \prod_{b=1}^{B_j} \pi(\mu_{jb} | T_j) \right].\end{aligned}$$

The first factorisation follows from (**A1**), the second from (**A2**), and the final factorisation from (**A3**). The resulting specification separates the prior into three components: the tree topology prior $\pi(T_j)$, the terminal node parameter prior $\pi(\mu_{jb} | T_j)$, and the residual variance prior $\pi(\sigma^2)$. The specific forms and calibrations of these priors are described in the following subsections.

2.2.4 The Tree Topology Prior $\pi(T_j)$

Chipman et al. (2010)’s proposed prior over tree structures, $\pi(T_j)$, heavily penalises deep tree structures to control the complexity of individual trees within the ensemble. The specification closely follows the branching process formulation originally proposed for Bayesian CART by Chipman et al. (1998). All trees in the ensemble are assumed to be i.i.d. under this prior, which is defined by three sequential generative components:

- (**T1**) The probability that a node at depth $d \in \{0, 1, 2, \dots\}$ is nonterminal (i.e., undergoes a split) is given by:

$$p_{\text{split}}(d) = \alpha(1 + d)^{-\beta}, \quad (2.13)$$

where $\alpha \in (0, 1)$ dictates the base probability of a split and $\beta \geq 0$ controls the rate of decay as depth increases.

- (**T2**) Given that a node splits, the distribution determining which of the p available

covariates is chosen for the splitting rule. This is typically specified as a uniform discrete distribution over all valid predictors, yielding a probability of $1/p_{\text{valid}}$.

- (T3) Given the chosen splitting variable, the distribution determining the threshold value. This is typically assigned a uniform discrete distribution over all valid available split points, giving each valid threshold a probability of $1/p_{\text{split}}$

Rather than requiring extensive cross-validation or pilot runs to tune the hyper-parameters in (T1), Chipman et al. (2010) recommend the default values $\alpha = 0.95$ and $\beta = 2$, which they find perform reliably across a wide range of applications. As illustrated in Figure 2.5, this parameterisation assigns substantial prior mass to shallow trees, making trees of depth four or greater highly unlikely. This strong regularisation limits the complexity of individual tree structures and discourages excessive recursive partitioning.

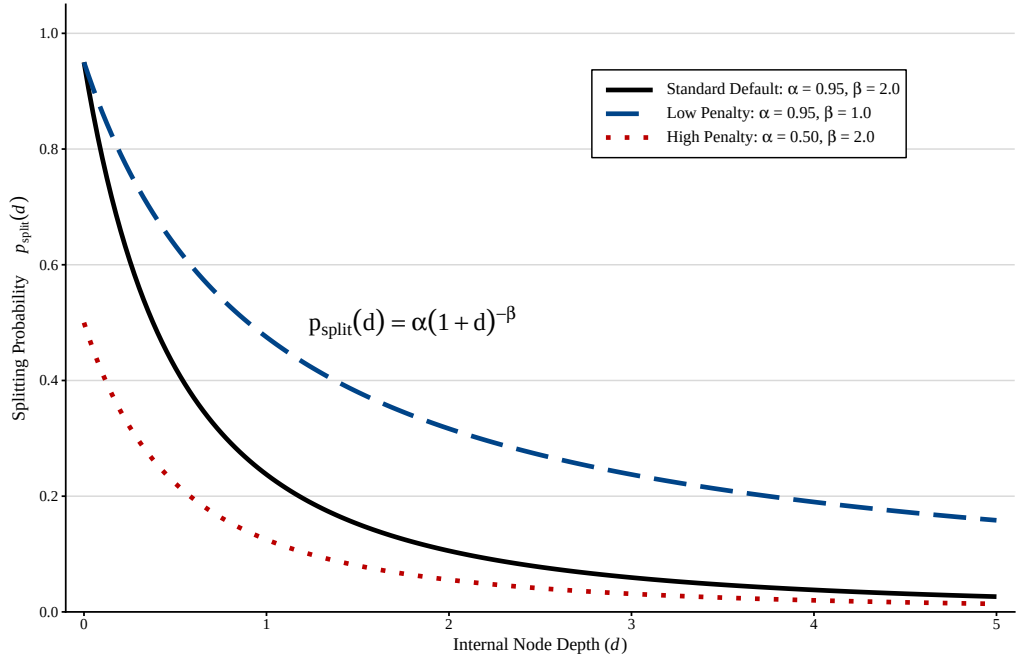


Figure 2.5: The prior probability of a node splitting, $p_{\text{split}}(d) = \alpha(1+d)^{-\beta}$, evaluated across tree depth $d \in \{0, \dots, 5\}$. The standard default specification ($\alpha = 0.95, \beta = 2.0$) heavily penalises depths beyond $d = 2$, structurally restricting individual trees within the ensemble to act as weak learners.

The distributions in (T2) and (T3) depend on the concept of valid splits. A **valid split** is one that partitions the observations in the current node into two non empty child nodes. For continuous variables, the threshold is selected uniformly from the set of distinct observed values within that node. For nominal categorical

variables with q levels, the variable is typically expanded into q binary indicators via one-hot-encoding, and the split is chosen uniformly from the available categories. Consequently, a variable becomes invalid for splitting if all remaining observations in the node share the same value for that predictor.

Let $\mathcal{I}_j$ denote the set of internal nodes and $\mathcal{L}_j$ denote the set of terminal leaves for a given tree T_j . Combining components **(T1)**, **(T2)**, and **(T3)**, the complete prior probability of a specific tree structure factorises as the product of the splitting probabilities and corresponding variable and threshold probabilities across all internal nodes, together with the stopping probabilities across all terminal nodes:

$$\pi(T_j) = \prod_{\eta \in \mathcal{I}_j} \left[\alpha(1 + d_\eta)^{-\beta} \frac{1}{p_{\text{valid}}(\eta)} \frac{1}{n_{\text{val}}(\eta)} \right] \prod_{\eta \in \mathcal{L}_j} \left[1 - \alpha(1 + d_\eta)^{-\beta} \right], \quad (2.14)$$

where d_η denotes the depth of node η , $p_{\text{valid}}(\eta)$ denotes the number of predictors that admit a valid split at node η , and $n_{\text{val}}(\eta)$ denotes the number of valid split points for the predictor selected at node η . The factors $1/p_{\text{valid}}(\eta)$ and $1/n_{\text{val}}(\eta)$ therefore represent the uniform prior probabilities assigned to the selected splitting variable and split point.

The uniform assignment in **(T2)** provides an impartial prior over the available predictors, assigning each an equal probability of being selected. This allows differences in variable importance to be driven by the data rather than by prior preferences. This assumption can be relaxed when prior knowledge suggests that some predictors should receive greater weight, in which case the uniform distribution can be replaced by a weighted distribution (Bleich et al., 2014a). Alternatively, as discussed in Section 2.2.6, methods such as Dirichlet Additive Regression Trees (DART) dynamically adapt the splitting probabilities to encourage variable selection (Linero, 2018).

2.2.5 The Terminal Node Prior $\pi(\mu_{jb} \mid T_j)$

While the tree topology prior $\pi(T_j)$ regularises the ensemble by restricting the complexity of the input space partitions, a tree may still become disproportionately influential if its terminal node values are permitted to grow excessively large. Recall that the conditional mean function $f(x) = \mathbb{E}[y \mid x]$ is constructed as the additive summation of the terminal node parameters μ_{jb} corresponding to the input x across all m trees. To constrain the magnitude of each tree’s contribution, BART imposes a regularising shrinkage prior on the leaf parameters. Specifically, conditional on

the tree structure T_j , each terminal node is assigned an i.i.d. distributed Gaussian prior:

$$\mu_{jb} \sim \mathcal{N}(\mu_\mu, \sigma_\mu^2). \quad (2.15)$$

Following the independence assumptions **(A2)** and **(A3)**, the additive structure of BART implies that the prior induced on the conditional mean function at any fixed evaluation point x is:

$$f(x) \sim \mathcal{N}(m\mu_\mu, m\sigma_\mu^2). \quad (2.16)$$

Chipman et al. (2010) suggest a data-informed calibration strategy. This approach relies on the assumption that the true conditional mean $f(x)$ is expected to lie within, or close to, the observed range of the response data $(y_{\min}, y_{\max})$. The strategy configures μ_μ and σ_μ such that the prior distribution over $f(x)$ assigns substantial probability mass to this interval. For a chosen bounding constant k , this is achieved by equating the interval limits to the k standard deviation bounds of the prior:

$$m\mu_\mu - k\sqrt{m}\sigma_\mu = y_{\min} \quad (2.17)$$

$$m\mu_\mu + k\sqrt{m}\sigma_\mu = y_{\max}. \quad (2.18)$$

Solving this system of equations simultaneously yields:

$$\mu_\mu = \frac{y_{\max} + y_{\min}}{2m}, \quad \sigma_\mu = \frac{y_{\max} - y_{\min}}{2k\sqrt{m}}. \quad (2.19)$$

To simplify this calibration, standard BART implementations rescale the response variable such that $y \in [-0.5, 0.5]$. Under this transformation, the prior mean becomes centred at zero, $\mu_\mu = 0$, and the prior standard deviation simplifies to:

$$\sigma_\mu = \frac{0.5}{k\sqrt{m}}. \quad (2.20)$$

This formulation concentrates the individual terminal node parameters around zero, limiting the contribution of any single tree to the overall ensemble. As the number of trees m increases, σ_μ decreases proportionally to $1/\sqrt{m}$, ensuring that the prior contribution of each individual tree becomes smaller as the ensemble grows. The hyperparameter k controls the degree of shrinkage, with larger values producing a more concentrated prior. Chipman et al. (2010) recommend a default value of $k = 2$, which places approximately 95% of the prior mass for $f(x)$ within the observed

response range. If required, alternative values of k can be selected through cross-validation on the training data.

Figure 2.6 displays the marginal prior distribution for different values of k . As expected, smaller values of k produce wider tails and therefore less aggressive shrinkage. Finally, only the response variable requires rescaling for the calibration of the terminal node prior. No analogous scaling is required for the predictors because tree-based splitting depends on the ordering of predictor values rather than their absolute scale. For a strictly monotonic transformation g , a split of the form $x_j < c$ is equivalent to $g(x_j) < g(c)$, so the same observations are assigned to each child node. Consequently, rescaling or standardising a continuous predictor does not alter the partitions available to the tree.

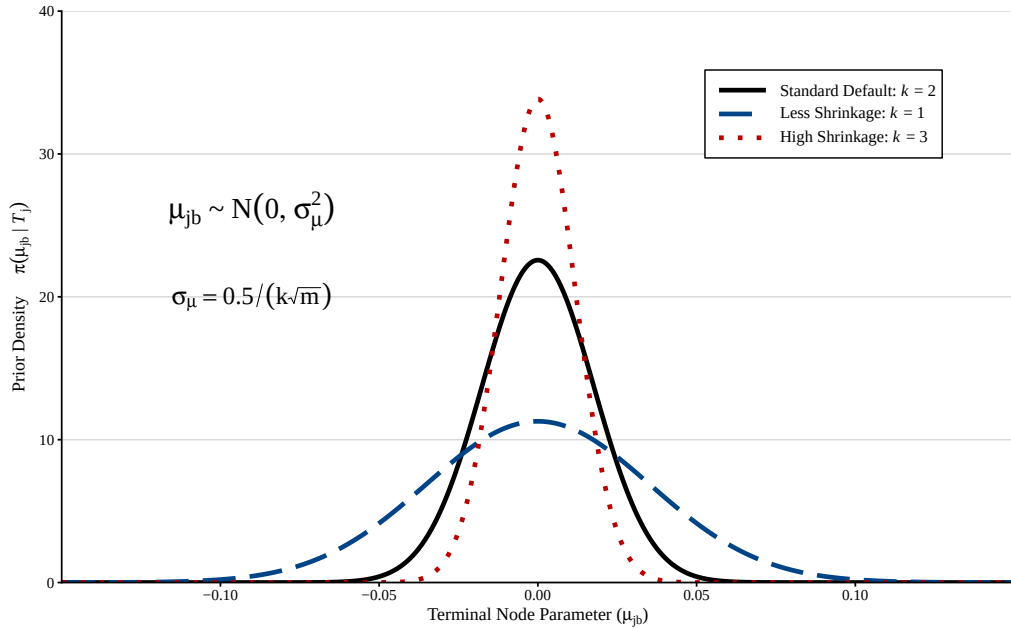


Figure 2.6: The marginal prior density of the terminal node parameters μ_{jb} for an ensemble of $m = 200$ trees across varying shrinkage hyperparameters k . The standard default specification of $k = 2$ applies substantial regularisation, concentrating the leaf parameters around zero.

2.2.6 The Error Variance Prior $\pi(\sigma^2)$

To model the residual variance σ^2 , BART employs a conjugate scaled inverse chi-squared distribution:

$$\sigma^2 \sim \frac{\nu\lambda}{\chi_\nu^2}, \quad (2.21)$$

where ν represents the degrees of freedom and λ denotes the scale parameter. Similar to the calibration of the terminal node parameters μ_{jb} , the hyperparameters ν

and λ are selected using a data-informed strategy. The objective is to place substantial prior probability below a conservative overestimate of the residual standard deviation, denoted by $\hat{\sigma}$. Rather than arbitrarily selecting this estimate, Chipman et al. (2010) propose obtaining a rough upper bound using one of the following two approaches:

Table 2.2: Methods for determining the conservative upper bound $\hat{\sigma}$

Method	Description
$\hat{\sigma}_{\text{SD}}$	The sample standard deviation of the response vector y , corresponding to the assumption that the observed variation is entirely attributable to residual noise.
$\hat{\sigma}_{\text{OLS}}$	The residual standard deviation from an ordinary least squares regression. Since a flexible BART model can capture nonlinearities and interactions that a linear model cannot, the OLS residual standard deviation provides a rough conservative overestimate of the residual variation.

Once $\hat{\sigma}$ is established, a value of ν is chosen and the scale parameter λ is selected such that the q^{th} quantile of the prior distribution for σ is equal to $\hat{\sigma}$:

$$\mathbb{P}(\sigma < \hat{\sigma}) = q. \quad (2.22)$$

The hyperparameter q controls the proportion of prior probability assigned below the conservative estimate $\hat{\sigma}$. It is typically selected from $\{0.75, 0.90, 0.99\}$, with larger values resulting in greater concentration of the prior towards smaller values of σ . The degrees of freedom parameter ν controls the shape of the prior and is typically chosen between 3 and 10. Chipman et al. (2010) recommend values within this range to avoid overly diffuse or concentrated prior specifications, with the default pairing $(\nu, q) = (3, 0.90)$ recommended for automatic use. If further calibration is required for a specific dataset, alternative values can be selected through cross-validation. In high-dimensional settings where $p > n$, the ordinary least squares estimate is unavailable, in which case the sample standard deviation provides the alternative estimate $\hat{\sigma}_{\text{SD}}$. Figure 2.7 illustrates the marginal prior density over σ for three distinct calibrations when the upper bound is $\hat{\sigma} = 1$: the conservative formulation $(10, 0.75)$, the standard default $(3, 0.90)$, and the more concentrated formulation $(3, 0.99)$.

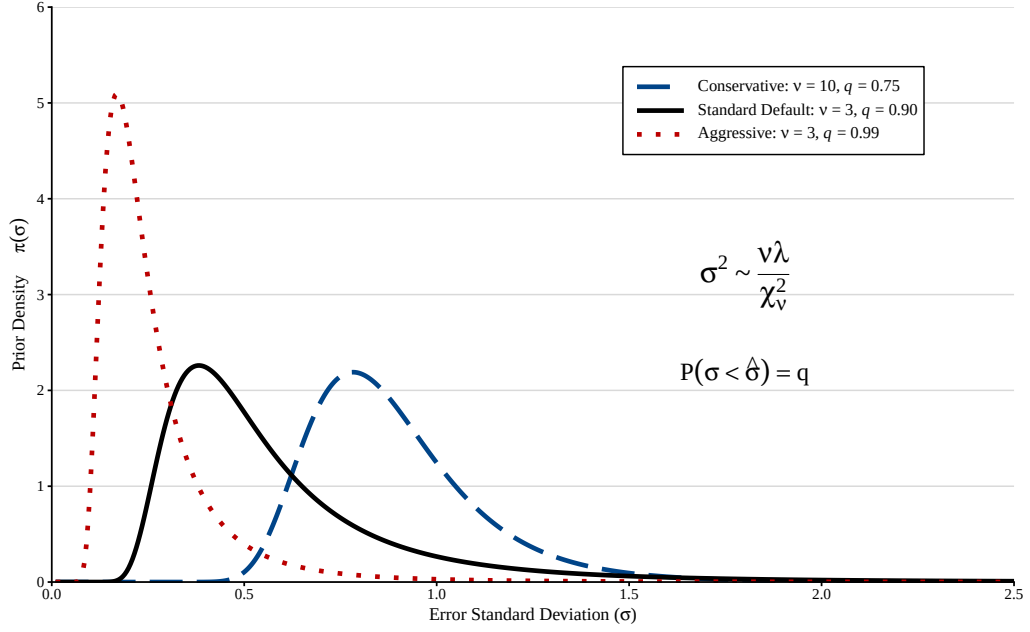


Figure 2.7: The marginal prior density of the residual standard deviation σ across three distinct calibration regimes, assuming a conservative upper bound of $\hat{\sigma} = 1$. The standard default specification ($\nu = 3, q = 0.90$) lies between the more diffuse $(10, 0.75)$ and more concentrated $(3, 0.99)$ specifications.

Alternative Prior Specifications

While the default priors of Chipman et al. (2010) established here remain the standard benchmark due to their useful conjugacy properties we will later see, several alternative prior formulations have emerged in the literature. To address the lack of smoothness inherent in tree-based models, Linero and Yang (2018a) introduced SoftBART, which replaces deterministic splitting rules with probabilistic, smooth decision boundaries. A bandwidth parameter controls the sharpness of these boundaries, with smaller bandwidths producing transitions that more closely resemble the hard splits of standard BART. The authors place a prior on these bandwidth parameters, allowing the model to adapt to the unknown smoothness of the underlying regression function. Ročková and Stephanie van der Pas (2019) developed the Spike and Forest prior framework, which combines tree-based modelling with sparsity-inducing priors to perform variable selection in high-dimensional settings. Furthermore, to relax the strict homoscedasticity assumption, Pratola et al. (2020) introduced a secondary sum of trees prior over log residual variance to accommodate input dependent variance $\sigma^2(x_i)$, while Du et al. (2019) introduced group Dirichlet priors to allow for processing of categorical predictors.

2.3 Posterior Inference for BART

Having fully defined the prior distributions over the tree topologies, terminal node parameters, and residual error variance, the remaining objective is to conduct Bayesian posterior inference. Specifically, we seek to generate dependent draws from the joint posterior distribution:

$$\pi((T_1, M_1), \dots, (T_m, M_m), \sigma^2 \mid x, y). \quad (2.23)$$

The size and complexity of the parameter space makes direct evaluation of this posterior distribution challenging. The parameter space comprises both continuous parameters and discrete tree topologies whose combinatorial complexity grows factorially with the number of predictors and possible split points across all m trees. Furthermore, the model is inherently transdimensional: modifying the topology of a tree T_j directly alters the dimensionality of its associated terminal parameter vector M_j . Consequently, the joint posterior does not admit a closed analytical form, necessitating iterative Markov Chain Monte Carlo (MCMC) sampling techniques. Posterior computation in BART relies on a Bayesian backfitting algorithm that integrates Gibbs sampling with Metropolis Hastings steps over tree structures. Readers unfamiliar with Gibbs sampling or the Metropolis Hastings algorithm are referred to Appendix A.2 and Appendix A.3 for theoretical reviews.

2.3.1 The BART Sampling Procedure

BART employs a Gibbs sampling strategy, leveraging the Bayesian backfitting framework originally developed for single tree models by Chipman et al. (1998), David G. T. Denison et al. (1998), and Wu et al. (2007). The objective is to produce a dependent sequence of samples from the joint posterior distribution:

$$\pi((T_1, M_1), \dots, (T_m, M_m), \sigma^2 \mid x, y). \quad (2.24)$$

For notational convenience, let $T_{(j)}$ denote the set of all tree structures within the ensemble excluding T_j . Similarly, define $M_{(j)}$ as the set of all terminal node parameter vectors excluding those associated with the j^{th} tree. The size of the parameter space makes obtaining an analytical closed form unfeasible, therefore inference proceeds via Gibbs sampling. The procedure cycles through the ensemble, executing m successive draws of (T_j, M_j) conditional on the remaining trees and the data, followed by a final draw of the variance parameter σ^2 . A single iteration of this standard sampler proceeds as follows:

$$\begin{aligned}
&1 : (T_1, M_1) \mid (T_{(1)}, M_{(1)}), \sigma^2, x, y \\
&\quad \vdots \\
&m : (T_m, M_m) \mid (T_{(m)}, M_{(m)}), \sigma^2, x, y \\
&m + 1 : \sigma^2 \mid (T_1, M_1), \dots, (T_m, M_m), x, y
\end{aligned}$$

Implementing this standard formulation directly is computationally challenging, as the full conditional distribution for the j^{th} tree requires explicitly evaluating the functional contributions of the other $m - 1$ trees. However, Chipman et al. (2010) note that the additive structure of the ensemble permits a simplification. Define the partial residual vector associated with the j th tree as:

$$R_j = y - \sum_{k \neq j} g(x; T_k, M_k). \quad (2.25)$$

This vector R_j isolates the portion of the response data that remains unexplained by all trees except T_j . Conditionally, the evaluation of (T_j, M_j) depends on the remaining trees $(T_{(j)}, M_{(j)})$ and the response y exclusively through this partial residual R_j . Consequently, rather than conditioning explicitly on the entire ensemble, inference for the j^{th} tree can be performed by treating R_j as the effective target response. The sampling iteration therefore simplifies to the following backfitting procedure:

$$\begin{aligned}
&1 : (T_1, M_1) \mid R_1, \sigma^2 \\
&\quad \vdots \\
&m : (T_m, M_m) \mid R_m, \sigma^2 \\
&m + 1 : \sigma^2 \mid (T_1, M_1), \dots, (T_m, M_m), x, y
\end{aligned}$$

Rather than fitting each tree to the full response vector y , the j th tree is sequentially updated to fit the residual variation left unexplained by the rest of the ensemble. As the algorithm cycles, the trees iteratively refine the residuals of the ensemble, improving the global approximation of the true conditional mean function.

Despite this improvement, drawing directly from the joint conditional $\pi((T_j, M_j) \mid R_j, \sigma^2)$ is not straightforward and requires a Metropolis-Hastings (MH) step. Difficulty arises since the model is transdimensional: the dimension of the parameter vector M_j is strictly determined by the topology of T_j . A standard MH step cannot compare two states whose parameter vectors have unequal dimension. In general, constructing a valid MCMC move between parameter spaces of differing dimension

requires the Reversible Jump MCMC (RJMCMC) framework of Green (1995). This complexity can be avoided here because M_j can be integrated out of the joint posterior in closed form, removing the dimension-varying component from the update before any MH step is taken. This decomposition is made explicit by the standard probability factorisation:

$$\pi(T_j, M_j \mid R_j, \sigma^2) = \pi(M_j \mid T_j, R_j, \sigma^2)\pi(T_j \mid R_j, \sigma^2). \quad (2.26)$$

This factorisation dictates a two step sampling procedure for each tree:

1. Draw the tree topology $T_j \sim \pi(T_j \mid R_j, \sigma^2)$.
2. Conditioned on the new topology, draw the leaf parameters $M_j \sim \pi(M_j \mid T_j, R_j, \sigma^2)$.

To execute the first step, the terminal node parameters M_j must be analytically integrated out of the joint distribution to yield the marginal posterior of the tree structure. By Bayes theorem and the law of total probability:

$$\pi(T_j \mid R_j, \sigma^2) \propto \pi(T_j) \int \pi(R_j \mid T_j, M_j, \sigma^2)\pi(M_j \mid T_j)dM_j. \quad (2.27)$$

This integral evaluates to a closed form solution because of the conjugate prior specifications established in Section 2.2.5. For the subset of observations falling into the b^{th} terminal node of tree T_j , the partial residuals follow a Gaussian distribution parameterised by the associated leaf value μ_{jb} :

$$R_{j,x \in \text{node } b} \mid \mu_{jb}, \sigma^2 \sim \mathcal{N}(\mu_{jb}, \sigma^2). \quad (2.28)$$

Simultaneously, the prior on the leaf parameter is also Gaussian:

$$\mu_{jb} \sim \mathcal{N}(0, \sigma_\mu^2). \quad (2.29)$$

Because the normal likelihood and normal prior form a conjugate pair, the vector M_j can be integrated out exactly. This analytical marginalisation yields the marginal posterior $\pi(T_j \mid R_j, \sigma^2)$ up to a normalising constant, which allows the tree structure to be updated independent of its leaf parameters and therefore removes the need for complex RJMCMC methods. Once a new tree T_j is drawn, the corresponding terminal node parameters are drawn from $\pi(M_j \mid T_j, R_j, \sigma^2)$, which is a standard Gaussian posterior. This decomposition transforms a complex transdimensional update into the following sequence of tractable steps:

BART MCMC MH-within-Gibbs Backfitting Algorithm

$$\begin{aligned}
 1 & : T_1 \mid R_1, \sigma^2 \\
 2 & : M_1 \mid T_1, R_1, \sigma^2 \\
 & : \vdots \\
 2m-1 & : T_m \mid R_m, \sigma^2 \\
 2m & : M_m \mid T_m, R_m, \sigma^2 \\
 2m+1 & : \sigma^2 \mid (T_1, M_1), \dots, (T_m, M_m), x, y
 \end{aligned}$$

This iteration above represents a single draw from the MH-within-Gibbs BART sampling procedure. While draws for the terminal parameters M_j and the variance σ^2 are taken from known exact posterior distributions, the structural update $\pi(T_j \mid R_j, \sigma^2)$ cannot be sampled from directly. Instead, it requires a MH step to propose and evaluate changes to the tree topology. The process of drawing from $\pi(T_j \mid R_j, \sigma^2)$ and $\pi(M_j \mid T_j, R_j, \sigma^2)$ will now be detailed in the following sections.

2.3.2 Sampling from $\pi(T_j \mid R_j, \sigma^2)$: Metropolis Hastings

We now consider steps $1, 3, \dots, 2m - 1$ of the backfitting algorithm, corresponding to structural draws from the marginal tree posterior $\pi(T_j \mid R_j, \sigma^2)$. As established in Section 2.3.1, this distribution is defined over the discrete space of all possible tree topologies. Consequently, whilst the posterior probability of a proposed tree can be evaluated up to a normalising constant, direct sampling would require evaluating this normalising constant over the entire tree space, which is computationally impractical for a large ensemble. To avoid this, structural updates are performed using an MH step.

Given the current state of the Markov chain T_j , a candidate tree T_j^* is proposed through a local structural modification drawn from a transition distribution $q(T_j^* \mid T_j)$. Table 2.3 outlines the set of permitted structural modifications, with visual illustrations provided in Figure 2.8. Following Chipman et al. (2010), the proposal mechanism selects one of four distinct transition types with specified probabilities: GROW a terminal node (0.25), PRUNE a pair of terminal nodes (0.25), CHANGE a non-terminal splitting rule (0.40), or SWAP a splitting rule between a parent and child node (0.10). Once a move type is selected and a candidate tree T_j^* is proposed, the modification is accepted with probability:

$$\alpha(T_j, T_j^*) = \min \left(1, \frac{\pi(T_j^* \mid R_j, \sigma^2) q(T_j \mid T_j^*)}{\pi(T_j \mid R_j, \sigma^2) q(T_j^* \mid T_j)} \right). \quad (2.30)$$

The marginal posterior of the tree structure is defined only up to a normalising constant. Since the MH acceptance probability depends on a ratio of posterior probabilities, this constant cancels. Applying Bayes' theorem gives

$$\frac{\pi(T_j^* \mid R_j, \sigma^2)}{\pi(T_j \mid R_j, \sigma^2)} = \frac{\pi(R_j \mid T_j^*, \sigma^2) \pi(T_j^* \mid \sigma^2)}{\pi(R_j \mid T_j, \sigma^2) \pi(T_j \mid \sigma^2)}. \quad (2.31)$$

Under the prior independence assumption (A1), the tree structure and residual variance are independent a priori, such that $\pi(T_j \mid \sigma^2) = \pi(T_j)$. The acceptance probability can therefore be expressed as

$$\alpha(T_j, T_j^*) = \min \left(1, \underbrace{\frac{q(T_j \mid T_j^*)}{q(T_j^* \mid T_j)}}_{\text{Proposal ratio}} \times \underbrace{\frac{\pi(R_j \mid T_j^*, \sigma^2)}{\pi(R_j \mid T_j, \sigma^2)}}_{\text{Marginal likelihood ratio}} \times \underbrace{\frac{\pi(T_j^*)}{\pi(T_j)}}_{\text{Tree prior ratio}} \right). \quad (2.32)$$

To execute the update, a random uniform variable $u \sim \mathcal{U}(0, 1)$ is drawn. If $u \leq \alpha(T_j, T_j^*)$, the candidate is accepted and the tree is updated such that $T_{j+1} = T_j^*$;

otherwise, the chain rejects the proposal and remains at T_j . It is worth noting that the four move probabilities apply only when the corresponding move is valid for the current tree T_j . For example, a PRUNE move cannot be proposed from a root stump. The proposal probabilities are therefore renormalised over the valid move types at each T_j , ensuring that $q(\cdot \mid T_j)$ remains a valid proposal distribution.

To summarise, a draw from the marginal tree distribution $\pi(T_j \mid R_j, \sigma^2)$ is obtained by first selecting a structural modification move, proposing a candidate tree, and evaluating the acceptance probability dictated by Equation 2.32. Executing this requires explicit analytical derivation of the transition ratio, likelihood ratio, and prior ratio for the specific move type. We now provide the explicit derivations for the GROW proposal. Analogous results for the PRUNE, SWAP, and CHANGE are widely documented and we therefore refer to Kapelner et al. (2014b) for derivations. Additionally, manual demonstration of these structural Metropolis Hastings modifications applied to a minimal dataset is provided in Appendix A.5.

GROW Transition Ratio

Recall from Equation (2.32) that the transition ratio for a proposed structural modification from state T to candidate state T^* is defined as:

$$\frac{q(T \mid T^*)}{q(T^* \mid T)}. \quad (2.33)$$

To derive the forward transition probability $q(T^* \mid T)$, we decompose the generation of a GROW proposal into four sequential, independent uniform choices:

1. Select the GROW proposal type with prior proposal probability $\mathbb{P}(\text{GROW})$.
2. Select a terminal node η uniformly at random from the b available terminal nodes in the current tree T , with probability $1/b$.
3. Select an eligible splitting variable x_k uniformly from the $p_{\text{adj}}(\eta)$ available candidate predictors that can be partitioned at node η , with probability $1/p_{\text{adj}}(\eta)$.
4. Select a split value uniformly from the $n_{k,\text{adj}}(\eta)$ available interior cutoff thresholds for predictor x_k at node η , with probability $1/n_{k,\text{adj}}(\eta)$.

Multiplying these independent components yields the complete forward transition probability:

$$q(T^* \mid T) = \mathbb{P}(\text{GROW}) \frac{1}{b} \frac{1}{p_{\text{adj}}(\eta)} \frac{1}{n_{k,\text{adj}}(\eta)}. \quad (2.34)$$

Conversely, the reverse transition probability $q(T \mid T^*)$ represents the probability of proposing a transition from candidate tree T^* back to the original structure T . Reversing a GROW modification requires selecting a PRUNE move and collapsing the two newly created children back into the original single terminal node η .

Let w_2^* denote the number of second generation internal nodes in T^* , defined as non-terminal nodes possessing two terminal child nodes. The reverse move selects the PRUNE proposal type with probability $\mathbb{P}(\text{PRUNE})$ and chooses uniformly among the w_2^* available prunable nodes. Hence, the reverse transition probability is:

$$q(T \mid T^*) = \mathbb{P}(\text{PRUNE}) \frac{1}{w_2^*}. \quad (2.35)$$

Taking the quotient of Equations (2.35) and (2.34) yields the analytical form of the GROW transition ratio:

$$\frac{q(T \mid T^*)}{q(T^* \mid T)} = \frac{\mathbb{P}(\text{PRUNE})}{\mathbb{P}(\text{GROW})} \frac{b p_{\text{adj}}(\eta) n_{k,\text{adj}}(\eta)}{w_2^*}. \quad (2.36)$$

Grow Marginal Likelihood Ratio

Recall from Equation (2.32) that the marginal likelihood ratio for a proposed structural modification is defined as:

$$\frac{\pi(R_j \mid T_j^*, \sigma^2)}{\pi(R_j \mid T_j, \sigma^2)}. \quad (2.37)$$

To evaluate this ratio, we exploit the conditional independence established by the tree topology. The tree structure T_j uniquely partitions the predictor space, allocating each observation into one of b disjoint terminal nodes. Given T_j , the subset of partial residuals within any terminal node is conditionally independent of the residuals in all other terminal nodes. Therefore, the joint marginal likelihood factorises across the leaves:

$$\pi(R_j \mid T_j, \sigma^2) = \prod_{\eta=1}^b \pi(R_{j,\eta} \mid \sigma^2), \quad (2.38)$$

where $R_{j,\eta}$ denotes the vector of n_η partial residuals allocated to terminal node η , such that the total number of observations is $n = \sum_{\eta=1}^b n_\eta$.

Evaluating the right hand side requires marginalising out the terminal node parameters. For any generic terminal node η , the marginal likelihood is obtained by integrating the joint Gaussian likelihood over the Gaussian prior assigned to the

leaf parameter μ_η :

$$\pi(R_{j,\eta} \mid \sigma^2) = \int_{\mathbb{R}} \pi(R_{j,\eta} \mid \mu_\eta, \sigma^2) \pi(\mu_\eta \mid \sigma_\mu^2) d\mu_\eta. \quad (2.39)$$

As derived previously in Section 2.3.1, the partial residuals falling into node η conditionally follow $\mathcal{N}(\mu_\eta, \sigma^2)$, and the conjugate prior on the leaf mean is $\mathcal{N}(0, \sigma_\mu^2)$. Substituting these densities yields:

$$\pi(R_{j,\eta} \mid \sigma^2) = \int_{\mathbb{R}} (2\pi\sigma^2)^{-\frac{n_\eta}{2}} \exp\left(-\frac{1}{2\sigma^2} \sum_{i=1}^{n_\eta} (R_{\eta,i} - \mu_\eta)^2\right) (2\pi\sigma_\mu^2)^{-\frac{1}{2}} \exp\left(-\frac{\mu_\eta^2}{2\sigma_\mu^2}\right) d\mu_\eta, \quad (2.40)$$

where $R_{\eta,i}$ denotes the i th residual observation in node η . Expanding the quadratic terms and completing the square for μ_η permits analytical integration, yielding the known closed form solution for the Gaussian marginal likelihood:

$$\pi(R_{j,\eta} \mid \sigma^2) = (2\pi\sigma^2)^{-\frac{n_\eta}{2}} \sqrt{\frac{\sigma^2}{\sigma^2 + n_\eta\sigma_\mu^2}} \exp\left(-\frac{1}{2\sigma^2} \sum_{i=1}^{n_\eta} R_{\eta,i}^2 + \frac{\sigma_\mu^2 \left(\sum_{i=1}^{n_\eta} R_{\eta,i}\right)^2}{2\sigma^2(\sigma^2 + n_\eta\sigma_\mu^2)}\right). \quad (2.41)$$

Because the likelihood is entirely determined by the terminal node assignments, the proposed tree likelihood $\pi(R_j \mid T_j^*, \sigma^2)$ differs from the current tree likelihood $\pi(R_j \mid T_j, \sigma^2)$ solely at the specific node selected for modification. Suppose a GROW move is applied to terminal node η in tree T_j , partitioning it into a new left child η_L and right child η_R in the candidate tree T_j^* . The marginal likelihoods of all unmodified terminal nodes strictly cancel out in the ratio, leaving only the components involved in the structural split:

$$\frac{\pi(R_j \mid T_j^*, \sigma^2)}{\pi(R_j \mid T_j, \sigma^2)} = \frac{\pi(R_{j,\eta_L} \mid \sigma^2) \pi(R_{j,\eta_R} \mid \sigma^2)}{\pi(R_{j,\eta} \mid \sigma^2)}. \quad (2.42)$$

Substituting Equation (2.41) into this quotient simplifies the expression significantly. Because the union of the newly grown child nodes perfectly reconstructs the original parent node, $n_{\eta_L} + n_{\eta_R} = n_\eta$ and $\sum R_{\eta_L,i}^2 + \sum R_{\eta_R,i}^2 = \sum R_{\eta,i}^2$. Consequently, the $(2\pi\sigma^2)^{-n_\eta/2}$ coefficient and the exponential sum of squared residuals identically cancel, isolating the structural improvement offered by the split. To streamline notation and mirror computational implementations, we express this ratio in terms of node specific sufficient statistics. For any generic node $c \in \{\eta, \eta_L, \eta_R\}$, define the sum of partial residuals S_c and the marginal variance factor V_c as:

$$S_c = \sum_{i=1}^{n_c} R_{c,i}, \quad (2.43)$$

$$V_c = \sigma^2 + n_c \sigma_\mu^2. \quad (2.44)$$

Substituting Equation (2.41) into Equation (2.42) three times, cancelling factors and substituting in the sufficient statistics yields the final likelihood ratio for the GROW modification:

$$\sqrt{\frac{\sigma^2 V_\eta}{V_{\eta_L} V_{\eta_R}}} \exp \left(\frac{\sigma_\mu^2}{2\sigma^2} \left[\frac{S_{\eta_L}^2}{V_{\eta_L}} + \frac{S_{\eta_R}^2}{V_{\eta_R}} - \frac{S_\eta^2}{V_\eta} \right] \right). \quad (2.45)$$

Grow Tree Structure Ratio

The final component required to evaluate the Metropolis Hastings acceptance probability is the tree structure ratio, occasionally referred to as the prior ratio, defined as $\pi(T_j^*)/\pi(T_j)$. Let $\mathcal{H}_{\text{terminal}}$ denote the set of terminal nodes and $\mathcal{H}_{\text{internal}}$ denote the set of internal nodes for a given tree. The prior probability of any tree structure T_j factorises over its nodes:

$$\pi(T_j) = \prod_{\eta \in \mathcal{H}_{\text{internal}}} \mathbb{P}_{\text{SPLIT}}(\eta) \mathbb{P}_{\text{RULE}}(\eta) \prod_{\eta \in \mathcal{H}_{\text{terminal}}} (1 - \mathbb{P}_{\text{SPLIT}}(\eta)). \quad (2.46)$$

Recall from Section 2.2.4 that the probability of proposing a split at node η is depth dependent, defined as $\mathbb{P}_{\text{SPLIT}}(\eta) = \alpha/(1 + d_\eta)^\beta$, where d_η is the depth of the node and α, β are regularisation hyperparameters. When a split occurs, the splitting rule probability $\mathbb{P}_{\text{RULE}}(\eta)$ consists of two independent uniform choices: selecting an eligible predictor x_k , and selecting a valid splitting threshold. Using the notation established in Section 2.3.2, this probability is:

$$\mathbb{P}_{\text{RULE}}(\eta) = \frac{1}{p_{\text{adj}}(\eta) n_{k,\text{adj}}(\eta)}. \quad (2.47)$$

Because the candidate tree T_j^* differs from the current tree T_j solely at the node η selected for growth, the prior probabilities of all unmodified nodes identically cancel in the ratio. In T_j , node η is terminal. In T_j^* , node η becomes internal, spawning two new terminal child nodes, η_L and η_R . The tree structure ratio therefore simplifies to:

$$\frac{\pi(T_j^*)}{\pi(T_j)} = \frac{(1 - \mathbb{P}_{\text{SPLIT}}(\eta_L))(1 - \mathbb{P}_{\text{SPLIT}}(\eta_R)) \mathbb{P}_{\text{SPLIT}}(\eta) \mathbb{P}_{\text{RULE}}(\eta)}{1 - \mathbb{P}_{\text{SPLIT}}(\eta)}. \quad (2.48)$$

Because the newly generated child nodes are one level deeper than their parent,

their depths satisfy $d_{\eta_L} = d_{\eta_R} = d_\eta + 1$. Expanding the split probabilities yields:

$$\begin{aligned} \frac{\pi(T_j^*)}{\pi(T_j)} &= \frac{\left(1 - \frac{\alpha}{(2+d_\eta)^\beta}\right)^2 \frac{\alpha}{(1+d_\eta)^\beta} \frac{1}{p_{\text{adj}}(\eta) n_{k,\text{adj}}(\eta)}}{1 - \frac{\alpha}{(1+d_\eta)^\beta}} \\ &= \left(1 - \frac{\alpha}{(2+d_\eta)^\beta}\right)^2 \frac{\alpha}{(1+d_\eta)^\beta} \cdot \frac{(1+d_\eta)^\beta}{(1+d_\eta)^\beta - \alpha} \cdot \frac{1}{p_{\text{adj}}(\eta) n_{k,\text{adj}}(\eta)}. \end{aligned} \quad (2.49)$$

Algebraically cancelling the $(1+d_\eta)^\beta$ terms isolates the final closed form expression for the GROW tree structure ratio:

$$\frac{\pi(T_j^*)}{\pi(T_j)} = \alpha \frac{\left(1 - \frac{\alpha}{(2+d_\eta)^\beta}\right)^2}{((1+d_\eta)^\beta - \alpha) p_{\text{adj}}(\eta) n_{k,\text{adj}}(\eta)}. \quad (2.50)$$

BART Tree Modification Moves

Move	Description
Grow	Add a split to a terminal node.
Prune	Remove a split, collapsing two leaves.
Change	Modify an existing split rule.
Swap	Exchange split rules between parent and child.

Table 2.3: BART tree modification moves

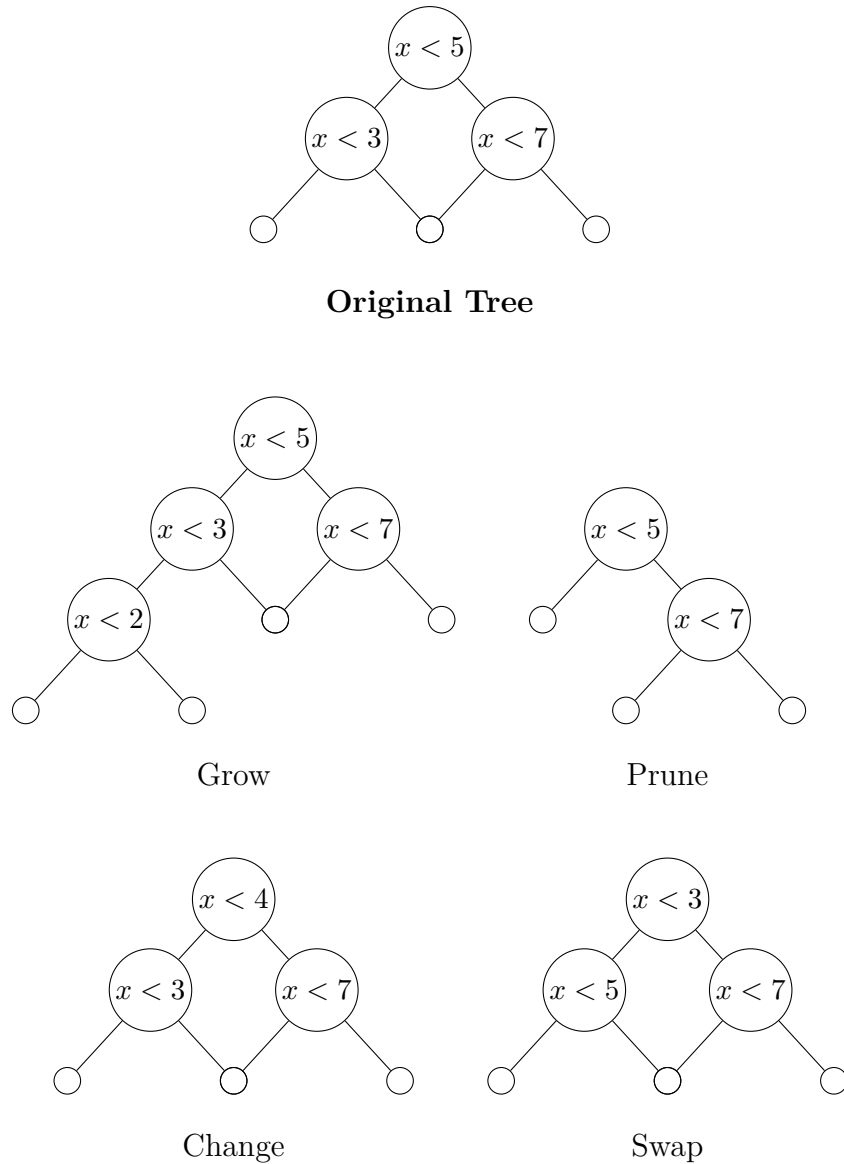


Figure 2.8: Original BART regression tree (top) and four modification moves (bottom).

2.3.3 Sampling from $\pi(M_j \mid T_j, R_j, \sigma^2)$

This section derives the marginal posterior distribution for the terminal node parameters, corresponding to the sequential updates executed in steps $2, 4, \dots, 2m$ of the BART Gibbs sampler. Once a structural update to the tree topology T_j is evaluated and either accepted or rejected via the MH step, the associated leaf parameter vector M_j must be drawn from $\pi(M_j \mid T_j, R_j, \sigma^2)$. Let η denote a terminal node of T_j , with associated leaf parameter μ_η , and let $R_{j,\eta}$ denote the n_η partial residuals allocated to this node. Since the likelihood factorises across the terminal nodes, the conditional posterior for μ_η depends only on $R_{j,\eta}$. From Section 2.2.5, $\mu_\eta \sim \mathcal{N}(0, \sigma_\mu^2)$, while $R_{\eta,i} \mid \mu_\eta, \sigma^2 \stackrel{\text{ind}}{\sim} \mathcal{N}(\mu_\eta, \sigma^2)$. Hence,

$$\pi(\mu_\eta \mid T_j, R_j, \sigma^2) \propto \exp\left(-\frac{1}{2\sigma^2} \sum_{i=1}^{n_\eta} (R_{\eta,i} - \mu_\eta)^2\right) \exp\left(-\frac{\mu_\eta^2}{2\sigma_\mu^2}\right). \quad (2.51)$$

Expanding the quadratic terms inside the exponential and grouping with respect to μ_η produces:

$$\pi(\mu_\eta \mid T_j, R_j, \sigma^2) \propto \exp\left(-\frac{1}{2} \left[\mu_\eta^2 \left(\frac{n_\eta}{\sigma^2} + \frac{1}{\sigma_\mu^2} \right) - 2\mu_\eta \left(\frac{\sum_{i=1}^{n_\eta} R_{\eta,i}}{\sigma^2} \right) \right] \right). \quad (2.52)$$

Completing the square with respect to μ_η reveals the functional form of a Gaussian kernel. To maintain notational consistency with the sufficient statistics defined in Section 2.3.2, let $S_\eta = \sum_{i=1}^{n_\eta} R_{\eta,i}$ denote the sum of the partial residuals in node η , and let $V_\eta = \sigma^2 + n_\eta \sigma_\mu^2$ denote the marginal variance factor. The conditional posterior distribution for the terminal node parameter therefore takes the following exact closed form:

$$\mu_\eta \mid T_j, R_j, \sigma^2 \sim \mathcal{N}\left(\frac{\sigma_\mu^2 S_\eta}{V_\eta}, \frac{\sigma^2 \sigma_\mu^2}{V_\eta}\right). \quad (2.53)$$

During the parameter update phase, a unique value μ_η is independently sampled from Equation (2.53) for every terminal node $\eta \in \{1, \dots, b\}$ across the tree T_j . This completely refreshes the parameter vector M_j , concluding the sequential Gibbs step for the j^{th} tree. A fully detailed version of the above derivation can be found in Appendix A.4. Additionally, a numerical demonstration of this terminal node parameter update is provided in Appendix A.5.

2.3.4 Posterior Inference

As discussed previously, a key advantage of BART is that its flexible regression framework is accompanied by a built-in mechanism for quantifying uncertainty. After discarding an initial burn-in period, the retained MCMC draws are treated as an approximate sample from the joint posterior distribution. Writing the k^{th} retained draw as:

$$\Theta^{(k)} = \left\{ (T_1^{(k)}, M_1^{(k)}), \dots, (T_m^{(k)}, M_m^{(k)}), \sigma^{(k)} \right\}, \quad k = 1, \dots, K,$$

This collection provides an approximate sample from the joint distribution

$$\pi((T_1, M_1), \dots, (T_m, M_m), \sigma^2 \mid x, y).$$

The primary utility of this joint posterior sample is the construction of the posterior ensemble of regression functions. Specifically, each draw defines a corresponding function evaluated at any covariate vector x :

$$f^{(k)}(x) = \sum_{j=1}^m g(x; T_j^{(k)}, M_j^{(k)}), \quad k = 1, \dots, K. \quad (2.54)$$

These pointwise evaluations provide direct Monte Carlo estimates of key posterior quantities. The posterior mean of the regression function is computed simply as the empirical average $\hat{f}(x) = \frac{1}{K} \sum_{k=1}^K f^{(k)}(x)$, while the associated posterior uncertainty is captured by the empirical standard deviation:

$$\widehat{\text{sd}}\{f(x)\} = \sqrt{\frac{1}{K-1} \sum_{k=1}^K \left(f^{(k)}(x) - \hat{f}(x) \right)^2}. \quad (2.55)$$

Furthermore, posterior credible intervals for the true underlying function are obtained directly from the empirical quantiles of the sampled values $f^{(1)}(x), \dots, f^{(K)}(x)$. Beyond estimating the underlying function, this ensemble naturally extends to forecasting future observations through the posterior predictive distribution. For prediction at a new covariate vector x_{new} , a predictive draw at iteration k incorporates both the estimated function and the residual variance:

$$y_{\text{new}}^{(k)} = f^{(k)}(x_{\text{new}}) + \varepsilon^{(k)}, \quad \varepsilon^{(k)} \sim \mathcal{N}(0, \sigma^{(k)}). \quad (2.56)$$

By extracting the empirical quantiles of these predictive draws, the resulting prediction intervals properly propagate both the structural uncertainty in the regression function and the fundamental observation error. Finally, the posterior samples offer insights into the structural composition of the model, specifically regarding variable

usage. Because covariates enter the BART model strictly through splitting rules, the frequency with which a predictor is selected provides a measure of its relative importance. Let $c_j^{(k)}$ denote the proportion of splitting rules in the k^{th} posterior draw that utilise predictor x_j . The posterior Variable Inclusion Proportion (VIP) is then defined as:

$$v_j = \frac{1}{K} \sum_{k=1}^K c_j^{(k)}, \quad j = 1, \dots, p. \quad (2.57)$$

This metric, v_j , measures the average frequency of predictor x_j appearing across the posterior ensemble. It is important to interpret this as a measure of relative variable usage rather than a formal posterior probability of relevance, given that irrelevant predictors may still receive non-zero inclusion proportions by chance during the MCMC exploration. BART’s ability to perform variable selection has seen significant development in recent years and will be considered in further detail later, forming the central focus of this thesis.

2.3.5 Extensions of BART

Since its introduction by Chipman et al. (2010), BART has developed beyond its original Gaussian regression formulation into a broader statistical methodology. Extensions have expanded the range of response types that can be modelled, including multinomial outcomes Murray (2021), while Soft BART replaces the hard partitions of standard BART with probabilistic splitting rules to provide greater flexibility for smooth regression functions Linero and Yang (2018a). BART has also been extended to survival analysis with censored outcomes Sparapani et al. (2016), causal inference and heterogeneous treatment effects Hill (2011), and heteroscedastic regression through a separate model for the residual variance Pratola et al. (2020).

These developments demonstrate how the success of BART has motivated a broader body of research aimed at establishing it as a comprehensive statistical methodology. Extensions addressing alternative response types and other modelling challenges have been important in expanding the applicability of the original framework. The same progression motivates similar developments for the AddiVortes model considered in this thesis. The following chapter introduces AddiVortes as an alternative sum-of-tessellations approach to BART, before the subsequent chapter develops a variable selection framework for the model. Drawing heavily on the principles underlying BART’s variable selection methodology, this extension aims to establish AddiVortes not only as a flexible regression model, but as a more complete statistical framework for high-dimensional modelling and variable selection.

Chapter 3

Bayesian Additive Voronoi Tessellations (AddiVortes)

A central strength of BART is its additive representation, in which a complex regression function is constructed by combining many simple, highly regularised weak learners. Rather than requiring a single model to capture the entire response surface, we have seen that BART distributes the signal across an ensemble of individual trees, allowing complex nonlinearities and interactions to emerge through their combined contribution. The individual learners used in BART are binary decision trees, which recursively partition the predictor space using axis-aligned splits. While this representation is highly flexible, its geometry imposes a fundamental constraint: every partition boundary is orthogonal to a coordinate axis. Consequently, when the underlying response surface contains diagonal, curved, or otherwise non-orthogonal structures, a tree must approximate these features through multiple axis-aligned partitions. Representing geometrically complex structures can therefore require highly intricate tree partitions, even though the underlying relationship itself may be relatively simple.

This motivates consideration of alternative weak learners that retain the additive structure and regularisation of BART while imposing less restrictive geometric constraints on the partition of the predictor space. In particular, a desirable alternative might allow non-orthogonal boundaries, providing a more direct representation of diagonal, curved, or irregular signals. Stone and Gosling (2025a) achieve this by replacing the tree-based partitions of BART with Voronoi tessellations, giving rise to the Bayesian Additive Voronoi Tessellation (AddiVortes) framework.

3.1 Voronoi Tessellations: Definition and Historical Overview

This chapter begins with a formal definition of Voronoi tessellations, followed by a review of their development as a statistical modelling framework. The review motivates the progression towards AddiVortes, after which the full model specification is presented, including its assumptions, prior structure, and posterior inference algorithm. The latter closely parallels the BART framework developed in the previous chapter, while replacing its tree-based engine with Voronoi tessellations.

3.1.1 Voronoi Tessellations: Formal Definition

Standard decision trees slice a predictor space using strict horizontal and vertical lines. In contrast, a Voronoi tessellation partitions space using a set of specific anchor points, known as centres. The entire space is then divided by assigning every possible location to its nearest centre. As illustrated in Figure 3.1, this approach allows the boundaries to form flexible polygons that can potentially adapt more naturally to the underlying data than the rigid rectangles of a traditional decision trees.

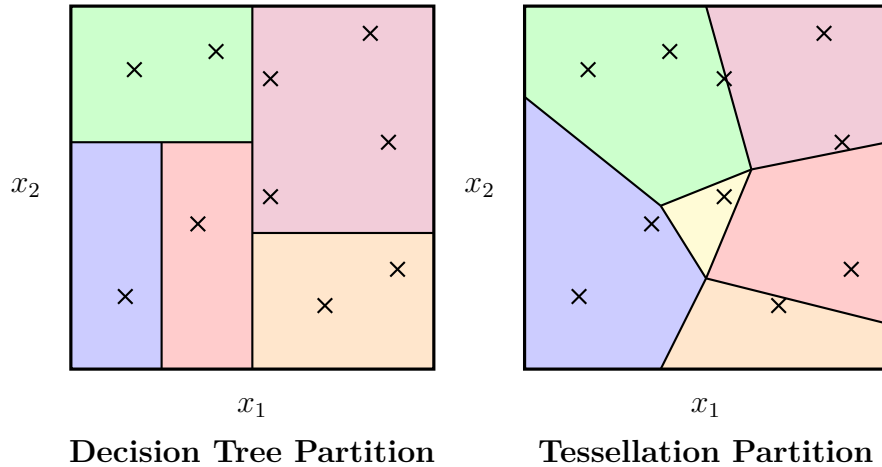


Figure 3.1: A comparison of how decision trees and Voronoi tessellations partition a 2D predictor space.

Formally, a Voronoi tessellation is defined over a metric space $(\mathcal{X}, d)$, where $\mathcal{X} \subseteq \mathbb{R}^p$ represents the predictor space and $d : \mathcal{X} \times \mathcal{X} \rightarrow \mathbb{R}_{\geq 0}$ is a valid distance metric, typically the Euclidean L_2 norm. Consider a finite collection of distinct generating sites, or centres, denoted by $C = (c_1, \dots, c_B) \subset \mathcal{X}$. The Voronoi tessellation associated with C partitions the predictor space into B mutually disjoint regions,

termed Voronoi cells. For each index $b \in \{1, \dots, B\}$, the corresponding Voronoi cell $\mathcal{A}_b$ associated with the generator c_b is defined mathematically as:

$$\mathcal{A}_b = \{x \in \mathcal{X} \mid d(x, c_b) \leq d(x, c_k) \text{ for all } k \neq b\}. \quad (3.1)$$

Every cell $\mathcal{A}_b$ contains all coordinates in $\mathcal{X}$ that are strictly closer to c_b than to any other generating site. The boundaries separating these regions consist of points equidistant to two or more generators. Under the Euclidean metric, these boundaries are linear hyperplanes, allowing the cells to form non-axis-aligned partitions of the predictor space. The construction of tessellations was first studied systematically by Dirichlet (1850) in two and three dimensions in connection with quadratic forms, before being generalised to arbitrary dimensions by Voronoi (1908).

3.1.2 Historical Progression of Tessellation-based methodologies

The geometric principles of Voronoi tessellations have underpinned distance based machine learning algorithms for decades. The evolution of these methods demonstrates a clear trajectory from rigid, deterministic classification to smooth spatial interpolation.

Nearest Neighbour Classification:

The classical 1-nearest neighbour algorithm, formalised by Cover et al. (1967), assigns any query point in $\mathbb{R}^p$ to the label of its closest training observation. The resulting decision boundaries are mathematically equivalent to a standard Voronoi tessellation in which every observed data point acts as a fixed generating site. While conceptually straightforward, this distance-based approach is highly susceptible to error if individual observations are mislabelled or poorly located, since a such a point generates its own Voronoi cell and will misclassify every query point that falls within it. Furthermore, restricting the generating sites strictly to the observed training coordinates prevents the algorithm from inferring optimal boundary placements. This results in rigid spatial partitions that cannot adapt flexibly to the underlying data generating process.

Centroid Based Clustering:

To introduce spatial adaptability, Lloyd (1982) proposed the K -means clustering algorithm. Rather than using all data points as fixed sites, K -means iteratively

partitions the covariate space into discrete Voronoi cells resulting from only K centroids. The algorithm alternates between two steps: each observation is assigned to its nearest centroid, inducing a Voronoi partition as before, and each centroid is then relocated to the mean of its assigned observations. While this allows the generating sites to adapt to the data, the partition is learned entirely independently of the response, since K -means never observes y_i . Its centroids are therefore positioned to represent the distribution of the predictors rather than the relationship between the predictors and the outcome.

Natural Neighbour Interpolation:

The transition from discrete classification to continuous spatial regression was achieved through natural neighbour interpolation, introduced by Sibson (1981). Rather than assigning a query point strictly to a single Voronoi cell, Sibson temporarily inserted the query location into the existing tessellation as a new generating site, inducing a new tessellation structure. Interpolation weights were then derived proportionally based on the geometric area the new cell “steals” from its neighbouring cells. This advancement showed that Voronoi geometry could be exploited to construct continuous spatial predictions rather than simple hard boundaries. However, Sibson’s method was strictly deterministic. It provided no statistical mechanism to quantify spatial uncertainty, and its reliance on fixed data points as generators meant it was restricted in learning the optimal positioning of the underlying partitions.

Probabilistic Generating Sites:

The limitations of deterministic spatial methods necessitated the development of probabilistic frameworks. The integration of Voronoi geometry into Bayesian statistics was pioneered by David G. T. Denison et al. (2002), who formulated the single Bayesian Voronoi partition model. Unlike classical geometric methods, David G. T. Denison et al. (2002) treated both the number of generating sites B , and their spatial coordinates C as unknown random variables governed by prior distributions. Inference was performed using Reversible Jump Markov Chain Monte Carlo (RJMCMC) (Green, 1995), enabling transdimensional state space exploration.

The algorithm dynamically altered the tessellation through BIRTH proposals (adding a new site), DEATH proposals (removing an existing site), and MOVE proposals (perturbing a site coordinate). This allowed the optimal partitioning to be learned from the data and whilst accounting for structural uncertainty. Within each Voronoi cell, the observations are modelled using a separate local probability distribution, with the parameters of these local models assumed independent across

regions. The flexibility of this approach was highly dependent on the statistical model specified within each Voronoi cell. While the RJMCMC framework allows the partition itself to adapt in its structure to learn the optimal partitioning, a restrictive local model may still limit the complexity of the response that can be represented within each region.

Voronoi Gated Gaussian Processes:

To address the need for more flexible local modelling within each cell, Pope et al. (2022) developed the Voronoi Gated Gaussian Process (VGGP). Standard Gaussian processes employ a single global covariance structure, which can be restrictive when the smoothness or variability of the underlying function changes across the predictor space. VGGP instead uses a tessellation resulting from the same RJMCMC procedure as David G. T. Denison et al. (2002) to partition the covariate space into a Voronoi Tessellation, but then assigns an independent Gaussian process with its own covariance hyperparameters to each cell. This allows different regions of the predictor space to exhibit different degrees of smoothness and variability, rather than forcing a single covariance structure to describe the entire response surface.

Despite this increased flexibility, VGGP remains based on a single tessellation that must represent the full structure of the response surface. As the dimensionality and complexity of the data increases, constructing one partition capable of capturing all relevant features becomes infeasible. This presents a trade-off identical to that seen earlier in tree-based methods such as Random forests: intricate partitions are needed to capture complex signals, but such partitions suffer from severe overfitting. To overcome this bottleneck, we have already seen that the BART framework provides a solution through its additive construction. Rather than representing the entire response surface through one increasingly complex tessellation, an alternative is to adopt the additive decomposition used by ensemble methods such as BART, where complex functions are constructed through the combination of many simpler learners. This motivates using Voronoi tessellations themselves as the individual components of an additive ensemble, allowing the overall response surface to be represented collectively rather than by a single partition.

3.2 The AddiVortes Framework

3.2.1 The Additive Tessellation Model

Let $\mathcal{X} \subseteq \mathbb{R}^p$ denote the predictor space. We observe training data comprising an $n \times p$ matrix X with rows $x_i \in \mathcal{X}$, and a continuous response vector $y \in \mathbb{R}^n$. The AddiVortes model, introduced by Stone and Gosling (2025a), assumes that the underlying data-generating mechanism is an additive ensemble of m distinct, piecewise constant geometric functions:

$$y_i = \sum_{j=1}^m g(x_i; T_j, M_j) + \varepsilon_i, \quad \varepsilon_i \sim \mathcal{N}(0, \sigma^2). \quad (3.2)$$

Unlike tree-based architectures that impose rigid orthogonal recursive partitions, each tessellation function g is governed by a topological structure T_j characterising a Voronoi tessellation over a specific subspace of $\mathcal{X}$. This topology is formally defined by two components. First, a dimensional subspace is selected, defined by an active index set $D_j \subseteq \{1, \dots, p\}$ with cardinality $d_j = |D_j|$. Second, a spatial partition is constructed using a geometric configuration of B_j generating centres $C_j = \{c_{j1}, \dots, c_{jb_j}\}$, where each centre exists strictly within the d_j dimensional Euclidean subspace $\mathbb{R}^{d_j}$.

Let $x_{i,D_j} \in \mathbb{R}^{d_j}$ denote the orthogonal projection of the predictor vector x_i onto the active subspace defined by D_j . The topology T_j partitions this subspace into B_j mutually disjoint Voronoi cells, denoted $\mathcal{A}_{j1}, \dots, \mathcal{A}_{jb_j}$. The geometric boundaries of the l^{th} cell are rigorously defined by a nearest neighbour routing mechanism under the Euclidean metric:

$$\mathcal{A}_{jl} = \{x \in \mathcal{X} \mid \|x_{D_j} - c_{jl}\|_2 \leq \|x_{D_j} - c_{jk}\|_2 \text{ for all } k \neq l\}. \quad (3.3)$$

Consequently, each cell $\mathcal{A}_{jl}$ captures all coordinates in the predictor space whose projection onto the active dimensions is closest to the generating centre c_{jl} . Coupled with this geometric topology is the parameter vector $M_j = (\mu_{j1}, \dots, \mu_{jb_j})^\top$, which assigns a continuous scalar value to each corresponding Voronoi cell. The individual regression function for the j th tessellation operates by mapping an input x_i to the parameter of its capturing cell. This mapping is formalised using indicator functions:

$$g(x_i; T_j, M_j) = \sum_{l=1}^{b_j} \mu_{jl} \mathbb{I}(x_i \in \mathcal{A}_{jl}), \quad (3.4)$$

where $\mathbb{I}(\cdot)$ evaluates to 1 if the condition is met and 0 otherwise. The final ensemble prediction for any observation is therefore the algebraic sum of the scalar parame-

ters assigned by the m capturing Voronoi cells across all independent tessellations. The integer m denotes the number of tessellations in the ensemble and is typically chosen sufficiently large that no individual component is required to explain the response surface independently. The additive composition mechanism is visually demonstrated in Figure 3.2. Similarly to BART, Stone and Gosling (2025a) suggest the default use of $m = 200$ for predictive purposes and demonstrate diminishing returns beyond this value. For the same reasoning as BART, the success of this additive representation depends on ensuring that no individual tessellation is allowed to dominate the overall fit. Each component is therefore regularised to act as a weak learner, so that complex structure is recovered through the accumulation of many small contributions across the ensemble. This regularisation is enforced through the prior distributions placed on the tessellation structures and their associated parameters, which are specified in the following sections.

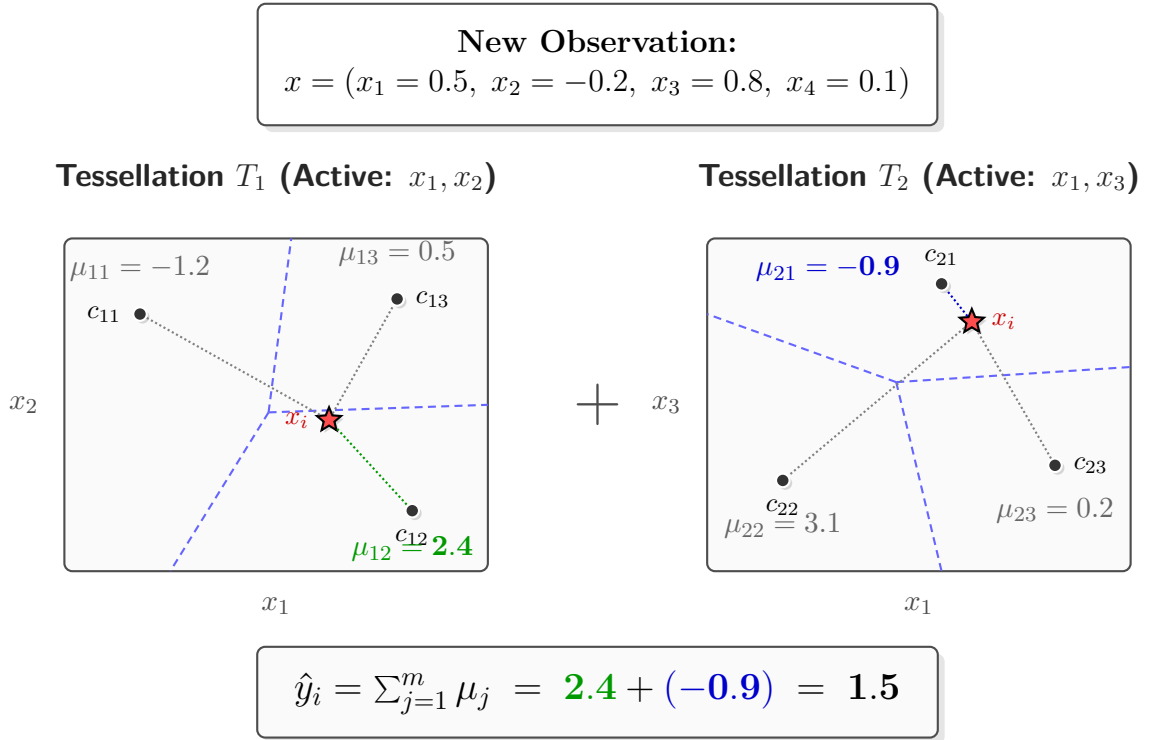


Figure 3.2: **The AddiVortes Routing Mechanism.** An observation x_i is evaluated across an ensemble of $m = 2$ tessellations. Within each active dimensional subspace, x_i is routed to the nearest generating centre via Euclidean distance. The final prediction $\hat{y}_i$ is the exact sum of the scalar parameters (μ) assigned to the Voronoi cells.

3.2.2 Prior Simplification and Independence Assumptions

Due to the size and complexity of the tessellation ensemble parameter space, specifying the full joint prior $\pi((T_1, M_1), \dots, (T_m, M_m), \sigma^2)$ is highly challenging for ensembles where m is large. To simplify the specification, the AddiVortes framework factorises this joint prior distribution through a series of assumptions:

$$\begin{aligned}
\pi((T_1, M_1), \dots, (T_m, M_m), \sigma^2) &= \pi((T_1, M_1), \dots, (T_m, M_m)) \pi(\sigma^2) \\
&= \left[\prod_{j=1}^m \pi(T_j, M_j) \right] \pi(\sigma^2) \\
&= \left[\prod_{j=1}^m \pi(M_j | T_j) \pi(T_j) \right] \pi(\sigma^2) \\
&= \left[\prod_{j=1}^m \prod_{b=1}^{b_j} \pi(\mu_{jb} | T_j) \pi(C_j | D_j) \pi(D_j) \right] \pi(\sigma^2). \quad (3.5)
\end{aligned}$$

The sequential decomposition in Equation (3.5) is achieved by applying four key independence assumptions. First, the global residual variance σ^2 is assumed to be independent of the tessellation structures and cell parameters **(A1)**, allowing the variance prior to be factored out. Second, the structural topologies and corresponding cell parameters for all m distinct Voronoi tessellations are treated as mutually independent *a priori* **(A2)**, breaking the complex joint distribution into a product of individual tessellation priors. Third, for any given tessellation j , the parameter vector M_j is assumed to depend on the topology T_j only through the total number of generated cells, denoted b_j . Furthermore, the individual scalar parameters assigned to these b_j Voronoi cells are mutually independent **(A3)**, completely separating the distribution of the cell values from their geometric boundaries. Finally, the Voronoi topology T_j separates into two components: the subset of active dimensions D_j and the spatial configuration of the generating centres C_j . Because the dimensionality of the generating centres must match the number of selected active covariates, C_j is conditionally dependent on D_j **(A4)**. These assumptions are summarised in Table 3.1.

Through these assumptions, it is now only necessary to define low dimensional prior distributions for the active covariate subset $\pi(D_j)$, the spatial point process governing the centres $\pi(C_j | D_j)$, the individual cell means $\pi(\mu_{jb} | T_j)$, and the global variance $\pi(\sigma^2)$. A comprehensive theoretical review of these specific prior distributions and their associated hyperparameter calibrations is provided in the

subsequent sections.

Table 3.1: A summary of AddiVortes independence assumptions

(A1)	$\{(T_1, M_1), \dots, (T_m, M_m)\} \perp\!\!\!\perp \sigma^2$
(A2)	$(T_j, M_j) \perp\!\!\!\perp (T_k, M_k) \quad \text{for all } j \neq k$
(A3)	$\mu_{jb} \perp\!\!\!\perp \mu_{jb'} \mid T_j \quad \text{for all } b \neq b'$
(A4)	$\pi(T_j) = \pi(C_j \mid D_j) \pi(D_j)$

3.2.3 Regularisation Priors

The factorised prior distribution established in Equation (3.5) requires explicit distributional forms for each of its components. As demonstrated, the specification of the AddiVortes prior simplifies to defining four lower dimensional distributions: $\pi(D_j)$, $\pi(C_j \mid D_j)$, $\pi(\mu_{jb} \mid T_j)$, and $\pi(\sigma^2)$. As with the BART framework, the specific choice of these prior distributions offers flexible control over the kind of tessellations that emerge within the ensemble. AddiVortes constructs its priors in such a way that they act as regularisers, ensuring that each tessellation functions strictly as a weak learner. By constraining the complexity and influence of any individual tessellation, the priors prevent any single model from becoming dominant therefore preventing overfitting. The priors for the global variance and cell means are identical to those previously for BART in Section 2.2.4 therefore we only provide an overview here.

It is standard practice to first centre and scale the observed response vector $y \in [-0.5, 0.5]$. A notable deviation from the BART framework is that AddiVortes also requires the covariate matrix X to be scaled to the interval $[-0.5, 0.5]$. While BART’s decision trees employ orthogonal partitions that are invariant to monotonic transformations of the covariates, the distance metrics for Voronoi tessellations are highly sensitive to the scale of the input variables. Scaling the covariate matrix ensures that distances are evaluated uniformly across dimensions and allows the hyperparameters governing the generating centres to be defined universally across datasets.

Global Error Variance $\pi(\sigma^2)$

The global residual variance σ^2 is assigned an Inverse Gamma prior distribution:

$$\sigma^2 \sim \text{IG}\left(\frac{\nu}{2}, \frac{\nu\lambda}{2}\right). \quad (3.6)$$

This specification is conditionally conjugate to the Gaussian likelihood of the regression error terms. As we will later see, this conjugacy is advantageous as it enables direct Gibbs sampling of the variance parameter. To inform the hyperparameters ν and λ , a data driven calibration strategy is employed. Identically to BART, the prior distribution is anchored around a conservative overestimate of the residual standard deviation, denoted as $\hat{\sigma}$. This bound is typically established using the residual standard error extracted from an ordinary least squares linear regression of y on X . In high dimensional settings where the number of predictors exceeds the number of observations, the OLS estimator becomes mathematically undefined (Tibshirani, 1996). In such cases, the conservative overestimate $\hat{\sigma}$ must strictly be defined as the sample standard deviation of the transformed response variable y . Once $\hat{\sigma}$ is established, the scale parameter λ is analytically derived such that a specific quantile q of the prior distribution exactly matches the overestimate, ensuring $\Pr(\sigma < \hat{\sigma}) = q$ for a selected quantile level $q \in (0.7, 0.99)$. Stone and Gosling (2025a), recommend the default settings of $\nu = 6$ and $q = 0.85$.

Cell Means $\pi(\mu_{jb} \mid T_j)$

To shrink the additive effects toward zero and systematically prevent any single tessellation from dominating the ensemble prediction, the scalar parameter assigned to each Voronoi cell is assigned a zero mean Gaussian prior:

$$\mu_{jb} \mid T_j \sim \mathcal{N}(0, \sigma_\mu^2). \quad (3.7)$$

Because the total ensemble prediction is formulated as the sum over m mutually independent tessellations, the induced prior on the expected response is itself normally distributed as $\mathcal{N}(0, m\sigma_\mu^2)$. To guarantee that the model output predominantly falls within the observed, scaled response interval of $[-0.5, 0.5]$, the variance is explicitly scaled by the ensemble size. The standard deviation is defined directly as:

$$\sigma_\mu = \frac{0.5}{k\sqrt{m}}, \quad (3.8)$$

where k is a specified hyperparameter (typically $k = 3$) that controls the absolute strength of the shrinkage. By constraining the parameter variance in this manner, the prior heavily penalises excessively large contributions from cells, effectively regularising the model and mitigating the risk of overfitting.

Tessellation Topology $\pi(T_j)$

The structural prior $\pi(T_j) = \pi(D_j)\pi(C_j \mid D_j)$ is defined by specifying generative mechanisms for the active dimensions and the spatial centres. The topology T_j is formulated through a sequence of four stochastic steps:

Step 1: Dimensionality - determining the number of active dimensions d_j .

Step 2: Variable Selection - identifying the subset of covariates D_j .

Step 3: Centre Cardinality - specifying the number of generating centres B_j .

Step 4: Spatial Allocation - allocating the coordinates of the generating centres C_j .

Step 1: Dimensionality (d_j)

The number of active dimensions d_j used by the j^{th} tessellation is assigned the shifted Binomial prior

$$d_j - 1 \sim \text{Binomial}\left(p - 1, \frac{\omega}{p}\right), \quad (3.9)$$

where p denotes the total number of available covariates and ω is a sparsity-controlling hyperparameter. Since

$$\mathbb{E}[d_j] = 1 + \frac{(p - 1)\omega}{p}, \quad (3.10)$$

the value of ω directly controls the prior complexity of each tessellation by determining the expected number of covariates on which it depends. Choosing ω small relative to p concentrates prior mass on low-dimensional tessellations, thereby regularising the ensemble towards simple main effects and low-order interactions. Higher-dimensional tessellations remain possible, but require sufficient support from the data.

Stone and Gosling (2025a) suggest the default specification $\omega = 3$ places most prior mass on tessellations involving only a small number of covariates, typically two or three, while still allowing moderate interactions to be represented. This provides

a balance between restricting the individual tessellations to weak, low-dimensional learners and retaining sufficient flexibility to capture interaction effects. The model has also been found to be relatively insensitive to small changes in ω . To ensure that ω/p defines a valid Binomial probability, $\omega < p$ is required.

Step 2: Variable Selection (D_j)

Conditional on the specified dimension count d_j , the active subset D_j is drawn uniformly over all $\binom{p}{d_j}$ possible combinations:

$$\pi(D_j \mid d_j) = \frac{1}{\binom{p}{d_j}}. \quad (3.11)$$

Equivalently, the covariates are sampled uniformly without replacement from the full set of available predictors. This prior is analytically straightforward and strictly noninformative, assigning equal prior probability to all covariates and avoiding subjective preference *a priori*. As demonstrated by Linero (2018), this assumption could be relaxed in future extensions by introducing a Dirichlet prior over the covariate inclusion probabilities, enabling adaptive variable selection capable of favouring informative predictors.

Step 3: Centre Cardinality.

Once the subset of active dimensions is defined, the spatial configuration of the centres must be established. The total number of generating centres B_j is assigned a shifted Poisson prior:

$$B_j - 1 \sim \text{Poisson}(\lambda_c). \quad (3.12)$$

The Poisson distribution provides a natural discrete prior for count variables. Because the standard Poisson random variable possesses support over $\{0, 1, 2, \dots\}$, modelling the shifted quantity $B_j - 1$ strictly ensures that $B_j \geq 1$, guaranteeing that every tessellation contains at least one generating centre. Selecting a relatively small rate parameter λ_c favours simpler tessellations with fewer cells, strictly discouraging unnecessary partitioning of the covariate space. Stone and Gosling (2025a) suggest the default of $\lambda_c = 25$ but note that in the case where the number of observations is small, such complex partitions may not be required and λ_c can be reduced further.

4: Spatial Allocation.

Conditional on B_j and the active dimensions D_j , the spatial coordinates of the Voronoi centres are assigned independent Gaussian priors. Specifically, for each dimension $k \in D_j$, the coordinate c_{jk} is assumed to follow a normal distribution:

$$c_{jk} \mid D_j \sim \mathcal{N}(0, \sigma_c^2), \quad (3.13)$$

where σ_c^2 controls the spatial dispersion of the centres. A zero mean prior is optimal as the covariates are preprocessed to lie strictly within the interval $[-0.5, 0.5]$. Following this standardisation. The variance parameter σ_c^2 is calibrated such that the majority of the prior mass remains within the transformed covariate range. Stone and Gosling (2025a) suggest the default setting of $\sigma_c = 0.8$ to ensure this. Since the Gaussian prior has an unbounded support, centres can still be sampled from beyond the bound of the training data. This permits the Tessellation to have some exploration ability whilst still remaining focused on the observed covariates range.

Table 3.2: Summary of AddiVortes Prior Distributions and Hyperparameter Calibrations

Parameter	Prior Distribution	Default Hyperparameters
Global Error Variance	$\sigma^2 \sim \text{IG}(\nu/2, \nu\lambda/2)$	$\nu = 6$, λ calibrated to data variance
Cell Mean	$\mu_{jb} \mid T_j \sim \mathcal{N}(0, \sigma_\mu^2)$	$\sigma_\mu = 0.5/(k\sqrt{m})$, with $k = 3$
Number of Dimensions	$d_j - 1 \sim \text{Binomial}(p - 1, \omega/p)$	$\omega = 3$
Subset of Dimensions	$D_j \mid d_j \sim \text{Uniform}\left(\binom{p}{d_j}\right)$	—
Number of Centres	$B_j - 1 \sim \text{Poisson}(\lambda_c)$	$\lambda_c = 25$
Centre Coordinates	$c_{jk} \mid D_j \sim \mathcal{N}(0, \sigma_c^2)$	$\sigma_c = 0.8$

3.3 Posterior Sampling in AddiVortes

Posterior inference for the AddiVortes framework requires sampling from the joint posterior distribution over the tessellation structures, their associated cell parameters, and the residual variance:

$$\pi((T_1, M_1), \dots, (T_m, M_m), \sigma^2 \mid x, y). \quad (3.14)$$

As with BART, direct sampling from this joint posterior is computationally challenging for large ensembles due to the vastness of the parameter space. To overcome this, the additive architecture of AddiVortes permits the use of the same Bayesian backfitting MH-within-Gibbs strategy described in Section 2.3.1. To avoid repetition, we provide only a brief overview of the general sampler here, focusing instead on the necessary modifications to the Metropolis-Hastings (MH) steps. Whereas BART iteratively updates each tree through a specific set of four modification moves, AddiVortes possesses its own distinct set of topological proposals to update the tessellations. These new modification moves require explicit MH acceptance ratios, which we derive in this section.

3.3.1 The AddiVortes Gibbs Sampler

Following the same procedure outlined in Section 2.3.1, we begin by constructing a Gibbs sampler for the target posterior in Equation (3.14). Explicitly, at each iteration, the m tessellations are sampled conditional on the current state of the remainder of the ensemble, followed by a draw of the residual variance. Let $(T_{(j)}, M_{(j)})$ denote the collection of all tessellation structures and cell parameters excluding the j th tessellation. The Gibbs sampler therefore cycles through

$$\begin{aligned} 1, \dots, m : & \quad (T_j, M_j) \mid (T_{(j)}, M_{(j)}), \sigma^2 \quad j = 1, \dots, m, \\ m + 1 : & \quad \sigma^2 \mid (T_1, M_1), \dots, (T_m, M_m), x, y \end{aligned} \quad (3.15)$$

For the j^{th} tessellation, the dependence on the remaining $m - 1$ tessellations is represented through the partial residual vector

$$R_j = y - \sum_{k \neq j} g(x; T_k, M_k), \quad (3.16)$$

which contains the variation in the response not currently explained by the remainder of the ensemble. The tessellation updates can therefore be written more compactly as

$$\begin{aligned}
1, \dots, m : & \quad (T_j, M_j) \mid R_j, \sigma^2 \quad j = 1, \dots, m, \\
m + 1 : & \quad \sigma^2 \mid (T_1, M_1), \dots, (T_m, M_m), x, y
\end{aligned} \tag{3.17}$$

As in BART, directly sampling from $\pi(T_j, M_j \mid R_j, \sigma^2)$ presents complications requiring reversible-jump MCMC between continuous spaces of varying dimension (Green, 1995). To overcome this, we observe that M_j can be marginalised out, permitting T_j and M_j to be sampled separately. We achieve this by analytically integrating the cell parameters from the conditional posterior. Under the conjugate Gaussian prior assigned to the cell means,

$$\pi(T_j \mid R_j, \sigma^2) \propto \pi(T_j) \int \pi(R_j \mid T_j, M_j, \sigma^2) \pi(M_j \mid T_j) dM_j. \tag{3.18}$$

Due to the conjugacy, this marginalisation can be obtained in a closed form up to a normalising constant, permitting two stage sampling of T_j and M_j :

Step 1: Structural Update. For each tessellation $j = 1, \dots, m$, construct the partial residual vector R_j and update T_j using an MH step targeting:

$$\pi(T_j \mid R_j, \sigma^2)$$

Step 2: Parameter Update. Conditional on the updated tessellation structure, sample the associated cell parameter vector from

$$\pi(M_j \mid T_j, R_j, \sigma^2)$$

Step 3: Variance Update. After all m tessellations have been updated, sample the residual variance from its conditional posterior

$$\pi(\sigma^2 \mid (T_1, M_1), \dots, (T_m, M_m), x, y). \tag{3.19}$$

The overall MH-within-Gibbs strategy is therefore identical in principle to that used by BART. The fundamental difference lies in the structural MH updates. BART explores the space of tree structures through GROW, PRUNE, CHANGE, and SWAP proposals, each of which modifies a tree in a specific way. AddiVortes instead employs a distinct set of moves that modify the dimensional and geometric structure of a Voronoi tessellation. These moves are introduced in the following section, and their corresponding MH acceptance probabilities must therefore be derived specifically for the AddiVortes model.

3.3.2 Sampling from $\pi(T_j \mid R_j, \sigma)$

To update the tessellation structure T_j conditional on the partial residual vector R_j , the AddiVortes framework employs a MH update step similar to BART. At each MCMC iteration, a modification of the current tessellation T_j is proposed to generate a candidate structure T_j^* . The proposed structure T_j^* is accepted with probability:

$$\alpha(T_j, T_j^*) = \min \left\{ 1, \underbrace{\frac{\pi(R_j \mid T_j^*, \sigma^2)}{\pi(R_j \mid T_j, \sigma^2)}}_{\text{Marginal likelihood ratio}} \times \underbrace{\frac{\pi(T_j^*)}{\pi(T_j)}}_{\text{Prior ratio}} \times \underbrace{\frac{q(T_j \mid T_j^*)}{q(T_j^* \mid T_j)}}_{\text{Proposal ratio}} \right\}. \quad (3.20)$$

Where $q(T_j^* \mid T_j)$ is the proposal distribution for transitioning between T_j and T_j^* . The proposal mechanism must allow the Markov chain to explore the support of the posterior through a sequence of valid structural transitions. Traditionally, proposals that modify the number of generating centres induce transitions between parameter spaces of different dimensions, requiring a Reversible Jump Markov Chain Monte Carlo (RJMCMC) construction following Green (1995). However, in the AddiVortes framework, the cell output values are analytically marginalised out of the joint posterior using their conjugate prior. This marginalisation entirely removes the need for RJMCMC because the continuous parameters are integrated out and therefore the algorithm avoids transitioning between continuous spaces of varying dimensions.

MCMC methods have previously been used to explore spatial partition structures. Early work in Bayesian spatial modelling used reversible jump methods to explore different partition configurations (Knorr-Held et al., 2000; David G T Denison et al., 2001; Kim et al., 2005). More recently, Pope et al. (2022) used MCMC to explore Voronoi partitions in a Voronoi-gated Gaussian process. Specifically, they use BIRTH, DEATH, MOVE, and CHANGE proposals to modify the Voronoi generating centres. In their approach, the active covariates remain fixed, with the MCMC sampler instead exploring changes in the number and locations of the centres.

The AddiVortes transition proposals expands this structural search. Rather than restricting the proposal mechanism solely to the spatial configuration of centres, AddiVortes jointly explores both the configuration of the generating centres and the active dimensions used to construct the distance metric of the tessellation. The proposal mechanism therefore consists of two complementary classes of structural updates:

Dimension Moves (Altering D_j)

These moves alter which covariates define the Euclidean distance routing of the Voronoi cells. Let $d_j = |D_j|$ denote the current number of active dimensions for tessellation j .

- **Add Dimension (AD)** ($d_j \rightarrow d_j + 1$): Propose T_j^* by adding a single new dimension selected uniformly at random from the set of currently inactive covariates.
- **Remove Dimension (RD)** ($d_j \rightarrow d_j - 1$): Propose T_j^* by deleting a single dimension selected uniformly at random from the currently active set D_j .
- **Swap Dimension (SD)** ($d_j \rightarrow d_j$): Propose T_j^* by selecting one active dimension uniformly at random for removal and replacing it with one inactive dimension selected uniformly at random, thereby strictly maintaining the overall dimensionality.

Centre Moves (Altering C_j)

These moves alter the spatial configuration of the generating centres across the active dimensional subspace defined by D_j . Let $B_j = |C_j|$ denote the current number of generating centres for tessellation j .

- **Add Centre (AC)** ($B_j \rightarrow B_j + 1$): Propose T_j^* by spawning a single new generating centre. Its d_j dimensional coordinate vector is generated by drawing each scalar independently from the zero mean Gaussian spatial prior, $\mathcal{N}(0, \sigma_c^2)$.
- **Remove Centre (RC)** ($B_j \rightarrow B_j - 1$): Propose T_j^* by deleting one of the existing generating centres chosen uniformly at random. The spatial coordinates of the selected centre are permanently discarded.
- **Change Centre (CC)** ($B_j \rightarrow B_j$): Propose T_j^* by selecting an existing centre uniformly at random and replacing its current coordinate vector with a newly generated vector drawn independently from the spatial prior $\mathcal{N}(0, \sigma_c^2)$.

Proposal Probabilities, Boundary Conditions

Let $\mathcal{P} = \{p_{AD} = 0.2, p_{RD} = 0.2, p_{SD} = 0.1, p_{AC} = 0.2, p_{RC} = 0.2, p_{CC} = 0.1\}$ represent the probabilities of choosing each respective tessellation modification with the values displaying the default recommendations of Stone and Gosling (2025a). These probabilities are defined universally, subject only to strict boundary conditions designed to prevent impossible configurations. For instance, proposal adjustments strictly ensure that $p_{RD} = 0$ if $d_j = 1$, $p_{AD} = 0$ if $d_j = p$ (where p denotes the total number of available covariates), and $p_{RC} = 0$ if $B_j = 1$.

A notable feature of these moves is that the likelihood ratio shares a universal form across all structural updates. Because the continuous cell parameters M_j are analytically integrated out, the marginal likelihood of the partial residuals, $\pi(R_j | T_j, \sigma^2)$, depends strictly on how the tessellation structure T_j partitions the data into distinct cells. The specific proposal mechanism used to generate a candidate tessellation T_j^* is completely irrelevant to the likelihood evaluation. This allows an identical marginal likelihood formulation to be applied uniformly across all moves, requiring only slight, move dependent modifications.

To simplify derivations, we factorise the full MH acceptance ratio into two distinct components: the marginal likelihood ratio, and a **structural ratio** that captures the move dependent prior and proposal probabilities:

$$\alpha(T_j, T_j^*) = \min \left\{ 1, \underbrace{\frac{\pi(R_j | T_j^*, \sigma^2)}{\pi(R_j | T_j, \sigma^2)}}_{\text{Marginal likelihood ratio}} \times \underbrace{\frac{\pi(T_j^*)q(T_j | T_j^*)}{\pi(T_j)q(T_j^* | T_j)}}_{\text{Structural ratio}} \right\}. \quad (3.21)$$

We begin by deriving the universal marginal likelihood ratio, followed by derivations of the structural ratios for the Add Dimension (AD) and the Add Centre (AC) moves. Derivations for the remaining structural modifications, Remove Dimension (RD), Swap Dimension (SD), Remove Centre (RC), and Change Centre(CC), can be found in Appendix B.2. To aid demonstration, Figure 3.3 and Figure 3.4 visually illustrate the proposal mechanisms for the AD and AC moves, respectively.

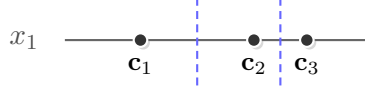
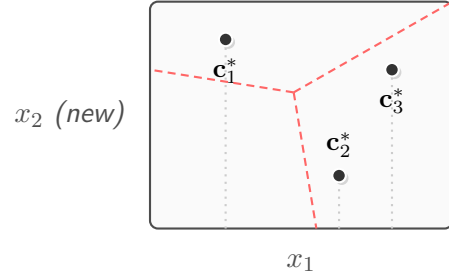
Current State T_j ($d = 1$)**Proposed State T_j^* ($d = 2$)**

Figure 3.3: **Add Dimension (AD)**. Transitioning from a 1D to a 2D topology strictly preserves the existing spatial coordinates along x_1 while drawing b new coordinates along the newly appended dimension x_2 from a zero-mean normal distribution.

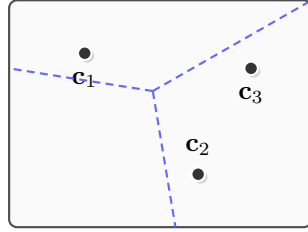
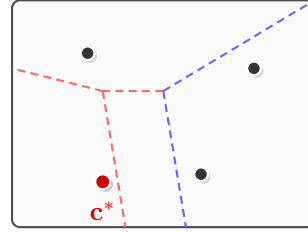
Current State T_j ($b = 3$)**Proposed State T_j^* ($b = 4$)**

Figure 3.4: **Add Centre (AC)**. A new spatial centre $\mathbf{c}^*$ is spawned by drawing a coordinate vector from a zero-mean normal distribution across the active dimensions. The addition is localized; it fractures the existing spatial topology, creating a new Voronoi cell while shrinking the jurisdiction of its immediate neighbors, without affecting distant regions. The exact mathematical symmetry between the spatial prior and the proposal distribution ensures detailed balance is preserved.

3.3.3 Deriving the MH Marginal Likelihood Ratio

The marginal likelihood of the partial residuals, conditional on the tessellation structure and the global variance, is obtained by integrating out the cell mean vector M_j from the joint conditional density:

$$\pi(R_j | T_j, \sigma^2) = \int \pi(R_j | M_j, T_j, \sigma^2) \pi(M_j | T_j) dM_j \quad (3.22)$$

Because the output values for each cell are assumed to be independent a priori, this joint probability factorises completely across the B_j individual cells:

$$\pi(R_j | M_j, T_j, \sigma^2) \pi(M_j | T_j) = \prod_{b=1}^{B_j} \pi(R_{jb} | \mu_{jb}, T_j, \sigma^2) \pi(\mu_{jb} | T_j) \quad (3.23)$$

where R_{jb} denotes the specific subset of partial residuals allocated strictly to cell b . Consequently, we can isolate the integration to a single generic cell b . The marginal likelihood for this individual cell is obtained by integrating its joint density over the cell mean:

$$\pi(R_{jb} | T_j, \sigma^2) = \int \pi(R_{jb}, \mu_{jb} | T_j, \sigma^2) d\mu_{jb} \quad (3.24)$$

Let cell b contain n_b observations, with R_{jbi} representing the i^{th} individual residual observation within this cell, such that $R_{jb} = (R_{jb1}, \dots, R_{jbn_b})^T$. Conditional on the cell mean μ_{jb} , the individual partial residuals are independent and identically distributed, and we assign a conjugate Gaussian prior to the cell mean:

$$\pi(R_{jb} | \mu_{jb}, T_j, \sigma^2) = (2\pi\sigma^2)^{-n_b/2} \exp\left(-\frac{1}{2\sigma^2} \sum_{i=1}^{n_b} (R_{jbi} - \mu_{jb})^2\right) \quad (3.25)$$

$$\pi(\mu_{jb} | T_j) = (2\pi\sigma_\mu^2)^{-1/2} \exp\left(-\frac{\mu_{jb}^2}{2\sigma_\mu^2}\right) \quad (3.26)$$

The joint density $\pi(R_{jb}, \mu_{jb} | T_j, \sigma^2)$ is the product of these two distributions. By defining the sufficient statistic as the sum of the residuals within the cell, $S_{jb} = \sum_{i=1}^{n_b} R_{jbi}$, we can expand the quadratic term $(R_{jbi} - \mu_{jb})^2$ and group the components by powers of μ_{jb} . The resulting expression for the integrand in Equation (3.24) is:

$$\begin{aligned} \pi(R_{jb}, \mu_{jb} | T_j, \sigma^2) &= (2\pi\sigma^2)^{-n_b/2} (2\pi\sigma_\mu^2)^{-1/2} \\ &\times \exp\left(-\frac{1}{2\sigma^2} \sum_{i=1}^{n_b} R_{jbi}^2 - \frac{1}{2} \left[\frac{n_b}{\sigma^2} + \frac{1}{\sigma_\mu^2}\right] \mu_{jb}^2 + \frac{S_{jb}}{\sigma^2} \mu_{jb}\right) \end{aligned} \quad (3.27)$$

To integrate out the cell mean μ_{jb} , we must complete the square for the terms in the exponent of the joint density that depend on μ_{jb} . Let us first isolate these terms:

$$-\frac{1}{2} \left[\mu_{jb}^2 \left(\frac{n_b}{\sigma^2} + \frac{1}{\sigma_\mu^2} \right) - 2\mu_{jb} \left(\frac{S_{jb}}{\sigma^2} \right) \right].$$

We begin by simplifying the coefficient of μ_{jb}^2 . By defining the marginal variance factor as $V_{jb} = \sigma^2 + n_b\sigma_\mu^2$, we can rewrite this coefficient over a common denominator:

$$\frac{n_b}{\sigma^2} + \frac{1}{\sigma_\mu^2} = \frac{n_b\sigma_\mu^2 + \sigma^2}{\sigma^2\sigma_\mu^2} = \frac{V_{jb}}{\sigma^2\sigma_\mu^2}.$$

Substituting this simplified coefficient back into the isolated exponent terms, we can factor out $\frac{V_{jb}}{\sigma^2\sigma_\mu^2}$:

$$\begin{aligned} -\frac{1}{2} \left[\frac{V_{jb}}{\sigma^2\sigma_\mu^2} \mu_{jb}^2 - \frac{2S_{jb}}{\sigma^2} \mu_{jb} \right] &= -\frac{V_{jb}}{2\sigma^2\sigma_\mu^2} \left(\mu_{jb}^2 - \frac{2\sigma^2\sigma_\mu^2 S_{jb}}{\sigma^2 V_{jb}} \mu_{jb} \right) \\ &= -\frac{V_{jb}}{2\sigma^2\sigma_\mu^2} \left(\mu_{jb}^2 - \frac{2\sigma_\mu^2 S_{jb}}{V_{jb}} \mu_{jb} \right). \end{aligned}$$

Next, we complete the square for the expression inside the parentheses:

$$\begin{aligned} -\frac{V_{jb}}{2\sigma^2\sigma_\mu^2} \left[\left(\mu_{jb} - \frac{\sigma_\mu^2 S_{jb}}{V_{jb}} \right)^2 - \left(\frac{\sigma_\mu^2 S_{jb}}{V_{jb}} \right)^2 \right] &= -\frac{V_{jb}}{2\sigma^2\sigma_\mu^2} \left(\mu_{jb} - \frac{\sigma_\mu^2 S_{jb}}{V_{jb}} \right)^2 + \frac{V_{jb}\sigma_\mu^4 S_{jb}^2}{2\sigma^2\sigma_\mu^2 V_{jb}^2} \\ &= -\frac{V_{jb}}{2\sigma^2\sigma_\mu^2} \left(\mu_{jb} - \frac{\sigma_\mu^2 S_{jb}}{V_{jb}} \right)^2 + \frac{\sigma_\mu^2 S_{jb}^2}{2\sigma^2 V_{jb}}. \end{aligned}$$

The second term in this result is now completely independent of μ_{jb} . We can move it into the constant part of the joint density's exponent. The full exponent of $\pi(R_{jb}, \mu_{jb} \mid T_j, \sigma^2)$ is now cleanly separated:

$$\left(-\frac{1}{2\sigma^2} \sum_{i=1}^{n_b} R_{jbi}^2 + \frac{\sigma_\mu^2 S_{jb}^2}{2\sigma^2 V_{jb}} \right) - \frac{V_{jb}}{2\sigma^2\sigma_\mu^2} \left(\mu_{jb} - \frac{\sigma_\mu^2 S_{jb}}{V_{jb}} \right)^2. \quad (3.28)$$

This separation reveals that the μ_{jb} dependent component is exactly the unnormalised kernel of a Gaussian distribution with variance $\sigma^2\sigma_\mu^2/V_{jb}$. Reintroducing the initial prefactors from the joint density, we can now integrate over μ_{jb} . The integral of this Gaussian kernel evaluates to its normalising constant, $\sqrt{2\pi\sigma^2\sigma_\mu^2/V_{jb}}$:

$$\begin{aligned} \pi(R_{jb} \mid T_j, \sigma^2) &= (2\pi\sigma^2)^{-n_b/2} (2\pi\sigma_\mu^2)^{-1/2} \exp \left(-\frac{1}{2\sigma^2} \sum_{i=1}^{n_b} R_{jbi}^2 + \frac{\sigma_\mu^2 S_{jb}^2}{2\sigma^2 V_{jb}} \right) \times \sqrt{\frac{2\pi\sigma^2\sigma_\mu^2}{V_{jb}}} \\ &= (2\pi\sigma^2)^{-n_b/2} \frac{\sqrt{2\pi\sigma^2\sigma_\mu^2}}{\sqrt{2\pi\sigma_\mu^2} \sqrt{V_{jb}}} \exp \left(-\frac{1}{2\sigma^2} \sum_{i=1}^{n_b} R_{jbi}^2 + \frac{\sigma_\mu^2 S_{jb}^2}{2\sigma^2 V_{jb}} \right). \quad (3.29) \end{aligned}$$

The constants cancel to leave a factor of $\sigma/\sqrt{V_{jb}}$, yielding the final, closed-form marginal likelihood for the single cell:

$$\pi(R_{jb} | T_j, \sigma^2) = (2\pi\sigma^2)^{-n_b/2} \frac{\sigma}{\sqrt{V_{jb}}} \exp\left(-\frac{1}{2\sigma^2} \sum_{i=1}^{n_b} R_{jbi}^2 + \frac{\sigma_\mu^2 S_{jb}^2}{2\sigma^2 V_{jb}}\right). \quad (3.30)$$

Since observations are routed to disjoint cells, the full marginal likelihood factorises across cells. Taking the product over $b = 1, \dots, B_j$ and writing $n = \sum_b n_b$ for the total sample size,

$$\pi(R_j | T_j, \sigma^2) = (2\pi\sigma^2)^{-n/2} \exp\left(-\frac{1}{2\sigma^2} \sum_{b=1}^{B_j} \sum_{i=1}^{n_b} R_{jbi}^2\right) \prod_{b=1}^{B_j} \frac{\sigma}{\sqrt{V_{jb}}} \exp\left(\frac{\sigma_\mu^2 S_{jb}^2}{2\sigma^2 V_{jb}}\right). \quad (3.31)$$

To construct the MH acceptance probability, we evaluate the quotient $\pi(R_j | T_j^*, \sigma^2)/\pi(R_j | T_j, \sigma^2)$. Let starred quantities $(B_j^*, n_b^*, S_{jb}^*, V_{jb}^*)$ denote the properties of the proposed tessellation T_j^* . Because the proposal mechanism simply regroups the exact same set of partial residuals into a new spatial partition, the global sum of squared residuals $\sum_b \sum_i R_{jbi}^2$ remains unchanged in T_j^* . Consequently, this exponential term, along with the $(2\pi\sigma^2)^{-n/2}$ coefficient, identically cancels out in the ratio. This isolation leaves the universal likelihood ratio:

$$\frac{\pi(R_j | T_j^*, \sigma^2)}{\pi(R_j | T_j, \sigma^2)} = \sigma^{B_j^* - B_j} \left(\frac{\prod_{b=1}^{B_j^*} V_{jb}^{*-1/2}}{\prod_{b=1}^{B_j} V_{jb}^{-1/2}} \right) \exp\left(\frac{\sigma_\mu^2}{2\sigma^2} \left[\sum_{b=1}^{B_j^*} \frac{(S_{jb}^*)^2}{V_{jb}^*} - \sum_{b=1}^{B_j} \frac{S_{jb}^2}{V_{jb}} \right]\right). \quad (3.32)$$

Every subsequent move-specific likelihood ratio is a special case of Equation 3.32, with the behaviour of the $\sigma^{B_j^* - B_j}$ depending on which class of move:

Constant centre count (AD, RD, SD and CC).

For structural modifications that preserve the total number of generating centres, we have $B_j^* = B_j$. The prefactor evaluates exactly to $\sigma^0 = 1$, and the ratio simplifies cleanly to:

$$\frac{\pi(R_j | T_j^*, \sigma^2)}{\pi(R_j | T_j, \sigma^2)} = \prod_{b=1}^{B_j} \sqrt{\frac{V_{jb}}{V_{jb}^*}} \exp\left(\frac{\sigma_\mu^2}{2\sigma^2} \left[\sum_{b=1}^{B_j} \frac{(S_{jb}^*)^2}{V_{jb}^*} - \sum_{b=1}^{B_j} \frac{S_{jb}^2}{V_{jb}} \right]\right). \quad (3.33)$$

Moves altering the centre count (AC and RC).

When proposing to add or remove a generating center, the total number of cells changes such that $B_j^* = B_j \pm 1$. In these cases, the prefactor equals $\sigma^{\pm 1}$ for an

addition move and σ^{-1} for a removal move.

3.3.4 Deriving The Add Dimension (AD) Structural Ratio

To isolate the components that vary across specific modifications from those that remain constant, we defined the structural ratio as the product of the prior ratio and the transition ratio:

$$\text{Structural Ratio} = \underbrace{\frac{\pi(T_j^*)}{\pi(T_j)}}_{\text{Prior Ratio}} \times \underbrace{\frac{q(T_j | T_j^*)}{q(T_j^* | T_j)}}_{\text{Transition Ratio}}. \quad (3.34)$$

Prior Ratio:

Let T_j be the current tessellation with d_j active dimensions and B_j generating centres. We propose T_j^* by adding a new dimension k^* , chosen uniformly at random from the $p - d_j$ currently inactive covariates. This increases the dimensionality from d_j to $d_j + 1$. We must also propose new coordinates for all B_j existing centres along this new dimension. Let D_j and D_j^* denote the subset of active dimensions in the current tessellation and proposed tessellation respectively. Similarly let C_j and C_j^* denote the set of coordinates for the centres in the current tessellation and proposed tessellation respectively.

To evaluate the prior ratio for a proposed topology T_j^* , the prior probability $\pi(T_j)$ must be decomposed into three hierarchical stochastic components. The first component, $\mathbb{P}(d_j)$, dictates the prior probability of the dimensionality of the active variable subspace. Conditional on this cardinality, the second component, $\mathbb{P}(D_j | d_j)$, governs the probability of selecting the specific active index set within D_j . Finally, conditioned on both the chosen subspace and the total number of cells B_j , the third component, $\mathbb{P}(C_j | B_j, D_j)$, defines the prior density for the spatial coordinates of the generating centres C_j . By conditional independence the overall prior ratio is expressed as:

$$\frac{\pi(T_j^*)}{\pi(T_j)} = \underbrace{\frac{\mathbb{P}(d_j + 1)}{\mathbb{P}(d_j)}}_{\text{Dimension Count}} \times \underbrace{\frac{\mathbb{P}(D_j^* | d_j + 1)}{\mathbb{P}(D_j | d_j)}}_{\text{Dimension Subset}} \times \underbrace{\frac{\mathbb{P}(C_j^* | B_j, D_j^*)}{\mathbb{P}(C_j | B_j, D_j)}}_{\text{Spatial Coordinates}} \quad (3.35)$$

1. Dimension Count:

The prior on the number of dimensions is defined as a shifted Binomial distribution, $d_j - 1 \sim \text{Binomial}(p - 1, \Omega/p)$. Letting $\pi = \Omega/p$, the ratio of the proposed $d_j +$

1 dimensions to the current d_j dimensions is derived by expanding the Binomial coefficients:

$$\frac{\mathbb{P}(d_j + 1)}{\mathbb{P}(d_j)} = \frac{\binom{p-1}{d_j} \pi^{d_j} (1 - \pi)^{p-1-d_j}}{\binom{p-1}{d_j-1} \pi^{d_j-1} (1 - \pi)^{p-d_j}} = \frac{p - d_j}{d_j} \left(\frac{\pi}{1 - \pi} \right) = \frac{p - d_j}{d_j} \left(\frac{\Omega}{p - \Omega} \right) \quad (3.36)$$

2. Dimension Subset:

Conditional on the number of active dimensions, the specific subset of covariates D_j is drawn uniformly from all possible combinations. The proposed state selects from $\binom{p}{d_j+1}$ combinations, while the current state selects from $\binom{p}{d_j}$:

$$\frac{p(D_j^* | d_j + 1)}{p(D_j | d_j)} = \frac{1/\binom{p}{d_j+1}}{1/\binom{p}{d_j}} = \frac{d_j + 1}{p - d_j} \quad (3.37)$$

3. Spatial Coordinates:

Because all spatial coordinates are drawn independently from $\mathcal{N}(0, \sigma_c^2)$, the joint spatial prior factors into a product of one-dimensional Gaussian distributions. Let $\phi(c_{lk}; \sigma_c)$ denote the Gaussian distribution from which the l^{th} generating center c_{lk} is drawn for the k^{th} dimension. Since the existing coordinates are perfectly preserved, their prior densities to cancel exactly in the ratio. The spatial prior ratio thus collapses to just the density of the B_j newly generated coordinates, $\{c_{1k^*}, \dots, c_{B_j k^*}\}$, mapped onto the new axis k^* :

$$\begin{aligned} \frac{\mathbb{P}(C_j^* | B_j, D_j^*)}{\mathbb{P}(C_j | B_j, D_j)} &= \frac{\left(\prod_{l=1}^{B_j} \prod_{k \in D_j} \phi(c_{lk}; \sigma_c^2) \right) \times \left(\prod_{l=1}^{B_j} \phi(c_{lk^*}; \sigma_c^2) \right)}{\prod_{l=1}^{B_j} \prod_{k \in D_j} \phi(c_{lk}; \sigma_c^2)} \\ &= \prod_{l=1}^{B_j} \phi(c_{lk^*}; \sigma_c^2) \end{aligned} \quad (3.38)$$

Final Prior Ratio:

After simplifying the product, we have the following expression for the overall Prior Ratio:

$$\frac{\pi(T_j^*)}{\pi(T_j)} = \frac{d_j + 1}{d_j} \left(\frac{\Omega}{p - \Omega} \right) \times \prod_{l=1}^{B_j} \phi(c_{lk^*}; \sigma_c^2) \quad (3.39)$$

Transition Ratio:

The transition ratio evaluates the asymmetry between the probability of proposing the new topology T_j^* from the current topology T_j (the forward move) and the probability of proposing T_j from T_j^* (the reverse move). We must decompose both the forward and reverse transition probabilities into a series of independent stochastic choices.

1. The Forward Move ($T_j \rightarrow T_j^*$):

The current state T_j has d_j active dimensions and B_j generating centres. Generating the proposed state T_j^* requires three sequential stochastic steps:

- **Move Selection:** The algorithm first selects the Add Dimension (AD) action from the set of available structural perturbations with probability p_{AD} .
- **Covariate Selection:** The algorithm must choose exactly which new dimension to add. It selects one dimension uniformly at random from the pool of $(p - d_j)$ currently inactive covariates. The probability of choosing the specific covariate k^* is $\frac{1}{p - d_j}$.
- **Spatial Coordinate Generation:** Because the tessellation is expanding into a new dimension k^* , all B_j existing generating centres must be assigned coordinates along this new axis. The algorithm draws B_j independent scalar values from a zero-mean normal distribution with variance σ_c^2 . The joint probability density of drawing these specific B_j coordinates, denoted as $\{c_{1k^*}, \dots, c_{B_j k^*}\}$, is the product of their individual Gaussian densities: $\prod_{l=1}^{B_j} \phi(c_{lk^*}; \sigma_c^2)$.

Combining these independent steps, the total forward transition probability density is:

$$q(T_j^* | T_j) = p_{AD} \times \frac{1}{p - d_j} \times \prod_{l=1}^{B_j} \phi(c_{lk^*}; \sigma_c^2) \quad (3.40)$$

2. The Reverse Move ($T_j^* \rightarrow T_j$):

To calculate the reverse probability, we must imagine the algorithm is currently at the proposed state T_j^* (which possesses $d_j + 1$ active dimensions and B_j generating centres) and attempts to perfectly reconstruct the original state T_j . This requires two sequential steps:

- **Move Selection:** The algorithm must select the Remove Dimension (RD) action, which is the exact structural inverse of AD, with probability p_{RD} .
- **Covariate Selection:** The algorithm must discard a dimension uniformly at random from its active set. Because T_j^* has $d_j + 1$ active dimensions, the probability that the algorithm randomly selects the exact dimension k^* for removal is $\frac{1}{d_j+1}$.

Note that removing a dimension deterministically deletes the coordinates along that axis; no stochastic generation of coordinates occurs during a reverse RD move. Therefore, the total reverse transition probability is simply:

$$q(T_j \mid T_j^*) = p_{RD} \times \frac{1}{d_j + 1} \quad (3.41)$$

Final Transition Ratio:

Dividing the reverse transition probability by the forward transition probability yields the complete transition ratio for the Add Dimension proposal:

$$\begin{aligned} \frac{q(T_j \mid T_j^*)}{q(T_j^* \mid T_j)} &= \frac{p_{RD} \times \frac{1}{d_j+1}}{p_{AD} \times \frac{1}{p-d_j} \times \prod_{l=1}^{B_j} \phi(c_{lk^*}; \sigma_c^2)} \\ &= \frac{p_{RD}}{p_{AD}} \times \frac{p-d_j}{d_j+1} \times \left(\prod_{l=1}^{B_j} \phi(c_{lk^*}; \sigma_c^2) \right)^{-1} \end{aligned} \quad (3.42)$$

Final Add Dimension Metropolis-Hastings Acceptance Ratio

The Structural Ratio isolates the non-data-driven components of the MH acceptance probability. Multiplying our derived components yeilds:

$$\begin{aligned} \text{Structural Ratio} &= \underbrace{\frac{d+1}{d} \left(\frac{\Omega}{p-\Omega} \right) \prod_{l=1}^b \mathcal{N}(c_{lk^*}; 0, \sigma_c^2)}_{\text{Prior Ratio}} \\ &\times \underbrace{\frac{p_{RD}}{p_{AD}} \left(\frac{p-d}{d+1} \right) \frac{1}{\prod_{l=1}^b \mathcal{N}(c_{lk^*}; 0, \sigma_c^2)}}_{\text{Transition Ratio}} \\ &= \frac{p_{RD}}{p_{AD}} \left(\frac{p-d}{d} \right) \left(\frac{\Omega}{p-\Omega} \right). \end{aligned} \quad (3.43)$$

Combining the Structural Ratio with the Likelihood Ratio, the final MH acceptance probability for the Add Dimension (AD) proposal evaluates to:

$$\alpha_{AD}(T_j \rightarrow T_j^*) := \min \left\{ 1, \frac{p(R_j | T_j^*, \sigma^2)}{p(R_j | T_j, \sigma^2)} \times \frac{p_{RD}}{p_{AD}} \left(\frac{p-d}{d} \right) \left(\frac{\Omega}{p-\Omega} \right) \right\} \quad (3.44)$$

3.3.5 Deriving The Add Centre (AC) Structural Ratio

Let T_j be the current tessellation with d_j active dimensions and B_j generating centres. We propose T_j^* by spawning a single new generating centre $\mathbf{c}^*$ in the active d_j -dimensional subspace. This increases the total number of centres from B_j to $B_j + 1$.

Prior Ratio:

The prior ratio for adding a centre decomposes into two components: the shifted Poisson prior on the total number of centres, and the continuous spatial prior on the newly generated coordinate vector. This decomposition arises from all other considerations around dimensionality remaining symmetric within T_j and T_j^* and therefore cancelling in the ratio.

- **Centre Count:** The prior over the number of centres is defined as a shifted Poisson distribution, $B_j - 1 \sim \text{Poisson}(\lambda_c)$. Transitioning from B_j to $B_j + 1$ centres evaluates to:

$$\frac{\mathbb{P}(B_j + 1)}{\mathbb{P}(B_j)} = \frac{\lambda_c^{B_j} e^{-\lambda_c} / B_j!}{\lambda_c^{B_j-1} e^{-\lambda_c} / (B_j - 1)!} = \frac{\lambda_c}{B_j} \quad (3.45)$$

- **Spatial Coordinates:** The new centre requires a scalar coordinate for each of the d_j active dimensions, denoted $\mathbf{c}^* = (c_1^*, \dots, c_{d_j}^*)$. Because the spatial partition is invariant to the labelling of the generating centres, they form an unordered set. The joint prior density for a configuration of B_j centres therefore includes a permutation factor $B_j!$. When proposing a new configuration with $B_j + 1$ centres, this factor becomes $(B_j + 1)!$. Since existing coordinates are preserved, their Gaussian densities cancel, leaving:

$$\frac{\mathbb{P}(C_j^* | B_j + 1)}{\mathbb{P}(C_j | B_j)} = \frac{(B_j + 1)!}{B_j!} \prod_{k=1}^{d_j} \phi(c_k^*; \sigma_c^2) = (B_j + 1) \prod_{k=1}^{d_j} \phi(c_k^*; \sigma_c^2) \quad (3.46)$$

Combining these components, the complete prior ratio for the AC proposal is:

$$\frac{\pi(T_j^*)}{\pi(T_j)} = \frac{\lambda_c}{B_j} (B_j + 1) \prod_{m=1}^{d_j} \phi(c_m^*; \sigma_c^2) \quad (3.47)$$

Transition Ratio:

To calculate $q(T_j | T_j^*)/q(T_j^* | T_j)$, we evaluate the probability of the forward move (spawning the centre) against the reverse move (deleting it).

- **The Forward Move ($T_j \rightarrow T_j^*$):** The algorithm selects the AC action with probability p_{AC} . It then generates the d_j -dimensional coordinate vector $\mathbf{c}^*$ by drawing each scalar independently from $\mathcal{N}(0, \sigma_c^2)$. The total forward transition probability density is:

$$q(T_j^* | T_j) = p_{AC} \times \prod_{m=1}^{d_j} \phi(c_m^*; \sigma_c^2) \quad (3.48)$$

- **The Reverse Move ($T_j^* \rightarrow T_j$):** To perfectly reconstruct the current state from the proposed state (which now possesses $B_j + 1$ centres), the algorithm must select the Remove Centre (RC) action with probability p_{RC} . It must then select the exact newly added centre $\mathbf{c}^*$ for deletion uniformly at random from the pool of $B_j + 1$ available centres. Because removing a centre deterministically deletes its coordinates, no stochastic spatial generation occurs. The total reverse transition probability is:

$$q(T_j | T_j^*) = p_{RC} \times \frac{1}{B_j + 1} \quad (3.49)$$

Dividing the reverse transition probability by the forward transition probability yields the final transition ratio:

$$\frac{q(T_j | T_j^*)}{q(T_j^* | T_j)} = \frac{p_{RC}}{p_{AC}(B_j + 1)} \times \left(\prod_{m=1}^{d_j} \phi(c_m^*; \sigma_c^2) \right)^{-1} \quad (3.50)$$

Final Add Centre (AC) Structural Ratio

The Structural Ratio isolates the non-data-driven components of the MH acceptance probability. Multiplying our derived components yields:

$$\begin{aligned}
 \text{Structural Ratio} &= \underbrace{\frac{\lambda_c}{B_j}(B_j + 1) \prod_{m=1}^{d_j} \phi(c_m^*; \sigma_c^2)}_{\text{Prior Ratio}} \\
 &\quad \times \underbrace{\frac{p_{RC}}{p_{AC}(B_j + 1)} \left(\prod_{m=1}^{d_j} \phi(c_m^*; \sigma_c^2) \right)^{-1}}_{\text{Transition Ratio}} \quad (3.51) \\
 &= \frac{p_{RC}}{p_{AC}} \left(\frac{\lambda_c}{B_j} \right).
 \end{aligned}$$

Combining the Structural Ratio with the Likelihood Ratio, the final MH acceptance probability for the Add Centre (AC) proposal evaluates to:

$$\alpha_{AC}(T_j \rightarrow T_j^*) := \min \left\{ 1, \frac{\pi(R_j \mid T_j^*, \sigma^2)}{\pi(R_j \mid T_j, \sigma^2)} \times \frac{p_{RC}}{p_{AC}} \left(\frac{\lambda_c}{B_j} \right) \right\} \quad (3.52)$$

3.3.6 Sampling from $\pi(M_j \mid T_j, R_j, \sigma^2)$

Following the structural update of the tessellation topology T_j via the MH step, the associated cell parameters $M_j = (\mu_{j1}, \dots, \mu_{jB_j})^\top$ must be updated. As in BART, this update follows directly from Gaussian conjugacy. The difference lies only in how observations are allocated to individual parameters. Whereas BART routes observations to terminal nodes through recursive binary splits, AddiVortes allocates observations to Voronoi cells according to their proximity to the generating centres. Once the data is geometrically partitioned into B_j disjoint cells, the conditional independence of the parameters applies identically, allowing for a mathematically equivalent closed form update to that seen in Section 2.3.3 for BART. Let $R_{jb} = (R_{j b_1}, \dots, R_{j b_{n_b}})^\top$ denote the subset of partial residuals in R_j , where n_b is the number of residuals allocated to the b^{th} cell with parameter μ_{jb} . Conditional on T_j , we have:

$$R_{jbi} \mid T_j, \mu_{jb}, \sigma^2 \sim \mathcal{N}(\mu_{jb}, \sigma^2), \quad \mu_{jb} \mid T_j \sim \mathcal{N}(0, \sigma_\mu^2).$$

The full conditional density of μ_{jb} is therefore given by:

$$\begin{aligned} \pi(\mu_{jb} \mid T_j, \sigma^2, R_{jb}) &\propto \pi(R_{jb} \mid T_j, \mu_{jb}, \sigma^2) \pi(\mu_{jb} \mid T_j) \\ &\propto \exp \left[-\frac{\sum_{i=1}^{n_b} (R_{jbi} - \mu_{jb})^2}{2\sigma^2} \right] \exp \left[-\frac{\mu_{jb}^2}{2\sigma_\mu^2} \right] \\ &\propto \exp \left[-\frac{(n_b \sigma_\mu^2 + \sigma^2) \mu_{jb}^2 - 2\sigma_\mu^2 \mu_{jb} \sum_{i=1}^{n_b} R_{jbi}}{2\sigma^2 \sigma_\mu^2} \right] \\ &\propto \exp \left[-\frac{\left(\mu_{jb} - \frac{\sigma_\mu^2 \sum_{i=1}^{n_b} R_{jbi}}{n_b \sigma_\mu^2 + \sigma^2} \right)^2}{\frac{2\sigma^2 \sigma_\mu^2}{n_b \sigma_\mu^2 + \sigma^2}} \right]. \end{aligned} \quad (3.53)$$

This is the kernel of a Normal distribution. Hence,

$$\mu_{jb} \mid T_j, \sigma^2, R_{jb} \sim \mathcal{N} \left(\frac{\sigma_\mu^2 \sum_{i=1}^{n_b} R_{jbi}}{n_b \sigma_\mu^2 + \sigma^2}, \frac{\sigma^2 \sigma_\mu^2}{n_b \sigma_\mu^2 + \sigma^2} \right). \quad (3.54)$$

Thus, conditional on the updated tessellation structure T_j , a cell parameter μ_{jb} is independently sampled for each $b = 1, \dots, B_j$, completing the Gibbs update of M_j .

3.3.7 Posterior Inference

Because AddiVortes shares the additive ensemble structure of BART, posterior inference proceeds in an almost identical manner to the methods described in Section 2.3.4. After discarding the initial burn-in period, the K retained MCMC iterations form an approximate sample from the joint posterior distribution:

$$\Theta^{(k)} = \left\{ (T_1^{(k)}, M_1^{(k)}), \dots, (T_m^{(k)}, M_m^{(k)}), \sigma^{(k)} \right\}, \quad k = 1, \dots, K.$$

Each draw defines a corresponding regression function evaluated at any covariate vector x :

$$f^{(k)}(x) = \sum_{j=1}^m g(x; T_j^{(k)}, M_j^{(k)}).$$

Pointwise posterior point estimates (such as the mean or median), standard deviations, and credible intervals are computed using the exact same empirical Monte Carlo estimators detailed previously for the tree-based framework. In practice, because the posterior may be highly multimodal or the proposal mechanism may mix slowly, the resulting Markov chain can exhibit autocorrelation. To mitigate this dependence, one may apply thinning, which involves retaining only every k^{th} sample from the chain (Owen, 2017). However, because discarding samples inherently reduces the total number of observations, thinning should be applied cautiously and is generally reserved for situations where memory constraints are applied.

Analogous to BART, the frequencies at which each variable is used within the ensemble can be used as variable importance measure. Stone and Gosling (2025a) provide a suggestion of this measure and show its ability to accurately separate important variables from noise variables. Since the work of Stone and Gosling (2025a) was to establish AddiVortes as a regression framework, they do not present a full variable selection framework for AddiVortes. The focus of the following chapter, and the main contribution of this thesis, is to provide a variable selection framework to the AddiVortes model.

3.3.8 Extensions of AddiVortes

Since the origination of AddiVortes by Stone and Gosling (2025a), there have been two main developments to the original algorithm. In particular, recent work has extended AddiVortes to handle binary response variables and non-constant error variance.

Binary AddiVortes

Introduced in Stone, Ogundimu, et al. (2025), the Binary AddiVortes framework adapts the continuous response architecture for classification tasks by integrating a probit link function within a data augmentation strategy. Rather than modelling the binary outcome directly, the model assumes an underlying continuous latent variable. The observed binary response is then determined via deterministic thresholding: if the latent variable is greater than zero, the response is classified as a positive outcome. This setup allows the framework to use the standard AddiVortes backfitting algorithm, requiring only an additional step to sample the latent variables from truncated normal distributions during posterior inference.

Heteroscedastic AddiVortes (H-AddiVortes)

The Heteroscedastic AddiVortes (H-AddiVortes) framework relaxes the assumption of constant error variance, allowing the conditional variance to be modelled as a function of the predictor space alongside the conditional mean. Stone and Gosling (2025b) achieve this by employing two separate Voronoi tessellation ensembles. The mean function is modelled using the standard additive sum-of-tessellations, whereas the variance function is modelled as the product of a distinct set of variance tessellations. This dual-ensemble structure ensures that the topological complexity and the active covariates driving the variance can differ entirely from those influencing the mean. By simultaneously modelling both components, H-AddiVortes can adaptively identify regions of high and low uncertainty, providing a more robust quantification of predictive variance in complex datasets.

Future Extensions

The AddiVortes framework presents several opportunities for methodological extension, similar to the development of BART. A primary direction is the development of a framework which uses probabilistic cell boundaries, comparable to SoftBART (Linero and Yang, 2018b). Standard AddiVortes partitions the covariate space using hard, discontinuous boundaries. A soft extension could introduce probabilistic cell assignments by weighting the contribution of each generating centre using a continuous distance metric. Further adaptations could parallel other established BART extensions. These include modifications for survival analysis, time-to-event data, and handling of missing data.

A significant limitation of the current AddiVortes formulation is the absence of a dedicated variable selection framework. Since its origination, BART has seen

significant exploration into its variable selection ability. The success of AddiVortes in regression tasks motivates the consideration that AddiVortes may also be able to leverage the improved flexibility of tessellations to further improve upon BART variable selection methods.

Chapter 4

Variable Selection using BART and AddiVortes

The previous chapters established the theoretical foundations of both BART and AddiVortes. We first demonstrated how BART achieves its non-parametric flexibility through an ensemble of weak regression trees. We then introduced AddiVortes as an alternative regression framework. By replacing the restriction of axis-aligned splitting rules with Voronoi tessellations, AddiVortes allows for highly flexible partitioning of the predictor space. This spatial flexibility enables the model to navigate complex data structures that may challenge tree-based partitioning. Although Stone and Gosling (2025a) focus on showing that this partition mechanism naturally improves regression ability, we theorise that the benefits of partitioning with tessellations extend directly to the domain of variable selection.

As BART has developed in recent years, a dedicated stream of literature emerged focused on an accompanying variable selection framework. These methodologies exploit the Bayesian ensemble structure to identify influential predictors in high-dimensional settings. In this chapter, we explore four of these primary BART variable selection frameworks. We first consider Variable Inclusion Proportions (VIP), which was originally introduced by Chipman et al. (1998) and later formalised into a threshold-based selection framework by Bleich et al. (2014a). To address inherent structural biases present in the VIP approach, we also examine the Metropolis Importance (MI) metric developed by Luo et al. (2021). Moving beyond permutation-based thresholds, we detail the spike-and-slab approach of the ABC algorithm proposed by Liu et al. (2021), and, finally, the sparsity-inducing Dirichlet prior of the DART model developed by Linero (2018).

Because AddiVortes shares a closely related Bayesian ensemble structure with BART, the underlying mechanisms of all four of these variable selection frame-

works can be naturally adapted to instead use the AddiVortes engine rather than BART. We detail the theoretical and practical extensions required to implement these methods within the AddiVortes framework. After detailing these adaptations, we conclude the chapter with a comprehensive simulation study to evaluate their selection accuracy. Our primary objective is to demonstrate the benefits of swapping the underlying partitioning engine of these established BART methods. By replacing the tree-based learners of BART with Voronoi tessellations, we aim to show that the flexibility of this new engine will improve variable selection performance.

4.1 Motivation, Definition and Historical Overview of Variable Selection

4.1.1 Motivating The Variable Selection Problem

Variable selection is a central problem in statistical modelling, inference, and prediction. Given a potentially large set of explanatory variables, the objective is to identify a subset of these variables that are genuinely relevant for explaining or predicting a response of interest. This problem arises naturally in a wide range of scientific disciplines, including genomics, economics and epidemiology, where technological advances increasingly generate datasets with a large number of candidate predictors (Donoho, 2000; Fan et al., 2006; Golub et al., 1999).

For instance, a single gene expression experiment can measure the activity of over 20,000 genes for each individual, while the study may include only a few hundred patients due to the costly data collection process. This abundance of covariates relative to the number of observations presents challenges for statistical inference and model stability, a phenomenon commonly referred to as the curse of dimensionality (Donoho, 2000). Figure 4.1 shows that as the number of predictors increases, the volume of the feature space expands exponentially, causing the available data to become increasingly sparse. In these settings, fitting statistical models leads to inflated variance, poor generalisation on unseen data, and a reduction in predictive accuracy (Hastie et al., 2009).

Once a model has been fit to high dimensional data, identifying the subset of truly informative variables provides critical benefits throughout the statistical modelling process. Firstly, it enables researchers to uncover meaningful relationships and learn the underlying mechanisms driving the response that might otherwise be obscured by noise. Furthermore, because each predictor often corresponds to a physical measurement that may be expensive, or time-consuming to collect, identifying

the subset of informative variables may reduce the cost and complexity of future data acquisition. Computationally, lower-dimensional models require less memory, train more efficiently, and demonstrate greater numerical stability. Removing irrelevant variables reduces model complexity and mitigates overfitting, frequently yielding improved predictive performance. Consequently, variable selection provides benefits across all aspects of scientific practice and is therefore a critical component of modern statistical analysis. Scenarios where the size of the covariate space is the primary structural challenge are typically characterised as $p \gg n$ problems, where n denotes the number of independent observations and p denotes the number of candidate covariates. Throughout this chapter, datasets possessing a large number of covariates relative to the number of observations will be formally referred to as **high-dimensional data**.

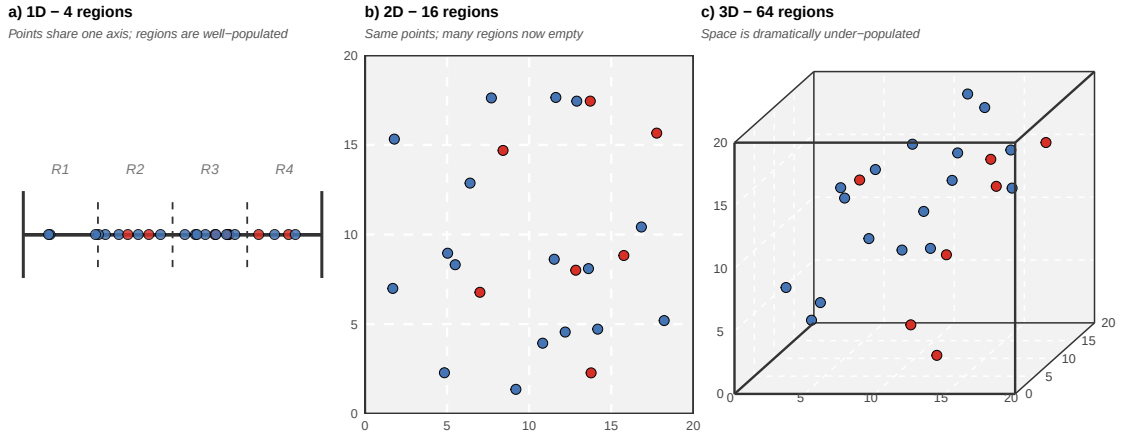


Figure 4.1: Illustration of the curse of dimensionality. As dimensionality increases, the volume of the feature space expands exponentially while the sample size remains fixed, causing the data to become increasingly sparse.

4.1.2 Defining The Variable Selection Problem

The preceding discussion establishes the necessity for methodologies that can identify the subset of predictors acting as the primary drivers of the response. We now provide a formal definition of the variable selection problem in the context of BART and AddiVortes. Consider a response variable $y \in \mathcal{Y}$ and a p -dimensional predictor vector $x = (x_1, \dots, x_p)^\top \in \mathcal{X} \subseteq \mathbb{R}^p$. While variable selection applies equally to discrete responses in classification, we focus on the standard non-parametric regression framework where $\mathcal{Y} \in \mathbb{R}$.

We assume the data generating mechanism is subject to additive noise such that:

$$y = f(x) + \varepsilon, \quad \varepsilon \sim \mathcal{N}(0, \sigma^2), \quad (4.1)$$

where $f : \mathcal{X} \rightarrow \mathbb{R}$ represents the unknown true conditional mean function, and σ^2 defines the residual error variance. While the noise distribution ε can theoretically take many forms, Gaussian errors remain the standard assumption within the BART and AddiVortes models discussed in this Chapter. Classical variable selection frameworks assume the existence of a strict **active set**, denoted as $\mathcal{S}^* \subseteq \{1, \dots, p\}$. In this case, the true regression function f depends on the full predictor vector x exclusively through the sparse subvector $x_{\mathcal{S}^*}$. Given $x_{\mathcal{S}^*}$, the response y is entirely independent of the remaining predictors, meaning their true influence on the response is exactly zero.

However, in highly flexible non-parametric frameworks such as those considered here, this strict binary classification of predictors is often an oversimplification. In reality, variables frequently exhibit a continuum of influence. Rather than a small set of covariates dictating the entire response, the true data generating mechanism is typically driven by a few primary predictors alongside many variables possessing marginal, non-zero effects. Consequently, for ensemble methods such as BART and AddiVortes, the objective fundamentally shifts from strict binary selection to quantifying **variable importance**. We therefore define the variable selection problem in this context as the construction of an estimated active set, $\hat{\mathcal{S}}$, that isolates the specific covariates most responsible for driving predictive performance. By doing so, we aim to separate the dominant structural signals from the weaker background noise.

4.1.3 Historical Development of Variable Selection

Variable selection methodologies have evolved to address competing demands for model flexibility, computational efficiency, and interpretability. Early approaches relied on discrete subset searches within classical linear models. Subsequent research developed continuous regularisation techniques before advancing to probabilistic selection within Bayesian ensembles. This progression illustrates the transition from rigid parametric assumptions to the adaptable non-parametric architectures central to this thesis.

Classical Discrete Subset Selection

Early approaches to evaluating variable importance were framed within the classical linear model:

$$\mathbb{E}[y \mid x] = \beta_0 + \sum_{j=1}^p \beta_j x_j$$

In this setting, the magnitude of each coefficient β_j provides a direct indication of variable importance, reflecting the expected change in the response for a one-unit increase in the predictor. To translate these rankings into a variable selection framework, a criterion is required to determine exactly which subset of predictors should be retained. Best subset selection achieves this by exhaustively evaluating all 2^p candidate sub-models. Rather than simply maximising training fit, which inherently favours larger models, it relies on complexity-penalising criteria such as AIC (Akaike, 1973) or BIC (Schwarz, 1978) to optimally balance goodness-of-fit against the number of included parameters. As the number of variables p increases, exhaustive subset search becomes computationally infeasible. This motivated the development of greedy procedures such as forward selection, backward elimination, and stepwise regression (Hastie et al., 2009). These methods, however, can be highly sensitive to small changes in the data, leading to different selected models and high variability in the resulting estimates.

Continuous Regularisation and the LASSO

The instability of discrete selection motivated the development of continuous regularisation methods, which control model complexity by penalising coefficient magnitudes rather than explicitly including or excluding predictors. More generally, regularised least squares estimates the coefficient vector by solving

$$\hat{\beta} = \arg \min_{\beta} \left\{ \frac{1}{2n} \sum_{i=1}^n \left(y_i - \beta_0 - \sum_{j=1}^p x_{ij} \beta_j \right)^2 + \mathcal{P}(\beta; \lambda) \right\}, \quad (4.2)$$

where the first term measures the discrepancy between the observed responses and the fitted values and $\mathcal{P}(\beta; \lambda)$ is a penalty function controlling the complexity of the fitted model. The tuning parameter $\lambda \geq 0$ determines the strength of regularisation, with the properties of the resulting estimators depending on the particular form of $\mathcal{P}$.

Hoerl et al. (1970) introduced Ridge regression, which uses the penalty $\mathcal{P}(\beta; \lambda) = \lambda \|\beta\|_2^2$. By penalising the squared L_2 norm of the coefficient vector, Ridge regression

shrinks coefficients towards zero and can stabilise estimation in the presence of multicollinearity. However, coefficients are generally not shrunk exactly to zero, meaning that an additional decision rule is required to determine which variables should be excluded.

The LASSO (Tibshirani (1996)) instead applies the L_1 penalty $\mathcal{P}(\beta, \lambda) = \lambda \|\beta\|_1$. The geometric structure of this constraint region includes vertices located exactly on the coordinate axes. The optimal solution frequently intersects this boundary at a corner, forcing specific coefficients to become identically zero. This structural property allows the LASSO to perform continuous shrinkage and discrete variable selection simultaneously.

Subsequent developments sought to address some of the limitations of the LASSO. The Elastic Net combines L_1 and L_2 penalties, encouraging correlated predictors to be selected together rather than arbitrarily favouring one over another (Zou and Hastie, 2005), while the Adaptive LASSO employs data-dependent penalty weights to allow different coefficients to be shrunk by different amounts (Zou, 2006). Despite these extensions, such approaches remain fundamentally point-estimation procedures and do not provide a natural framework for quantifying uncertainty.

The Bayesian Approach

Bayesian variable selection addresses the uncertainty inherent in choosing a single model by placing a posterior distribution over competing variable subsets. One Bayesian analogue of regularisation is the Bayesian LASSO of Park et al. (2008), obtained by assigning independent Laplace priors to the regression coefficients. Although this maintains the regularisation ability of LASSO, the continuous Laplace prior places substantial density near zero but assigns zero probability to $\beta_j = 0$. Consequently, unimportant coefficients are shrunk towards zero rather than setting them exactly to zero, meaning that an additional decision rule is still required to determine which variables should be excluded.

The spike-and-slab framework instead incorporates variable selection directly into the prior (Mitchell et al., 1988; George et al., 1993). Introducing an inclusion indicator $\gamma_j \in \{0, 1\}$ for each predictor, the coefficient prior is given by

$$\pi(\beta_j \mid \gamma_j) = (1 - \gamma_j)\pi_{\text{spike}}(\beta_j) + \gamma_j\pi_{\text{slab}}(\beta_j), \quad (4.3)$$

where the spike represents an excluded variable and is concentrated at or near zero, while the slab allows the coefficient to take a range of non-zero values, representing an included variable. The spike may be a point mass at zero or a narrow

continuous distribution. Assigning a prior to γ_j , for example $\gamma_j \sim \text{Bernoulli}(\theta)$, allows posterior inference to quantify the evidence for each variable through its marginal posterior inclusion probability,

$$\mathbb{P}(\gamma_j = 1 \mid \mathbf{y}). \quad (4.4)$$

Unlike methods that return a single selected model, these probabilities retain uncertainty about variable inclusion. To select variables and obtain a single model, a popular threshold is the median probability model which includes variables with posterior inclusion probabilities of at least 0.5 (Barbieri et al. (2004)).

Non-parametric Selection and Tree Ensembles

The methods considered so far are primarily based on linear models, and can therefore struggle when predictor effects are non-linear or arise only through interactions with other variables. In such settings, a variable may appear unimportant when considered through a linear effect despite contributing substantially to the response through more complex structure. Tree-based methods provide a natural alternative by partitioning the predictor space without imposing a fixed functional form.

The most prominent example is the Random Forest, which combines many decorrelated decision trees to capture non-linear relationships and interactions. Variable relevance is typically assessed using measures such as Mean Decrease in Impurity (MDI) or permutation-based importance. These quantities provide useful rankings, but they do not directly quantify uncertainty about variable inclusion. Moreover, MDI is known to favour predictors with many potential splitting points, including continuous and high-cardinality variables (Strobl, Boulesteix, Zeileis, et al., 2007).

These limitations motivated the development of a variable selection framework for BART (Chipman et al., 2010), which recasts the tree ensemble within a fully probabilistic framework. As seen in Chapter 1, BART models the response surface as a sum of regression trees and places priors over tree structure and terminal node parameters. As a result, variable relevance can then be read directly from the posterior distribution of splitting rules as suggested in Chipman et al. (2010). As we will see in this chapter, Bleich et al. (2014b) formalised this into variable inclusion proportions, and subsequent work has built sparsity directly into the tree prior itself, for example through Dirichlet priors on splitting probabilities Linero (2018) and Spike-and-Forest priors Ročková and Stephanie van der Pas (2019).

4.2 The Variable Inclusion Proportion (VIP) Metric

The foundational variable selection framework for BART originates in the methodology introduced by Chipman et al. (2010). To quantify predictor relevance, the authors propose variable inclusion proportions (VIP), which measure the proportion of splitting rules across the sampled tree ensembles that involve each predictor. The underlying intuition is that predictors useful for explaining variation in the response should be accepted more frequently during the MH proposal update, whereas uninformative predictors should be selected less often. Consequently, genuinely relevant predictors are expected to have higher variable inclusion proportions.

Definition 3.1: BART Variable Inclusion Proportion (VIP)

Let K denote the total number of retained posterior samples drawn from a BART ensemble. For each predictor index $j \in \{1, \dots, p\}$ and posterior sample $k \in \{1, \dots, K\}$, let c_{kj} denote the frequency at which the predictor x_j is used as an internal splitting variable across all trees in the ensemble. Furthermore, let $c_{k\cdot} = \sum_{j=1}^p c_{kj}$ represent the total number of internal splitting rules within the entire k^{th} posterior sample. The BART variable inclusion proportion (VIP) for predictor x_j is formally defined as:

$$v_j^{\text{VIP}} = \frac{1}{K} \sum_{k=1}^K \frac{c_{kj}}{c_{k\cdot}}. \quad (4.5)$$

The metric v_j^{VIP} quantifies the average frequency with which the BART ensemble selects x_j for partitioning across the posterior distribution. As established in Section 2.2.4, the standard BART specification places a uniform discrete prior over the available splitting variables at any given node. Consequently, the MH proposal mechanism considers all candidate predictors with equal theoretical probability when generating proposed structural updates. As a result, the frequency at which a variable makes it into the ensemble relies entirely upon the marginal likelihood ratio within the MH acceptance probability. Therefore, covariates that yield greater reductions in residual variance are accepted more frequently during the MH step, directly inflating their corresponding v_j^{VIP} value. Section C.1 details the exact calculation of this metric across a simplified example.

4.2.1 Effect of Ensemble Size on The BART VIP Metric

To optimise the VIP metric for variable selection, Chipman et al. (2010) recommend reducing the ensemble size hyperparameter, m , from its standard default of $m = 200$ to a substantially smaller value (e.g., $m = 20$ or $m = 50$). When the ensemble size is large, the algorithm is afforded abundant opportunities to generate splits across the MCMC iterations. Once the dominant structural effects of the true signal variables have been captured by a subset of trees, the remaining trees in the ensemble are left to model increasingly marginal residuals. These excess learners will frequently propose and accept splits on purely random variables to capture idiosyncrasies and local noise within the training data. Consequently, the raw splitting counts of the true signals are diluted as the inclusion proportions of the noise variables are artificially inflated. By reducing the number of trees in the ensemble, the model is granted less opportunities to split upon variables, and therefore variables must compete for inclusion.

We conduct a simulated experiment to investigate the effect of the ensemble size, m , on the VIP metric. We consider the classical Friedman (1991) dataset with $n = 250$ observations, $p = 10$ predictors, and noise level $\sigma = 1$. An overview of this data-generating process and parameter descriptions is provided in Appendix C.2. Note that in this data-configuration only $\{x_1, \dots, x_5\}$ contribute to the response and the remaining variables act purely as noise. We fit ten independent BART models with all other hyperparameters kept as standard, retaining 1000 posterior samples for the calculation of the VIPs. Figure 4.2 compares the VIPs obtained under varying ensemble sizes. When m is large ($m = 200$ and $m = 500$), the distinction between the true signal variables ($x_1, \dots, x_5$) and noise variables ($x_6, \dots, x_{10}$) deteriorates substantially. Conversely, restricting the ensemble size ($m = 10$ and $m = 50$) induces greater competition among candidate predictors, resulting in a clearer separation between signal and noise variables. In particular, the noise variables are assigned low inclusion proportions, while the true signals remain substantially above the theoretical baseline. These results confirm that a relatively small ensemble is preferable when using the VIP metric to assess variable importance. We fix $m = 20$ throughout the remainder of this chapter, unless stated otherwise, in line with previous studies (Chipman et al., 2010; Bleich et al., 2014b).

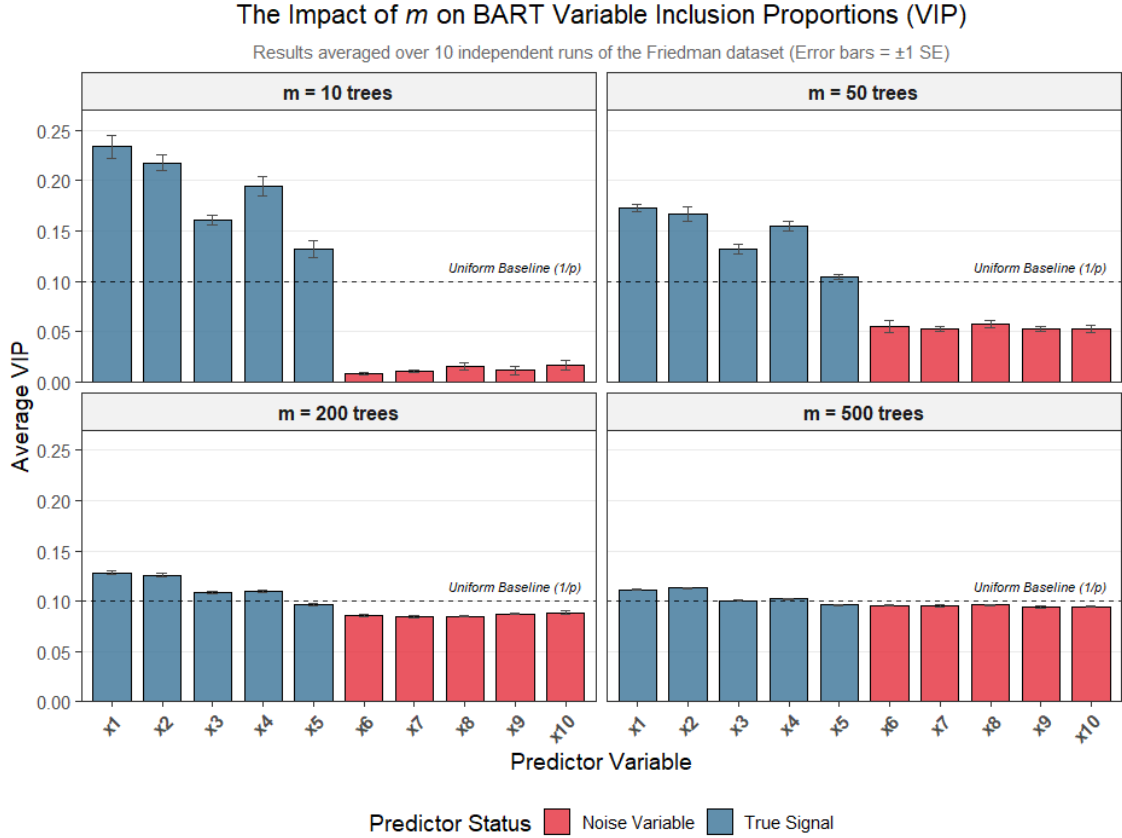


Figure 4.2: The effect of ensemble size (m) on BART Variable Inclusion Proportions (VIP). Bars display the average VIP scores for true signal variables (x_1 – x_5) versus noise variables (x_6 – x_{10}) across four tree capacities ($m \in \{10, 50, 200, 500\}$), utilising simulated Friedman data. Error bars denote ± 1 standard error calculated over 10 independent replicates. The dashed horizontal line represents the theoretical uniform prior baseline ($1/p = 0.10$).

4.2.2 Cardinality Bias in Tree Based Methods

We now examine a fundamental structural limitation of the BART VIP metric: its inherent bias against predictors with low cardinality. A **low-cardinality predictor** is formally defined as a discrete variable possessing a strictly limited set of unique values, such as a binary indicator. Bias against low-cardinality predictors is a widely documented observation within tree-based learning. For instance, Strobl, Boulesteix, Zeileis, et al. (2007) observe a systematic preference for high-cardinality predictors within Random Forests. More generally, variable selection mechanisms relying on Gini impurity or empirical split frequencies consistently exhibit this **cardinality bias** (Nicodemus et al., 2010). These studies demonstrate that recursive partitioning algorithms induce a functional preference for continuous or multi-level categorical variables simply because they offer a larger combinatorial space of po-

tential cutpoints. To illustrate the origin of this bias, consider a simplified regression scenario comprising two covariates: a binary predictor $x_1 \in \{0, 1\}$ and a continuous predictor $x_2 \in \mathbb{R}$. Figure 4.3 depicts a plausible decision tree constructed from these variables, perhaps within a BART ensemble.

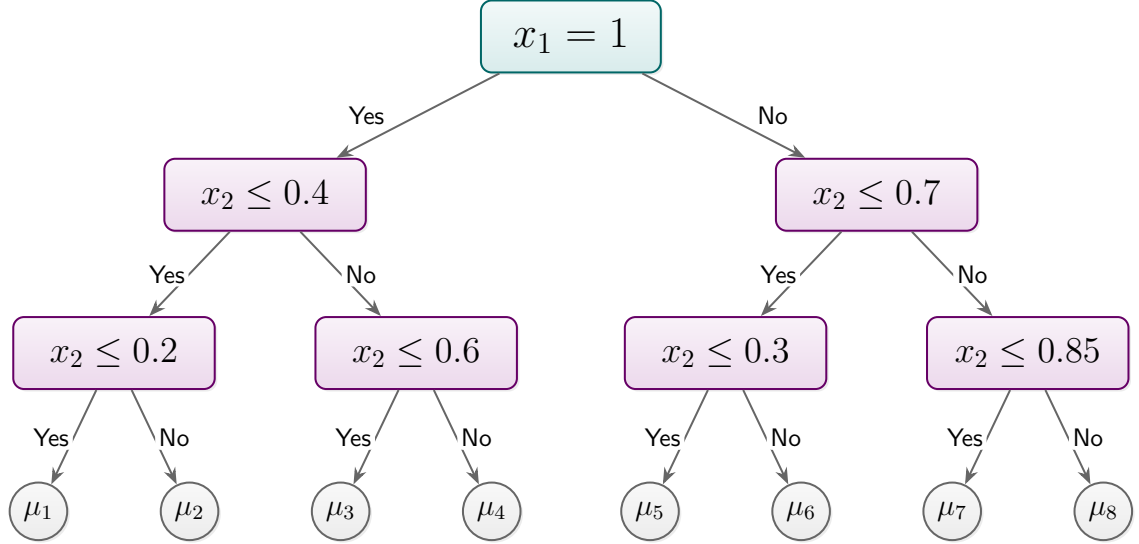


Figure 4.3: A visual demonstration of cardinality bias in a depth-three decision tree. The binary predictor x_1 exhausts its single cutpoint at the root, allowing the continuous predictor x_2 to dominate the subsequent partitions due to its abundance of available cutpoints.

The binary predictor x_1 forms the critical root partition, dictating the global topology of the tree. However, because x_1 is a binary variable, it instantly exhausts its only valid cutpoint along any given branch. Consequently, the recursive partitioning algorithm is structurally forced to draw subsequent splits exclusively from the high-cardinality predictor x_2 , which offers a functionally infinite set of available cutpoints. This example highlights how variable importance metrics based on splitting frequencies within trees are inherently biased towards higher cardinality variables. Although x_1 being selected for the root node suggests it is an important variable, it exhausts its only cut-point at the root and therefore accounts for only one of the seven total splits ($1/7 \approx 14\%$). Conversely, x_2 entirely dominates metrics based on inclusion frequencies despite capturing increasingly marginal residuals in the deeper nodes. This bias has been extensively proved to occur in Random Forests (Hothorn et al., 2006; Strobl, Boulesteix, Zeileis, et al., 2007) for variable importance metrics based on split frequencies. Consequently, we theorise that BART may also inherit this tendency to artificially inflate the importance of high-cardinality variables.

Beyond the structural mechanics visualised in Figure 4.3, the cardinality bias inherent to the VIP metric can be formally understood by recognising it as a constrained variant of the Gini Importance metric. Gini Importance is the standard variable evaluation criterion in Random Forests and has been extensively proven to systematically favour high-cardinality variables (Hothorn et al., 2006). For a derivation of the VIP metric as a special case of Gini Importance see Appendix C.3.

4.2.3 Empirical Experiment: Assessing Cardinality Bias in BART VIP

To empirically test for cardinality bias within the BART VIP metric, we design a simulated experiment to evaluate whether the ensemble systematically favours high-cardinality predictors over low-cardinality covariates when both possess a strictly controlled functional influence on the response variable.

Experimental Configuration A.1

We simulate a synthetic dataset comprising $n = 500$ observations and $p = 20$ candidate predictors. The predictor space is partitioned into two groups:

- **Binary Predictors** ($x_1, \dots, x_{10}$): Representing low-cardinality inputs, sampled from a Bernoulli distribution with $P(x_j = 1) = 0.5$.
- **Continuous Predictors** ($x_{11}, \dots, x_{20}$): Representing high-cardinality inputs, sampled from a standard uniform distribution, $\mathcal{U}(0, 1)$.

These covariates are passed through a standard Friedman function, $f_0(x)$, defined by a non-linear combination of five active predictors (for a comprehensive review of the standard Friedman data generation process, see Appendix C.2):

$$f_0(x) = 10 \sin(\pi x_1 x_{11}) + 20(x_{13} - 0.5)^2 + 10x_2 + 5x_{12}. \quad (4.6)$$

The observed continuous response y is generated by adding Gaussian noise to $f_0(x)$, such that $\varepsilon \sim \mathcal{N}(0, 1)$. This experimental design has been carefully chosen to isolate specific cardinality effects. Observe that the binary predictor x_2 possesses greater structural importance than the continuous predictor x_{12} , dictated by its coefficient being twice as large (10 versus 5). If we then see that x_2 has a VIP less than that of x_{12} , it will suggest there is bias. Furthermore, the remaining 15 variables act as purely stochastic noise.

Analysis of Configuration A.1 Results

We fit BART models using a restricted ensemble of $m = 20$ trees across 2,000 MCMC iterations, discarding the initial 500 iterations as burn-in. All other BART hyperparameters are set to their standard settings seen in Section 2.2.5. To guarantee algorithmic stability, we fit 10 independent BART runs, with the resulting VIP scores averaged across all simulations.

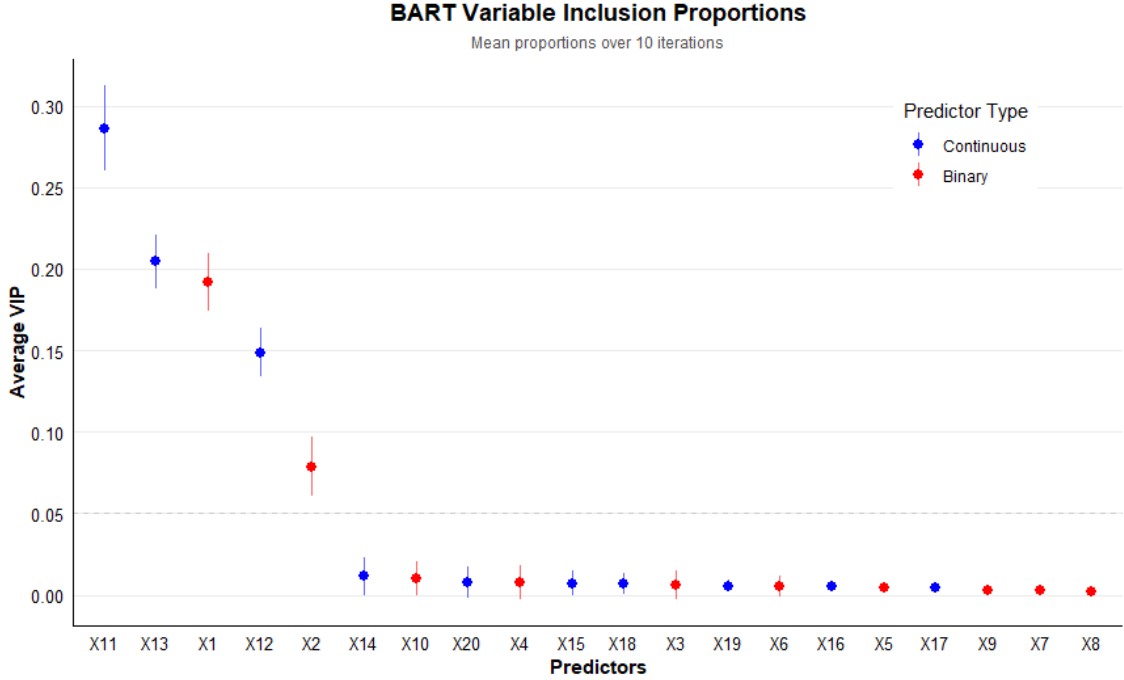


Figure 4.4: BART VIP bias against low-cardinality predictors. Despite strictly controlled functional importance, the metric ranks high-cardinality continuous predictors above their binary counterparts.

The results, visualised in Figure 4.4, demonstrate a seemingly successful variable selection procedure. The metric correctly identifies the five active predictors, maintaining their inclusion proportions well above the theoretical noise threshold of $1/p = 0.05$, while effectively suppressing the fifteen purely stochastic noise variables. However, upon closer inspection of the rankings among the active set, cardinality bias becomes evident. The response function is explicitly designed such that the variable x_2 exerts twice the functional importance on the response compared to the continuous variable x_{12} (possessing coefficients of 10 and 5, respectively). Nevertheless, the VIP metric fails to reflect this, assigning a higher inclusion score to x_{12} . Following Luo et al. (2021), we suggest that this occurs because the BART algorithm exploits the abundance of cut-points provided by the continuous variable, inflating the selection frequency of these predictors.

Experimental Configuration A.2

To explore the persistence of this algorithmic bias, we modify the Friedman experiment seen in Luo et al. (2021). This modification is inspired by the “power setting” simulations introduced by Strobl, Boulesteix, Zeileis, et al. (2007), which systematically scaled the functional importance of specific predictors to ascertain whether cardinality bias in Random Forests could entirely mask genuine underlying signals. In Configuration A.1, standard VIP favoured the continuous predictor x_{12} despite x_2 having a coefficient of twice the magnitude. We theorised this occurred because the functionally infinite geometric splitting space of continuous variables artificially inflates their inclusion frequency, overpowering the true functional signal of the binary covariate. To determine the persistence of this bias, we introduce a deterministic sensitivity parameter, α , to replace the fixed coefficient of x_2 . The generative response model is redefined as:

$$f_0(x) = 10 \sin(\pi x_1 x_{11}) + 20(x_{13} - 0.5)^2 + \alpha x_2 + 5x_{12} + \varepsilon, \quad (4.7)$$

where $\varepsilon \sim \mathcal{N}(0, 1)$. By systematically scaling the magnitude of α (e.g., $\alpha \in \{5, 50, 100, 250\}$), we can dynamically scale the functional importance of the binary predictor x_2 relative to its continuous counterpart x_{12} , which remains strictly fixed at a coefficient of 5. Theoretically, for a sufficiently large α , the VIP score of x_2 must eventually overcome the cardinality bias and dominate the VIP of x_{12} .

This α -scaling framework allows us to identify the severity of the cardinality bias in the BART VIP metric. We aim to find the exact signal-to-noise ratio at which the true functional importance of a low-cardinality variable finally overcomes the algorithm’s inherent preference for high-cardinality variables. The results of this sensitivity analysis are presented in Figure 4.5. For visual clarity, the plot exclusively displays the inclusion proportions for the five active variables: $\{x_1, x_2, x_{11}, x_{12}, x_{13}\}$.

Analysis of Configuration A.2 Results

The empirical results visualised in Figure 4.5 demonstrate the severity of the algorithmic cardinality bias. At the baseline condition ($\alpha = 5$), the binary predictor x_2 and the continuous predictor x_{12} possess identical coefficients in the response function, yet the VIP metric assigns x_{12} a score approximately three times larger. When the coefficient of x_2 is raised to $\alpha = 50$, we observe a marginal improvement as the VIP of x_2 and x_{12} become closer. However this result is still unsatisfying considering that we have now given the binary variable x_2 a coefficient which is ten times the magnitude of the x_{12} coefficient, and yet BART VIP metric still favours

the continuous variable. Increasing α beyond this, we see no further improvement, confirming the topological exhaustion mechanism. Regardless of how important the variable may be, a low-cardinality variable's inclusion frequency is strictly bounded by its limited valid cut-points. Once x_2 forms the primary structural partitions, it becomes topologically invalid for descendant splits, forcing the algorithm to draw from the functionally infinite cut-points of x_{12} to model remaining residuals.

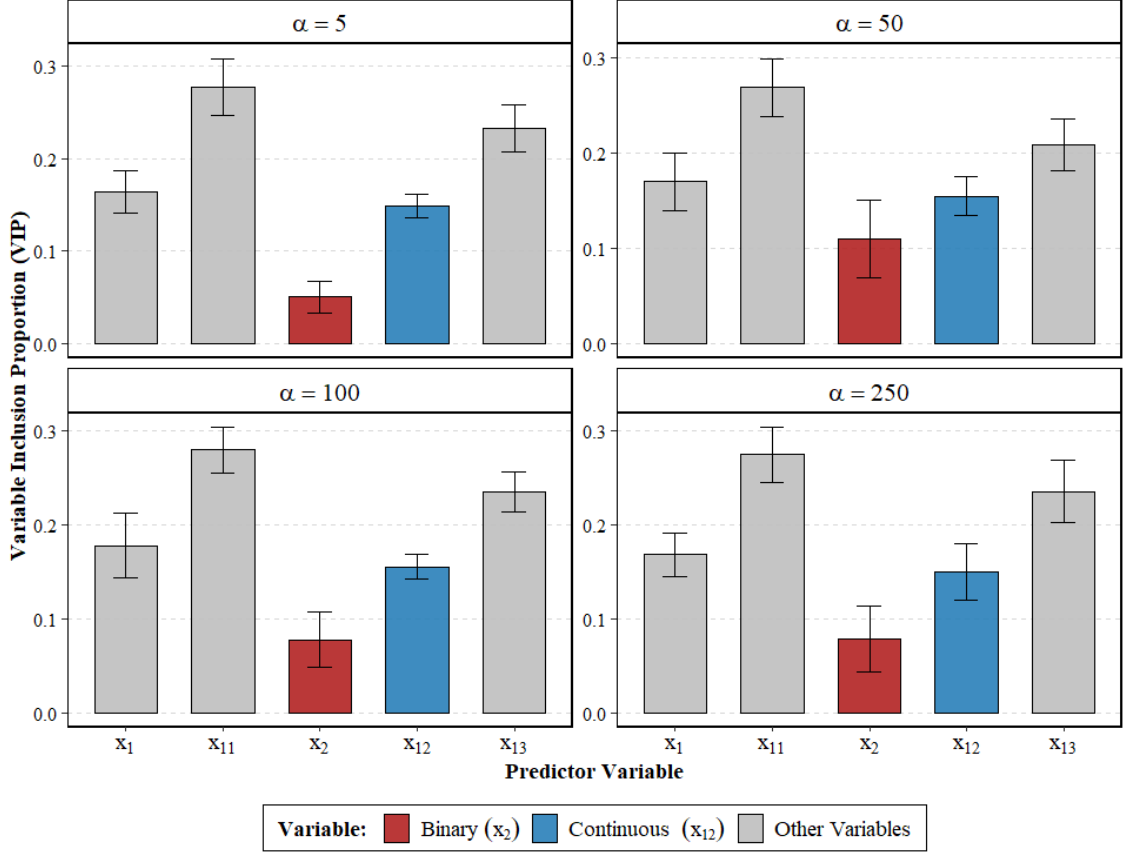


Figure 4.5: **Sensitivity of BART Variable Inclusion Proportions to Signal Magnitude.** Mean VIP scores (± 1 standard deviation) across 10 independent BART models utilizing the modified α -scaled Friedman equation. To quantify the severity of the cardinality bias, the true coefficient (α) of the active binary predictor x_2 is systematically scaled from 5 to 250, while the active continuous predictor x_{12} remains fixed at a coefficient of 5. At $\alpha = 5$, despite possessing equal mathematical weight in the underlying data generating process, the continuous predictor x_{12} heavily dominates due to its effectively infinite splitting space. Even in the extreme structural case of $\alpha = 250$, the binary signal x_2 fails to mathematically overcome the algorithmic bias toward the continuous variable.

4.3 The Metropolis-Hastings Information (MI) Metric

The preceding experiments demonstrate a fundamental limitation of the VIP metric: a systematic bias against low-cardinality predictors. This is particularly problematic in applied settings, where datasets commonly contain covariates with heterogeneous cardinalities. Clinical and genomic applications, for example, often combine binary indicators, such as treatment assignment or genetic mutations, with continuous measurements such as blood pressure. A variable importance metric that systematically downweights discrete predictors because they admit fewer potential splits may therefore fail to identify genuinely informative predictors, treating meaningful structure as noise. This bias places the standard VIP framework at a disadvantage compared to established variable selection methods whose measures of predictor relevance are not based on splitting frequencies. The LASSO, for example, evaluates predictors through continuous coefficient shrinkage (Tibshirani, 1996), while Conditional Random Forests use conditional permutation importance to account for biases arising from the split-selection mechanism (Strobl, Boulesteix, Kneib, et al., 2008). Neither approach inherently penalises a predictor simply because it admits fewer potential cutpoints. A comparable variable selection metric is therefore required for BART if predictor relevance is to reflect genuine association with the response rather than the cardinality of the predictor.

To address this limitation, Luo et al. (2021) propose the **Metropolis-Hastings Information (MI)** metric. Unlike the standard VIP, which records only whether a predictor appears in an accepted splitting rule, MI contextualises each appearance using the MH acceptance probabilities generated during the BART MCMC sampling. These probabilities quantify the posterior support for proposed structural modifications and therefore provide information about how strongly the model favours a proposed split involving a particular predictor. Splitting frequencies distinguish only between predictors that are used often and those that are used rarely, they do not account for the degree to which individual splits are supported by the posterior. MI weights variable usage according to the corresponding MH acceptance probabilities, shifting the assessment of variable importance from the frequency with which a predictor is used to the strength of posterior support for its proposed partitions. The key innovation of Luo et al. (2021) extends beyond accounting for the strength of individual variable inclusions. As demonstrated in the following section, incorporating this information also mitigates the cardinality bias inherent in standard inclusion frequencies.

4.3.1 Metropolis Information: Derivation and Definition

We first recall the MH-within-Gibbs sampler established in Section 2.3.2. During each MCMC transition, the algorithm proposes a topological modification to a tree, drawn from a discrete set of possible modifications: GROW, PRUNE, CHANGE, or SWAP. Once a structural proposal is generated, the algorithm computes the MH acceptance ratio, which is evaluated against a standard uniform random variable to dictate acceptance.

The standard VIP metric evaluates variable importance solely through the frequency with which a predictor is selected for splitting during this MH update. Although this provides an intuitive measure of feature relevance, it assigns equal inferential value to every inclusion. This ignores the fact that each split can contribute very differently to the fitted model. For example, a variable added by a GROW proposal deep within a tree may capture only a small, localised feature of the residual structure. Conversely, a split near the root can induce a substantial partition of the predictor space and contribute more broadly to the model fit. By assigning equal weight to every observed inclusion, the standard VIP does not account for the context in which a split appears. A more informative metric should therefore move beyond raw inclusion frequencies and account for both:

- the topological depth (d_η) at which the proposed split occurs
- the marginal likelihood improvement associated with the proposed structural update.

To introduce this contextual evaluation for each split, Luo et al. (2021) propose constructing an importance metric based on $\pi_{\text{Grow}}(\eta)$, the MH acceptance probability for a GROW proposal at a terminal node η of tree T_j . This probability is bounded and defined as:

$$\pi_{\text{Grow}}(\eta) = \min\{1, r(\eta)\}. \quad (4.8)$$

As derived in Section 2.3.2, the unbounded acceptance ratio $r(\eta)$ for a GROW proposal decomposes into the product of the transition ratio, the tree structure ratio, and the marginal likelihood ratio:

$$r(\eta) = \frac{q(T_j \mid T_j^*)}{q(T_j^* \mid T_j)} \cdot \frac{\pi(T_j^*)}{\pi(T_j)} \cdot \frac{\pi(R_j \mid T_j^*, \sigma^2)}{\pi(R_j \mid T_j, \sigma^2)}. \quad (4.9)$$

Substituting the explicit forms derived for these components in Section 2.3.2 yields:

$$\begin{aligned}
r(\eta) &= \left(\frac{\mathbb{P}(\text{PRUNE})}{\mathbb{P}(\text{GROW})} \frac{b \cdot p_{\text{adj}}(\eta) \cdot n_{j,\text{adj}}(\eta)}{w_2^*} \right) \\
&\quad \cdot \left(\alpha \frac{\left(1 - \frac{\alpha}{(2+d_\eta)^\beta}\right)^2}{((1+d_\eta)^\beta - \alpha) p_{\text{adj}}(\eta) n_{j,\text{adj}}(\eta)} \right) \\
&\quad \cdot \frac{\pi(R_m \mid T_m^*, \sigma^2)}{\pi(R_m \mid T_m, \sigma^2)}
\end{aligned} \tag{4.10}$$

$$= \frac{b}{w_2^*} \cdot \alpha \frac{\left(1 - \frac{\alpha}{(2+d_\eta)^\beta}\right)^2}{(1+d_\eta)^\beta - \alpha} \cdot \frac{\pi(R_m \mid T_m^*, \sigma^2)}{\pi(R_m \mid T_m, \sigma^2)}. \tag{4.11}$$

where w_2^* represents the number of internal nodes with exactly two terminal children in the proposed tree T_j^* , and b denotes the total number of terminal nodes in the current tree T_j . The simplification in the final equality relies on the assumption that, $\mathbb{P}(\text{GROW}) = \mathbb{P}(\text{PRUNE}) = 0.25$. Following Luo et al. (2021), the acceptance ratio isolates three distinct structural drivers:

$$\text{Nodes Ratio} = \frac{b}{w_2^*}, \tag{4.12}$$

$$\text{Depth Ratio} = \alpha \frac{\left(1 - \frac{\alpha}{(2+d_\eta)^\beta}\right)^2}{(1+d_\eta)^\beta - \alpha}, \tag{4.13}$$

$$\text{Likelihood Ratio} = \frac{\pi(R_m \mid T_m^*, \sigma^2)}{\pi(R_m \mid T_m, \sigma^2)}. \tag{4.14}$$

The contribution of each term to the MH acceptance ratio can now be considered in turn. We begin with the nodes ratio, b/w_2^* , which depends on the topology of the proposed tree and therefore varies across MCMC iterations. Its range is, however, restricted. A GROW proposal converts the terminal node η into an internal node with two terminal children, ensuring that $w_2^* \geq 1$. Consequently,

$$\frac{b}{w_2^*} \leq b. \tag{4.15}$$

Since the BART priors strongly favour small trees, b is typically small and the nodes ratio can contribute only a small factor. Even under the generous bound of $b \leq 10$, the largest possible nodes ratio value is 10. The depth ratio operates on a significantly different scale. As illustrated in Figure 4.6, it falls from approximately 11.05 for a proposal at the root ($d_\eta = 0$) to 0.058 at $d_\eta = 3$, before declining further to 0.019 at $d_\eta = 6$. As a result, the depth ratio will significantly dominate the nodes

ratio in the full product. The dominance of the depth ratio in the structural product is what enables the MI metric to mitigate cardinality bias. High-cardinality predictors can be used repeatedly for redundant splits deep within a tree, inflating their standard VIP through additional inclusion counts. The depth ratio heavily down-weights these deep partitions, ensuring that a variable importance metric based on $r(\eta)$ places greater emphasis on the earlier, more consequential splits whilst down-weighting redundant splits deeper within the tree.

Finally, the full acceptance ratio $r(\eta)$ also contains the likelihood ratio, which compares the marginal likelihood of the proposed and current tree structures. This term weights a proposed split according to the support it receives from the observed data, complementing the structural weighting imposed by the nodes and depth ratios. A variable importance metric based on $r(\eta)$ will therefore evaluate splits according to both their position within the tree and their contribution to model fit, rather than naively treating every inclusion as equally informative. Because the raw acceptance ratio $r(\eta)$ is unbounded and can exhibit substantial positive skew, Luo et al. (2021) instead construct the MI metric using the bounded acceptance probability $\pi_{\text{GROW}}(\eta)$ defined in Equation (4.8).

Definition 3.2: Metropolis Importance (MI): Let K denote the total number of retained posterior samples drawn from a BART ensemble. For each candidate predictor x_i ($i = 1, \dots, p$) and posterior sample $k \in \{1, \dots, K\}$, define $\tilde{u}_{ki}$ as the average Metropolis acceptance probability across all internal splitting nodes utilising x_i :

$$\tilde{u}_{ki} = \frac{\sum_{j=1}^m \sum_{\eta \in \phi_{jk}} \mathbb{I}(v(\eta) = i) \pi_{\text{GROW}}(\eta)}{\sum_{j=1}^m \sum_{\eta \in \phi_{jk}} \mathbb{I}(v(\eta) = i)}, \quad (4.16)$$

where m is the total number of trees in the ensemble, ϕ_{jk} defines the set of all internal splitting nodes within the j^{th} tree of the k^{th} posterior sample, and $v(\eta)$ is a function returning the index of the splitting variable at node η . If a predictor x_i is not selected for any split within the k^{th} posterior sample, $\tilde{u}_{ki}$ is defined as zero. The BART Metropolis Importance (MI) for predictor x_i is formally defined as the normalised acceptance probability averaged across the K posterior samples:

$$v_i^{\text{MI}} = \frac{1}{K} \sum_{k=1}^K \frac{\tilde{u}_{ki}}{\sum_{l=1}^p \tilde{u}_{kl}}. \quad (4.17)$$

The formulation is similar to the standard VIP in that variable importance is still determined by aggregating information across the splitting rules of the posterior tree ensembles. The key difference is that VIP treats each observed inclusion as

equally informative, whereas MI weights each inclusion according to its associated MH acceptance probability. MI therefore retains information about the degree of posterior support for a proposed inclusion, allowing variable importance to reflect not only how frequently a predictor is used, but also how strongly the posterior favours the partitions it produces. This provides a more informative measure of predictor relevance in general, while the depth-dependent weighting of the MH acceptance probabilities additionally mitigates the cardinality bias that can arise when high-cardinality predictors are repeatedly selected for redundant splits.

Furthermore, MI can be computed as a computationally cost-free extension of the standard BART algorithm, since the MH acceptance ratios required for its construction are already evaluated during MCMC sampling and would otherwise be discarded. As noted by Luo et al. (2021), effective application of the MI metric also requires a reduced ensemble size, for the same reasons discussed for VIP in Section 4.2.1. Note that all other hyperparameters can be kept as standard for evaluating MI. A detailed example illustrating the calculation of MI across is provided in Appendix C.4.

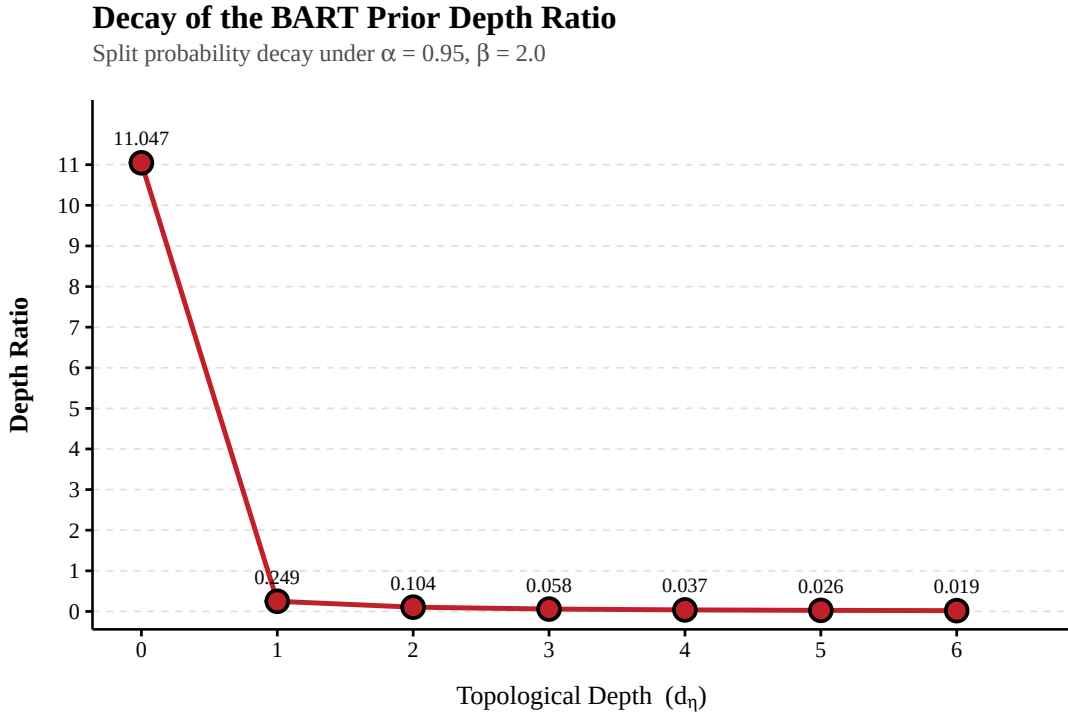


Figure 4.6: **Asymptotic Decay of the Depth Ratio.** The depth ratio is evaluated across increasing topological depths (d_η) using standard BART hyperparameters ($\alpha = 0.95$, $\beta = 2$). The function enforces a strict structural penalty on deeper partitions, decaying from 11.05 at the root node ($d_\eta = 0$) to 0.06 by the third generation ($d_\eta = 3$).

4.3.2 Assessing Cardinality Bias in BART MI

Having established the motivation for the MI metric, we now assess whether it mitigates the cardinality bias observed under standard VIP. To provide a direct comparison, we revisit Configuration A.1, using the same BART models previously fitted for the VIP analysis and extracting both VIP and MI scores from the resulting posterior samples. In this configuration, the comparison of primary interest is between x_2 and x_{12} . Both variables contribute to the response, but x_2 is binary with coefficient 10, whereas x_{12} is continuous with coefficient 5. The data-generating mechanism therefore assigns a stronger functional effect to x_2 , and a variable importance metric unaffected by cardinality bias should rank x_2 above x_{12} . As shown in the previous section, the standard VIP fails to recover this ordering, instead assigning greater importance to the continuous predictor x_{12} .

The results in Figure 4.7 show how this comparison changes under MI. While VIP ranks the weaker continuous signal x_{12} above the stronger binary signal x_2 , MI reverses this ordering and assigns greater importance to x_2 . This is consistent with the preceding analysis as the additional deep, redundant splits available to the continuous predictor contribute to its VIP, but receive substantially less weight under the MI metric. By incorporating the posterior support associated with each inclusion, MI reduces the advantage otherwise obtained from the greater number of available cutpoints.

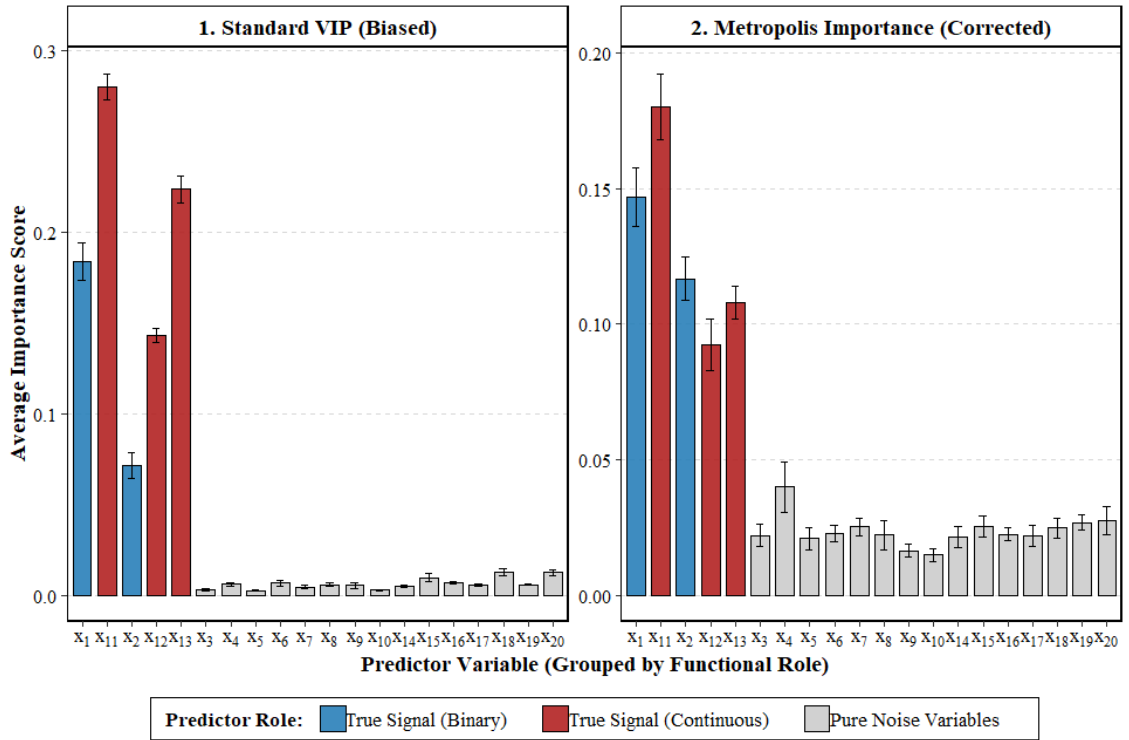


Figure 4.7: **Comparison of Variable Inclusion Proportions and Metropolis Importance.** Standard VIP (left) and MI (right) for the modified Friedman configuration. The comparison of interest is between the binary predictor x_2 , with coefficient 10, and the continuous predictor x_{12} , with coefficient 5. Standard VIP assigns greater importance to x_{12} despite its weaker contribution to the response, whereas MI recovers the expected ordering and ranks x_2 above x_{12} .

4.4 Permutation-Based Variable Selection Thresholds

Having established two measures of variable importance within a BART ensemble, the standard Variable Inclusion Proportion (VIP) proposed by Chipman et al. (2010) and the Metropolis Importance (MI) metric of Luo et al. (2021), we now turn to the problem of variable selection. An importance score alone provides a relative measure of predictor relevance, but does not determine whether a predictor should ultimately be retained or excluded. This requires an inclusion threshold and raises the fundamental question:

How large must a VIP or MI score be before a predictor can be considered important?

A natural starting point is to consider a fixed threshold, whereby predictors are selected whenever their importance score exceeds some predetermined value. Such a rule is simple and provides a direct link between the importance metric and the final selected model. Its validity, however, depends on the distribution of the importance scores being sufficiently stable across datasets and BART models. The following analysis shows that this assumption does not hold. We show that the variability of these metrics depends on the data and the fitted ensemble. Consequently, the same importance score may represent strong evidence of relevance in one setting but be consistent with the level of importance obtained by irrelevant predictors in another. We therefore first examine this variability and show why a universal threshold is insufficient. This motivates the permutation-based approach of Bleich et al. (2014a), which replaces a fixed threshold with a dataset-specific null distribution obtained by removing the relationship between the predictors and response. The resulting procedure provides an adaptive reference against which the observed VIP or MI scores can be evaluated, thereby extending these variable importance metrics into complete variable selection frameworks.

4.4.1 The Limitations of a Fixed Threshold

A natural starting point for answering the question above is a fixed threshold. In this case, a predictor is chosen whenever its importance score exceeds a predetermined constant. Such a rule is simple, but its validity depends on the distribution of the importance metric being sufficiently stable for the same numerical cut-off to be meaningful across different datasets. We first examine this question under a **null**

setting, in which the response is independent of every candidate predictor. In this case, no predictor contains genuine information about the response, so BART is fit to data where there is no basis for favouring one variable over another. Under the standard BART uniform prior (see Section 2.2.4), the expected VIP of each of the p predictors is therefore $1/p$. A fixed threshold based on this expectation would only be useful if the VIP's of irrelevant predictors remained sufficiently close to this baseline.

To examine if this behaviour holds true, we simulate a null dataset with $n = 250$ observations and $p = 40$ predictors, where both the covariates $X \in \mathbb{R}^{n \times p}$ and the continuous response $y \in \mathbb{R}$ are generated independently from standard normal distributions. A BART model with $m = 20$ trees is fitted to the data, generating 10,000 posterior samples. For each predictor, the cumulative VIP is recorded across the MCMC iterations. The resulting VIPs of all predictors are shown in Figure 4.8.

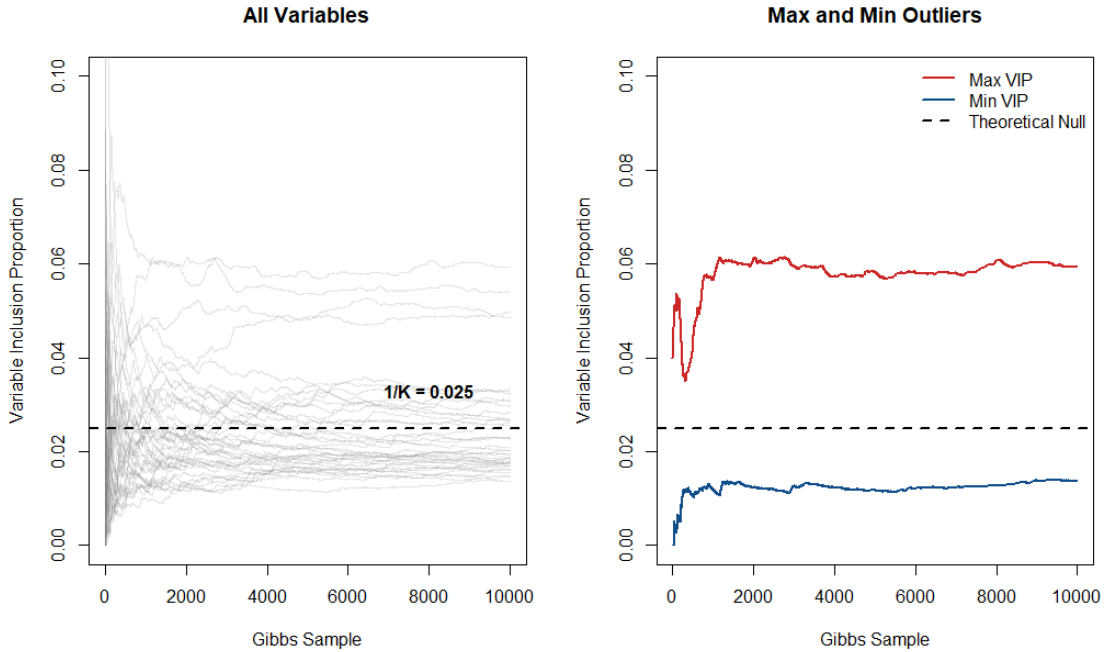


Figure 4.8: **Cumulative BART VIP under a null setting.** The model is fitted to $n = 250$ observations and $p = 40$ purely stochastic predictors. Despite the absence of any association with the response, several empirical inclusion proportions stabilise above the theoretical baseline of $1/p$. The right panel highlights the most extreme deviations.

The behaviour differs substantially from the theoretical expectation. Although all predictors are purely noise, several VIPs stabilise well above $1/p$ rather than converging to a common value. These deviations persist throughout the MCMC run and are reproduced across independent simulations.

This has a direct implication for fixed-threshold selection. A cutoff close to $1/p$ would incorrectly select some irrelevant predictors, whereas a sufficiently high threshold to exclude the largest VIP could remove genuinely weak signals in other datasets. The appropriate threshold therefore depends on the level of variation generated under the particular data and model configuration, rather than being determined by a universal constant. Before introducing an adaptive threshold, it is necessary to understand the source of this variation. The observed divergence in Figure 4.8 may reflect incomplete MCMC mixing, or it may arise from BART overfitting to random noise structures within a particular dataset. The following subsection examines these competing explanations. For brevity, the following analysis focuses on VIPs, although we find that all subsequent conclusions extend to MI.

4.4.2 Investigating the Instability of BART VIPs in the Null Setting

The presence of outlying VIPs observed in Figure 4.8 creates a risk of Type I errors, since variables with no association with the response may be incorrectly selected. This raises an important question about the source of the observed instability. It may arise from the stochastic behaviour of the MCMC sampler, or from BART fitting coincidental patterns present in the particular dataset. These explanations have different implications for determining a threshold. If the instability is algorithmic, averaging VIPs across independent MCMC chains should reduce it. If it arises from data-level variation, repeated chains fitted to the same dataset will reproduce the same associations, and averaging cannot remove them. In this case, no universal fixed threshold can reliably distinguish genuine signals from noise. Following Bleich et al. (2014a), these sources of variation can be separated into two components:

1. **Data-Level Variability.** The flexibility of BART allows chance associations between noise predictors and the response to be represented as genuine signals. In high-dimensional data, some irrelevant predictors may exhibit relatively strong associations with the response purely by chance.
2. **Algorithmic Variability.** The MCMC sampler may not explore the posterior distribution uniformly. Different chains can therefore produce different estimates of variable importance, particularly if they explore different regions of the posterior.

To isolate which mechanism drives the instability, we use the nested variance decomposition of Bleich et al. (2014a). We generate 100 independent null datasets, $\mathcal{D}_1, \dots, \mathcal{D}_{100}$, each with $n = 250$ observations and $p = 40$ predictors. For each dataset we fit BART 50 times using independent random initialisations. Let v_{ijk} denote the VIP for predictor k on run j of dataset i , and define

$$\bar{v}_{i \cdot k} = \frac{1}{50} \sum_{j=1}^{50} v_{ijk}, \quad \bar{v}_{\cdot \cdot k} = \frac{1}{100} \sum_{i=1}^{100} \bar{v}_{i \cdot k}, \quad \bar{v}_{\dots} = \frac{1}{p} \sum_{k=1}^p \bar{v}_{\cdot \cdot k}.$$

These correspond respectively to the mean VIP for predictor k within dataset i , the mean VIP for predictor k across all datasets, and the global mean VIP across all predictors, runs, and datasets. We decompose the total variance into three components, given in Table 4.1.

Table 4.1: Nested Variance Decomposition of Variable Inclusion Proportions

Variance Component	Statistical Interpretation
$s_{ik}^2 = \frac{1}{50} \sum_{j=1}^{50} (v_{ijk} - \bar{v}_{i \cdot k})^2$	Algorithmic Instability. Variability of VIP for predictor k across runs within a fixed dataset i .
$s_k^2 = \frac{1}{100} \sum_{i=1}^{100} (\bar{v}_{i \cdot k} - \bar{v}_{\cdot \cdot k})^2$	Data Level Overfitting. Variability of predictor k 's mean VIP across different null datasets.
$s^2 = \frac{1}{p} \sum_{k=1}^p (\bar{v}_{\cdot \cdot k} - \bar{v}_{\dots})^2$	Global Variation. Variability between predictors overall, indicating a systematic preference for particular variables.

Case 1: If the deviation is caused by data-level overfitting, BART exploits the same noise correlations on every run of a given dataset, producing a low within-dataset variance s_{ik}^2 . A new dataset has a different noise profile, so the model overfits to a different set of predictors, producing a high across-dataset variance s_k^2 . Data-level overfitting therefore implies a low s_{ik}^2 and a high s_k^2 .

Case 2: If the deviation is caused by MCMC instability, repeated runs on the same dataset explore different posterior paths and settle in different local modes, producing a high within-dataset variance s_{ik}^2 . Since this instability is not tied to any particular noise structure, it does not systematically favour the same predictors across datasets, producing a low across-dataset variance s_k^2 . Algorithmic instability therefore implies a high s_{ik}^2 and a low s_k^2 .

Results and Implications

Figure 4.9 presents the results of the variance decomposition. Across the 40 predictors, the across-dataset variance dominates the within-dataset variance, with a ratio of $s_k^2/s_{ik}^2 = 3.51$. Repeated MCMC runs on a fixed dataset produce nearly identical VIP values, so s_{ik}^2 is small. Mean VIPs vary substantially from dataset to dataset, so s_k^2 is large.

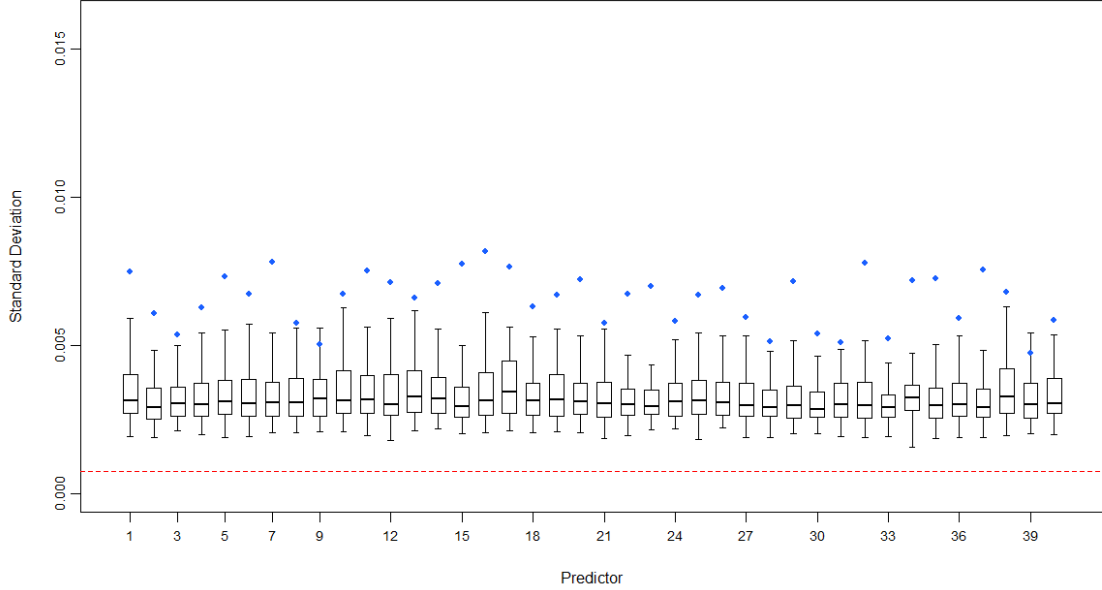


Figure 4.9: Nested variance decomposition of BART variable inclusion proportions across 100 null datasets ($n = 250$, $p = 40$, $m = 20$ trees). Boxplots show the within-dataset standard deviation s_{ik} across 50 independent runs. Circles show the across-dataset standard deviation s_k of the mean VIPs. The dashed line shows the overall standard deviation s after averaging over runs and datasets. The ratio $s_k^2/s_{ik}^2 = 3.51$ indicates that variability in VIP scores is driven primarily by dataset specific noise rather than MCMC instability.

Following the diagnostic logic above, a low s_{ik}^2 and a high s_k^2 indicate data-level overfitting. Averaging across MCMC runs would not remove this instability, since the inflated VIP values are not a consequence of poor chain mixing. Instead, the random variation within each dataset can make some predictors appear associated with the response purely by chance. BART’s flexibility allows these chance associations to be repeatedly identified, causing predictors that are important only by chance to receive persistently inflated VIP values. These results imply that a fixed threshold is not suitable, since a fixed cut-off can only be correct if the amount of overfitting is the same across datasets and predictors, and the decomposition shows the opposite: the inflation of VIP scores is idiosyncratic to each sample. What is needed instead

is a way to measure, for this specific dataset and this specific predictor, how large a VIP can become purely from BART overfitting to noise.

We now introduce a permutation-based procedure to construct this dataset-specific reference distribution. The basic idea is to generate datasets in which any genuine relationship between the predictors and response has been removed, while retaining the structure of the predictors. This is achieved by randomly permuting the response vector while holding the predictor matrix, $\mathbf{X}$, fixed. BART is then fit upon the resulting dataset, so any importance assigned to a predictor must arise from chance patterns rather than genuine signal. Repeating this process generates an empirical null distribution for each predictor, describing the VIP values that can arise through fitting noise in that particular dataset. The observed VIP can then be compared against this dataset-specific reference distribution rather than a universal fixed threshold.

4.4.3 Permutation-Based Thresholding for Variable Inclusion Proportions

We now introduce a permutation procedure to characterise the dataset-specific noise profile against which the observed VIPs can be assessed. A similar approach was proposed by Altmann et al. (2010) for Random Forest variable importance. Their procedure removes the relationship between the predictors and response by permuting the response, refits the model, and uses the resulting importance scores to characterise the level of importance generated by chance alone. Bleich et al. (2014a) adapt this idea to BART, using the resulting null distribution to determine whether an observed VIP exceeds the threshold that can be generated by fitting noise in the particular dataset.

The procedure begins by permuting the response vector $\mathbf{y}$ while holding the predictor matrix $\mathbf{X}$ fixed, producing S permuted datasets $\{\mathbf{X}, \mathbf{y}_{(1)}^*\}, \dots, \{\mathbf{X}, \mathbf{y}_{(S)}^*\}$. Permuting the response removes any genuine association between the predictors and the response while preserving the correlation structure of $\mathbf{X}$, including the sample-specific structure through which BART may overfit to noise. BART models are then fit on each permuted dataset, and the resulting VIP for predictor k on permutation s is denoted $v_{s,k}^*$. Since the predictor-response relationship has been removed, the collection of VIP scores across the S permutations defines the empirical null distribution for predictor k :

$$\mathbf{v}_k^* = (v_{1,k}^*, \dots, v_{S,k}^*). \quad (4.18)$$

This distribution describes the range of VIP values that predictor k can attain through fitting chance associations alone. The observed VIP, denoted $\bar{v}_k$, is obtained by averaging VIPs across n_{runs} independent BART fits. Figure 4.10 summarises the full procedure.

Given the observed VIP $\bar{v}_k$ and its corresponding empirical null distribution, variable selection reduces to determining whether the observed importance of x_k is unusually large relative to the values that can arise through fitting noise alone. This requires a decision rule specifying how the observed VIP should be compared with the null distribution. Bleich et al. (2014a) propose three such rules, which differ in how the null distributions are used and therefore in the degree of conservativeness of the resulting selection procedure.

Local Threshold. Predictor x_k is selected if $\bar{v}_k$ exceeds the $(1 - \alpha)$ quantile of its own null distribution $\{v_{1,k}^*, \dots, v_{S,k}^*\}$. Since each predictor is judged only against its own null distribution, no multiplicity correction is applied across predictors, making this the least conservative of the three rules. It is appropriate when a relatively large number of predictors are expected to be active.

Global Maximum Threshold. A single threshold is derived from the maximum inclusion proportion observed across all K predictors within each permutation, $v_{\max,s}^* := \max\{v_{s,1}^*, \dots, v_{s,K}^*\}$. Predictor x_k is selected only if $\bar{v}_k$ exceeds the $(1 - \alpha)$ quantile of $\{v_{\max,1}^*, \dots, v_{\max,S}^*\}$. Conditioning on the maximum across predictors makes this the most conservative rule, appropriate in highly sparse settings where Type I errors carry a high cost.

Global Standard Error Threshold. This rule combines the mean m_k and standard deviation s_k of each predictor's null distribution with a single multiplier C^* , chosen as the smallest value satisfying $(1 - \alpha)$ coverage simultaneously across all predictors,

$$C^* = \inf_{C \in \mathbb{R}^+} \left\{ \forall k, \frac{1}{S} \sum_{s=1}^S \mathbf{1}(v_{s,k}^* \leq m_k + C \cdot s_k) > 1 - \alpha \right\}.$$

Predictor x_k is selected if $\bar{v}_k > m_k + C^* s_k$. Because the threshold adapts to the shape of each predictor's null distribution while enforcing a common multiplier, this rule sits between the local and global maximum thresholds in conservativeness.

This permutation approach to variable selection requires no assumptions beyond those already embedded in the BART model. It is worth noting that this method comes at a computational cost, since BART must be refit on every permuted dataset. However since these fits are independent of one another, they can be easily paral-
 elised. A simple example demonstrating this permutation methodology is provided in Appendix C.5.

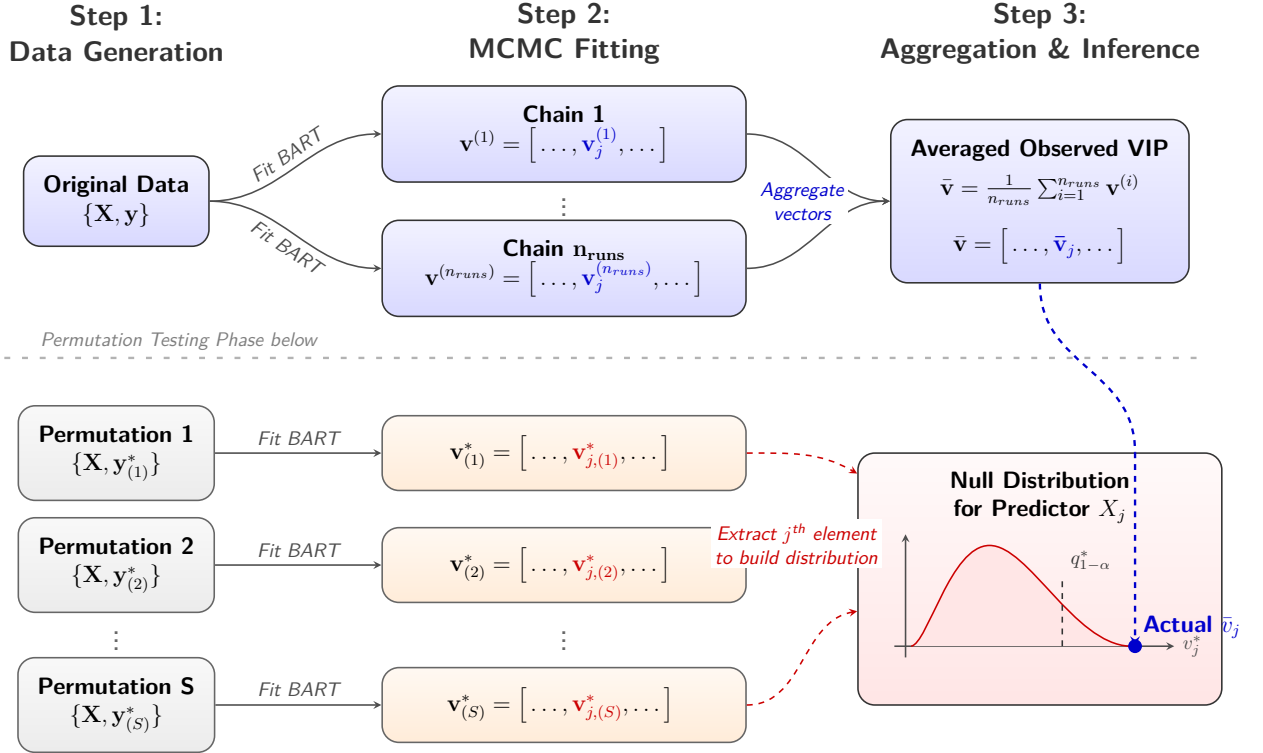


Figure 4.10: The permutation-based variable selection framework for a single predictor x_j . The observed VIP $\bar{v}_j$ is obtained by averaging over n_{runs} independent chains (top panel). The response vector is permuted S times and BART is refit on each permutation to build a predictor-specific null distribution (bottom panel). The observed VIP is then evaluated against this null distribution.

4.4.4 Extension to MI and Determining n_{runs}

Although the variance decomposition in Figure 4.9 showed that data-level variation is the dominant source of VIP instability, repeated BART runs on the same dataset still exhibit some algorithmic variation. We therefore estimate the observed importance of predictor x_j by averaging its VIP across n_{runs} independent BART fits. The remaining question is how large n_{runs} must be for this average to stabilise. We investigate this using the standard Friedman data-generating process

with $n = 250$ observations and five active predictors (see Appendix C.2). The experiment considers a 3×3 grid of dimensionalities, $p \in \{100, 500, 1000\}$, and noise levels, $\sigma \in \{1, 2, 5\}$. For each configuration, a single dataset is generated and BART is fitted independently 25 times. The cumulative average VIP is then recorded after each run.

Figure 4.11 shows that averaging substantially reduces the variation present in individual BART runs. Across all nine configurations, the cumulative VIPs have largely stabilised by $n_{\text{runs}} = 10$, with further runs producing little change in the estimated importance of either signal or noise predictors. We therefore use $n_{\text{runs}} = 10$ independent BART runs and $S = 50$ permutations throughout the subsequent variable selection experiments, consistent with the settings adopted by Bleich et al. (2014a) and Luo et al. (2021). Although this section has been presented in terms of the VIP metric, the same stability analysis and permutation-based thresholding procedure apply directly to MI. Both metrics are derived from the same BART posterior samples and therefore share the same sources of algorithmic variation and sample-specific overfitting. The only change required to adapt the thresholding methodology above to the MI metric is that rather than taking the average over n_{runs} BART models, Luo et al. (2021) suggest taking the median.

Stabilization of BART Variable Inclusion Proportions (VIP)

Cumulative average across 25 chains. The vertical dotted line at 10 runs indicates the heuristic threshold for sufficient algorithmic stabilization.

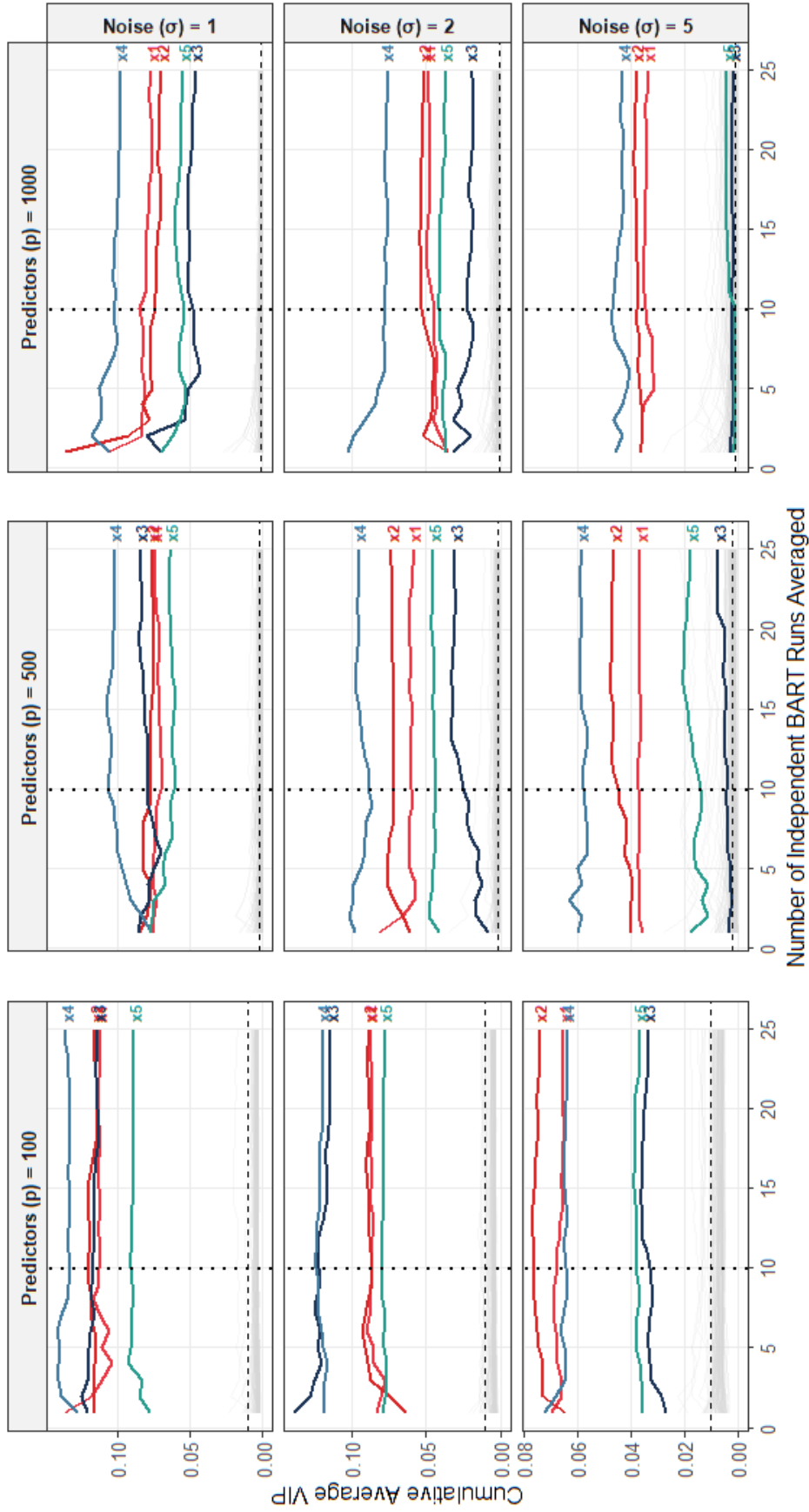


Figure 4.11: Stabilisation of cumulative average VIPs across 25 independent BART runs under varying dimensionality (p) and noise levels (σ). True signal variables are highlighted in colour and noise variables in grey. The dashed horizontal line denotes the theoretical null baseline $1/p$. The vertical dotted line marks $n_{\text{runs}} = 10$.

4.5 Sparsity-Inducing Priors

The preceding sections considered variable selection as a problem of interpreting a fitted BART model. The Variable Inclusion Proportion (VIP) and Metropolis Importance (MI) metrics summarise how frequently, and with what degree of posterior support, predictors are used across the sampled tree ensembles. Permutation-based thresholding then determines whether the observed importance of each predictor is sufficiently large relative to that expected from fitting noise. Under this approach, however, the standard BART model is left unchanged: the prior construction remains the same, and variable relevance is assessed only after the posterior has been sampled. Throughout this section, we use the term sparsity to mean that only a relatively small subset of the available predictors is needed to explain the response.

An alternative approach is to embed variable selection ability directly within the prior specification of the model. This follows the same general principle as regularisation methods such as the LASSO. The LASSO modifies the least-squares objective by adding an L_1 penalty on the regression coefficients, so that increasing the complexity of the fitted model incurs a direct cost. The resulting optimisation therefore balances improvement in model fit against the inclusion of additional non-zero coefficients, with sufficiently weak predictors shrunk exactly to zero during estimation. Variable selection is thus a consequence of the estimation procedure itself rather than a separate decision applied afterwards.

Sparsity-inducing priors introduce an analogous mechanism within the Bayesian construction of BART. Rather than penalising coefficient magnitudes, the prior is specified to favour tree ensembles that concentrate their splitting activity on a relatively small subset of predictors. The posterior sampling procedure must therefore balance improvements in model fit against this prior preference for sparse predictor usage. The remainder of this section examines two specific implementations of this approach. The first approach is ABC Bayesian Forests of Liu et al. (2021). This framework applies an **external** spike-and-slab prior to the global predictor set. It explicitly models the active or inactive status of each candidate variable before the tree structures are proposed. The second approach is Dirichlet Additive Regression Trees (DART) of Linero (2018). This framework embeds an **internal** sparsity-inducing Dirichlet prior over the splitting probabilities. It encourages the algorithm to concentrate all spatial partitions on a restricted subset of predictors. Both methods redesign the prior architecture of BART to execute variable selection natively.

4.5.1 Variable selection with ABC Bayesian forests

The permutation framework seen so far evaluates variable selection as a post-hoc diagnostic applied to an unconstrained BART model. An alternative methodology incorporates sparsity directly into the prior distribution. This structural change ensures that variable selection is executed natively as part of the generative model. Ročková and Stéphanie van der Pas (2020) formalise this approach by introducing the **spike-and-forest** prior, with Liu et al. (2021) later extending the methodology to perform variable selection. Their formulation augments the standard BART architecture by imposing a spike-and-slab prior over the space of candidate predictor subsets. Rather than allowing every predictor to be available for splitting throughout the ensemble, the method first samples an active subset of predictors,

$$S \sim \pi(S), \quad S \subseteq \{1, \dots, p\}, \quad (4.19)$$

and the entire BART ensemble is then conditioned exclusively on this subset,

$$\{T_j, M_j\}_{j=1}^m, \sigma^2 \mid S \sim \text{BART prior}. \quad (4.20)$$

In practice, the prior $\pi(S)$ is constructed using a beta-binomial spike-and-slab formulation. A global inclusion probability θ is drawn from a $\text{Beta}(a, b)$ hyperprior, and each predictor is assigned an independent indicator $\gamma_j \mid \theta \sim \text{Bernoulli}(\theta)$. These indicators determine the active subset $S = \{j : \gamma_j = 1\}$. A predictor with $\gamma_j = 0$ belongs to the *spike* and is excluded from S , making it unavailable for splitting in every tree. A predictor with $\gamma_j = 1$ belongs to the *slab* and is available to the BART prior for tree construction. The spike-and-slab mechanism therefore acts externally by determining which predictors are available to the trees, rather than shrinking individual tree parameters.

Formal Bayesian inference over the active set S requires evaluating the marginal likelihood $\pi(Y \mid S)$ which is analytically intractable because it involves integrating over the large combinatorial space of all possible tree structures (see Liu et al., 2021). Consequently, exact posterior probabilities cannot be computed, and standard BART MCMC methods are unavailable. To resolve this mathematical limitation, Liu et al. (2021) apply an Approximate Bayesian Computation (ABC) methodology. Rather than attempting to evaluate exact likelihoods, the ABC framework evaluates candidate models empirically. The algorithm first proposes a candidate subset of predictors. It then constructs a lightweight, restricted BART model utilising exclusively that specific subset. This restricted model is subsequently eval-

uated on a held-out testing dataset. If the model generates sufficiently accurate predictions, the candidate subset is accepted. By repeating this procedure across thousands of proposals, the algorithm compiles a collection of successful candidate subsets. Variable selection is then achieved by analysing the predictor composition of these accepted models. This empirical filtering process learns the structural makeup of highly predictive models, entirely bypassing the need to directly calculate the intractable posterior distribution.

The ABC Bayesian Forest Algorithm

The algorithm proceeds as follows, for $j = 1, \dots, N_{\text{ABC}}$.

1. **Split the data.** At iteration j , randomly draw a subsample $\mathcal{I}_j \subset \{1, \dots, n\}$ of size s without replacement. The observations indexed by $\mathcal{I}_j$ form the training set $Y_{\mathcal{I}_j}^{\text{obs}}$, while the complementary observations $Y_{\mathcal{I}_j^c}^{\text{obs}}$ are reserved for ABC rejection.
2. **Propose a subset.** Draw a candidate active set $S_j \sim \pi(S)$ from the spike-and-slab prior. Empty draws are discarded and resampled, so that every candidate model contains at least one predictor.
3. **Fit a restricted BART model.** Fit BART using only the predictors in S_j and only the training observations $Y_{\mathcal{I}_j}^{\text{obs}}$. A full MCMC run at every one of the N_{ABC} iterations is computationally prohibitive, so Li (2020) instead suggest using a small number of trees $m \in \{10, 20\}$ and retaining a single posterior draw after a short burn-in of around 200 iterations.
4. **Generate pseudo-data on the held-out set.** Using the fitted model $\hat{f}^{(j)}$, predict the held-out responses and add noise sampled from the fitted posterior error variance,

$$y_i^* = \hat{f}^{(j)}(x_i) + \varepsilon_i, \quad \varepsilon_i \sim N(0, \sigma_j^2), \quad i \in \mathcal{I}_j^c.$$

5. **Compute the discrepancy.** The discrepancy between the pseudo-data and the observed held-out responses is the Euclidean L_2 distance,

$$\epsilon_j = \left\| Y_{\mathcal{I}_j^c}^* - Y_{\mathcal{I}_j^c}^{\text{obs}} \right\|_2.$$

6. **Repeat and accept.** Steps 1 to 5 are repeated across N_{ABC} independently sampled subsets, producing a reference table $\{(S_j, \hat{f}^{(j)}, \epsilon_j)\}_{j=1}^{N_{\text{ABC}}}$. Candidate

models are ranked by ϵ_j , and only the best-performing fraction with $\epsilon_j < \epsilon$ are retained as the set of ABC-accepted models.

Marginal Posterior Variable Inclusion Probabilities

The algorithm above returns a reference table $\{(S_j, f^{(j)}, \epsilon_j)\}_{j=1}^{N_{\text{ABC}}}$, recording the candidate subset, fitted model, and discrepancy for every ABC iteration. Variable selection proceeds by filtering and summarising this table. First, a tolerance ϵ is fixed and the table is filtered down to the accepted models, those with $\epsilon_j < \epsilon$. Each accepted fit $f^{(j)}$ was built using only the predictors in its proposed subset S_j , but does not necessarily use every predictor in S_j when constructing its trees, since some proposed predictors may go unused in the resulting splits. Write S_j^{actual} for the predictors actually used by the accepted fit $f^{(j)}$.

For each predictor x_k , the **marginal posterior variable inclusion probability (MPVIP)** is the proportion of accepted models in which x_k appears in S_j^{actual} , that is, the fraction of accepted fits that genuinely split on x_k at least once,

$$\pi_k(\epsilon) = \frac{1}{M(\epsilon)} \sum_{j: \epsilon_j < \epsilon} \mathbb{I}\{x_k \in S_j^{\text{actual}}\}, \quad (4.21)$$

where $M(\epsilon)$ is the number of accepted models at tolerance ϵ . This is computed separately for every predictor, using the same filtered set of accepted models, giving one MPVIP per predictor from a single pass over the table. Once each predictor has an MPVIP, variable selection reduces to a threshold rule: predictor x_k is selected if $\pi_k(\epsilon)$ exceeds a pre-specified cutoff, commonly 0.5, corresponding to the median probability model (Barbieri et al., 2004), or a more liberal 0.25 when sensitivity to weak signal is preferred over precision.

Hyperparameter Configuration

The choice of tolerance ϵ determines how closely the pseudo-data must agree with the observed data. Smaller values impose a more stringent acceptance criterion, so fewer ABC samples are retained. Liu et al. (2021) recommend choosing ϵ as an empirical quantile of the observed discrepancies. In their simulations, the closest 10% of candidate models are retained rather than fixing a universal numerical threshold. Furthermore, the ABC framework uses a small ensemble, with the forest restricted to 10 or 20 trees. This keeps the repeated BART fits computationally feasible while limiting the number of available splitting opportunities and increasing competition among candidate predictors, as seen for the VIP metric. Since a full BART MCMC run for every ABC proposal would be computationally prohibitive, each of the N_{ABC}

BART fits is run for only 200 burn-in Gibbs iterations, after which the next posterior draw is retained. This short burn-in is intended to capture the main variables and tree structure early in the chain while keeping the cost of repeatedly fitting BART manageable. Finally, the beta-binomial spike-and-slab prior controls the size of the proposed active subset. The hyperparameters of the Beta distribution determine the prior distribution of the subset size, with the standard implementation using the uninformative hyperprior $\theta \sim \text{Beta}(1, 1)$.

4.5.2 Motivating a Sparsity-Adaptive Approach

ABC Bayesian Forests impose sparsity through an external subset-selection procedure. Candidate active sets are drawn from the spike-and-forest prior and restricted BART models are fitted independently to each proposed subset. Information obtained from one candidate subset does not directly influence the proposal of the next. The method must therefore explore the model space through a large collection of independently generated restricted BART fits, which creates a substantial computational burden. This motivates a different approach in which sparsity is incorporated directly into the BART sampling mechanism. Rather than repeatedly proposing a discrete subset of available predictors before fitting the model, the splitting probabilities themselves can be treated as unknown parameters and learned jointly with the tree ensemble. Predictors are then not made strictly available or unavailable before the trees are constructed. Instead, the posterior distribution can progressively concentrate splitting probability on a smaller subset of predictors supported by the data.

Standard BART does not possess such a mechanism. Under its default prior, each of the predictors is assigned equal probability of being proposed as a splitting variable. These probabilities remain fixed throughout the MCMC algorithm, irrespective of the evidence accumulated for individual predictors. A sparsity-adaptive formulation instead allows the splitting distribution to be learned as part of the model. As the posterior concentrates, splitting probability can become concentrated on a small number of predictors, reducing the probability of repeatedly partitioning on irrelevant variables without imposing a fixed active subset in advance.

This is the approach adopted by Dirichlet Additive Regression Trees (DART) Linero (2018). DART replaces the fixed uniform distribution over splitting variables with a vector of splitting probabilities assigned a Dirichlet prior. The splitting probabilities are subsequently updated during posterior sampling, allowing the ensemble to concentrate its partitions on predictors receiving substantial posterior support.

In this way, DART replaces the hard inclusion or exclusion mechanism of ABC Bayesian Forests with an adaptive distribution over the candidate predictors.

4.5.3 Variable Selection with Dirichlet Additive Regression Trees (DART)

Standard implementations of Bayesian Additive Regression Trees assign a uniform probability to all candidate predictors. Let s_j denote the probability of selecting predictor x_j to construct a splitting rule at any internal node. The default BART architecture fixes this probability at $s_j = 1/p$ across all candidate variables (Chipman et al., 2010), where p denotes the number of available covariates in the data. Linero (2018) show that the uniform prior imposes a preference for highly non-sparse ensembles, forcing the predictive signal to be distributed across many variables. This guarantees the recurrent inclusion of noise predictors, leading to overfitting and corruption of inclusion frequency based variable selection metrics.

Linero (2018) resolve this limitation by abandoning the fixed uniform prior. Instead, the splitting probabilities are modelled as a latent, continuous vector $s = (s_1, \dots, s_p)$ that is estimated sequentially from the data. This framework effectively maps the discrete combinatorial problem of variable selection onto a continuous probability simplex, allowing sparsity to emerge natively during the MCMC tree-growing process.

The Sparsity-Inducing Dirichlet Prior

To induce sparsity natively within the tree-growing process, DART places a symmetric Dirichlet prior over the splitting probability vector:

$$s \sim \mathcal{D}\left(\frac{\alpha}{p}, \dots, \frac{\alpha}{p}\right), \quad (4.22)$$

where p is the total number of available candidate predictors and α is a positive hyperparameter. In this formulation, the ratio α/p acts as the concentration parameter of the Dirichlet distribution. The mechanics of this concentration parameter dictate how the algorithm achieves sparsity in practice. When $\alpha/p \geq 1$, the probability mass is concentrated toward the center of the p -dimensional simplex, encouraging models where all variables share roughly equal selection probabilities. However, in high-dimensional settings where p is large, the concentration parameter α/p becomes strictly less than one. This forces the density of the Dirichlet distribution to concentrate heavily at the vertices and edges of the simplex. Figure 4.12

displays this concentration of mass in the vertices as α/p changes. A draw from this sparse prior results in a highly skewed probability vector s . A very small subset of predictors is assigned a substantial probability of being selected for a split, while the probabilities for the remaining predictors are suppressed to values close to zero. This creates a natural probabilistic competition: because the vector s must sum exactly to one, increasing the selection probability of important covariates forces the probabilities of irrelevant covariates toward zero.

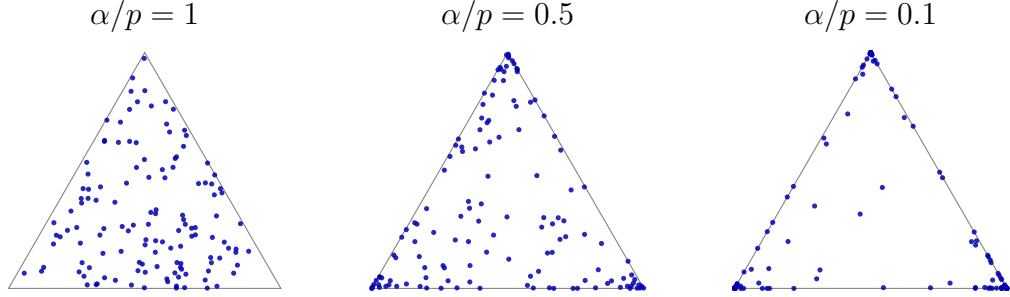


Figure 4.12: Simulated draws from a $\mathcal{D}(\alpha/p, \alpha/p, \alpha/p)$ prior on the 3-dimensional simplex. When the concentration parameter α/P is equal to 1, probability mass is distributed uniformly. As α/p decreases below 1, the mass is driven strictly toward the edges and vertices, representing sparse probability vectors.

This induces a softer form of sparsity than the spike-and-slab approach used by ABC Bayesian Forests. In DART, predictors are not explicitly removed from the model space. A variable with $s_j \approx 0$ has a very small probability of being selected for a split, but it remains available to the tree-growing algorithm. Its splitting probability can therefore increase if later exploration of the posterior space supports its inclusion.

Conjugate Updating and Sequential Learning

The continuous probability vector s enables the model to learn the structural importance of variables sequentially across successive iterations of the MCMC algorithm. Following Linero (2018), the Dirichlet prior is conjugate to the multinomial distribution of variable usage counts and therefore the vector s admits a closed-form Gibbs update at the end of each MCMC cycle. Conditioned on the current tree structures across the ensemble, the posterior update for the splitting probabilities is given by:

$$s \sim \mathcal{D}\left(\frac{\alpha}{P} + m_1, \dots, \frac{\alpha}{P} + m_P\right), \quad (4.23)$$

where m_j represents the total number of splits on predictor x_j across all internal nodes in the current ensemble.

This formulation establishes a powerful sequential feedback loop. During a given MCMC iteration, predictors that successfully partition the data and improve model fit are selected more frequently, generating a higher observed count m_j . When s is subsequently updated at the end of the cycle, these elevated counts subsequently increase the corresponding selection probabilities s_j for the next iteration. Consequently, as the chain progresses, the algorithm iteratively concentrates its splitting behaviour strictly onto the active covariates, learning the sparsity structure directly from the data.

Adaptive Sparsity

The degree to which the Dirichlet prior concentrates its mass on a subset of predictors is governed by the hyperparameter α . Rather than fixing α deterministically, Linero (2018) place a hyperprior on this parameter, allowing the algorithm to adaptively learn the true underlying sparsity directly from the data. This is implemented via a Beta distribution on a transformed scale:

$$\frac{\alpha}{\alpha + \rho} \sim \text{Beta}(a, b). \quad (4.24)$$

By setting $a = 0.5$ and $b = 1$, the resulting prior on α exhibits heavy, Cauchy-like tails with a median of ρ . Setting the scaling parameter $\rho = P$ conservatively aligns the baseline expected sparsity with the total number of available predictors. This prevents overly aggressive variable pruning unless the empirical evidence strongly dictates a sparse structure. If the underlying signal is genuinely sparse, the posterior for α concentrates near zero, forcefully activating the Dirichlet penalty. Conversely, if the true regression function uses a large proportion of the available covariates, the heavy-tailed prior allows α to diverge, effectively recovering the uniform splitting behaviour of standard BART. As a result, Linero (2018) show that due to DART’s ability to adapt to the sparsity level of the data, it can be used as an alternative to BART for regression in both sparse and non-sparse settings.

Variable Selection Inference

Within the DART framework, the latent splitting probabilities s_j themselves can be extracted and interpreted as continuous measures of variable importance. Because s_j are informed by the frequency with which a predictor is selected across the ensemble, examining the posterior mean of these splitting parameters provides a direct assessment of covariate importance.

For variable selection, Linero (2018) instead suggest using the Marginal Posterior Variable Inclusion Probability (MPVIP) seen in Section 4.5.1. This metric calculates the posterior probability that a given predictor appears in at least one splitting rule anywhere within the ensemble. Variable selection is subsequently performed by using the median probability model, formalised by Barbieri et al. (2004). In particular, any predictor with a MPVIP exceeding 0.5 is retained in the final model. The DART authors show that this selection strategy provides competitive variable selection performance whilst entirely avoiding the computational overhead required by permutation-based resampling methodologies seen previously in this chapter.

4.6 Extension of BART Variable Selection Methods to AddiVortes

This section develops a framework for variable selection using AddiVortes by extending the variable selection methods reviewed in this chapter to instead use the AddiVortes engine in replacement of BART. The overarching Bayesian ensemble architecture, including the MH-within-Gibbs sampler, is retained entirely. The conceptual shift consists solely of replacing the internal base learner from a decision tree to a Voronoi tessellation. Due to this structural alignment, the majority of standard BART variable selection techniques can be directly adapted to utilise the AddiVortes engine without requiring new theoretical derivations, necessitating only slight adjustments in interpretation. We establish this framework here and subsequently present a simulation study to evaluate whether the geometric flexibility of Voronoi tessellations improves upon the variable selection accuracy of BART methodologies.

AddiVortes Variable Inclusion Proportions (VIP)

Under the AddiVortes architecture, the Variable Inclusion Proportion (VIP) requires no modification, although its interpretation differs. As seen previously, in a standard decision tree a single variable can be selected repeatedly across multiple branches. Let D_l denote the set of active dimensions in the l^{th} tessellation. In AddiVortes, the inclusion of variables is strictly defined by the membership of the set $j \in D_l$. A predictor is either active in a tessellation or is not; it can be selected at most once per learner.

Since a variable can now appear in a tessellation at most once, the components of the BART VIP metric established in Section 4.2 must be reinterpreted. It is still justified to assume that frequent use within the ensemble suggests some degree of variable importance, however these usage frequencies are now more confined. Since a variable can appear at most once in a tessellation, c_{kj} becomes the count of distinct tessellations in the ensemble that have activated predictor j , strictly bounded by the ensemble size m . The denominator becomes the total number of active dimension slots across the entire ensemble. For a posterior sample k , the VIP is derived as:

$$c_{kj} = \sum_{l=1}^m \mathbb{I}(j \in D_l^{(k)}) \quad c_{k\cdot} = \sum_{l=1}^m d_l^{(k)} \quad \text{VIP}_j^{(k)} = \frac{c_{kj}}{c_{k\cdot}}$$

where $d_l^{(k)}$ is the size of the active dimension set $D_l^{(k)}$. Because a predictor can enter D_l only once, it cannot artificially inflate its own inclusion count by being split upon multiple times within the same learner. Consequently, we theorise that this

architectural shift may help to mitigate the cardinality bias inherent to standard BART VIP. For identical reasons as BART established in Section 4.2, AddiVortes VIP performs optimally for variable selection when the ensemble size is reduced. We suggest the default use of $m = 20$ to aid comparison with competing BART methods. Beyond the ensemble size, no alterations to the standard hyperparameters are required.

AddiVortes Metropolis Importance (MI)

As discussed in Section 4.3, Metropolis Importance (MI) extends the VIP metric by weighting each inclusion according to the MH acceptance probability associated with the proposal that introduced it. Within BART, this weighting additionally penalises inclusions occurring at greater tree depths, since deeper structural proposals are typically subject to stronger regularisation. This interpretation is specific to the tree-based representation. Voronoi tessellations possess no hierarchical notion of depth, and within AddiVortes the MH acceptance probability is instead interpreted more generally as the degree of posterior support for a proposed structural modification relative to its additional complexity.

In AddiVortes, this interpretation arises naturally through the Add Dimension (AD) move. When a proposed move includes a predictor within the active dimension set of a tessellation, its MH acceptance probability reflects the improvement in model fit obtained from including that predictor relative to the accompanying complexity penalty imposed by the prior. A high acceptance probability therefore indicates that the proposed inclusion is strongly favoured, whereas a low probability suggests that the additional predictor provides comparatively little support for the proposed tessellation.

Consequently, the AddiVortes analogue of MI rewards predictors that are not only frequently activated across the ensemble, but whose activations are consistently supported by favourable MH evaluations. We hypothesise that MI may be particularly beneficial for AddiVortes, where variable inclusions are inherently less frequent because a predictor can appear at most once within each tessellation. The resulting VIP distribution may therefore be comparatively flatter, making it more difficult to distinguish genuinely important predictors using inclusion frequency alone. By incorporating the MH acceptance probability, MI provides additional information about the strength of each inclusion and may help separate the most important predictors from variables that are selected with only weak support.

For posterior sample k , let $\pi_{\text{AD},klj}$ denote the acceptance probability associated with the Add Dimension move that activates predictor j in tessellation l . The corresponding MI metric is defined by

$$c_{kj} = \sum_{l=1}^m \mathbb{I}(j \in D_l^{(k)}) \quad c_{k\cdot} = \sum_{l=1}^m d_l^{(k)} \quad \text{MI}_j^{(k)} = \frac{\sum_{l=1}^m \pi_{\text{AD},klj} \mathbb{I}(j \in D_l^{(k)})}{c_{k\cdot}}$$

MI therefore contextualises the frequency-based information provided by VIP. Two predictors may be activated equally often, while receiving substantially different MI scores if the proposed inclusions of one predictor are consistently more strongly supported by the MH evaluations. In this sense, MI measures not only how frequently the model includes a predictor, but also the extent to which the sampler favours its inclusion. As with VIP, extracting optimal variable selection performance using MI requires reducing the ensemble size m .

Permutation-Based Thresholding

The permutation-based thresholding procedure introduced in Section 4.4.3 transfers directly to AddiVortes. Since the wrapper operates solely on the outputted variable importance scores and requires only that the underlying model can be refitted to data with a permuted response, it remains agnostic to the internal structure of the base learner. No modification to the thresholding procedure is therefore required.

For the empirical implementation, we adopt the same settings established for BART by Bleich et al. (2014a) and Luo et al. (2021), using 10 independent runs to estimate the observed importance scores and 50 permuted-response refits to construct the empirical null distribution. Variable selection is then performed using the three threshold rules introduced previously: the global maximum threshold, the local threshold, and the maximum local threshold. These settings are adopted unchanged for AddiVortes both because the thresholding procedure is independent of the underlying learner and to facilitate direct comparison with the corresponding BART variable selection results.

ABC Bayesian Forests Extended to AddiVortes

The ABC variable selection framework introduced in Section 4.5.1 transfers directly to AddiVortes. The procedure operates externally to the underlying learner, proposing candidate predictor subsets and evaluating the resulting restricted models according to their predictive performance on held-out data. Consequently, replacing

the restricted BART fits with AddiVortes requires no further theoretical modification to the ABC procedure.

For each proposed subset, a lightweight AddiVortes model is fitted using only the available predictors before its predictive discrepancy is evaluated on held-out observations. Following the computational strategy used for ABC Bayesian forests by Liu et al. (2021), these restricted fits employ a small ensemble of either $m = 10$ or $m = 20$ tessellations and are run for approximately 200 MCMC iterations. Candidate subsets are then retained according to the ABC tolerance described in Section 4.5.1, and variable importance is summarised through the marginal posterior variable inclusion probabilities of the accepted models. As in the original ABC Bayesian forest framework, we evaluate variable selection using MPVIP thresholds of 0.25 and 0.5, following Luo et al. (2021).

Dirichlet Additive Voronoi Tessellations (DAVT)

The standard AddiVortes framework assigns equal prior probability to every available predictor when constructing the active dimension set of a tessellation. Let

$$s = (s_1, \dots, s_p)$$

denote the vector of variable selection probabilities, where s_j represents the probability associated with selecting predictor x_j . Under the default AddiVortes specification, these probabilities are fixed uniformly as

$$s_j = \frac{1}{p}, \quad j = 1, \dots, p, \quad (4.25)$$

where p denotes the total number of available predictors. Unlike this uniform specification, the sparsity adaptive Dirichlet prior proposed by Linero (2018) allows the variable selection probabilities to adapt to evidence accumulated during MCMC sampling. We refer to the resulting extension to AddiVortes as Dirichlet Additive Voronoi Tessellations (DAVT), with

$$s \sim \mathcal{D}\left(\frac{\alpha}{p}, \dots, \frac{\alpha}{p}\right), \quad (4.26)$$

where α controls the concentration of the prior and therefore the degree of sparsity encouraged across the predictor space. As discussed in Section 4.5.3, conditional on the structural usage counts $m_1, \dots, m_p$, the posterior distribution takes the form

$$s \mid T_1, \dots, T_m \sim \mathcal{D} \left(\frac{\alpha}{p} + m_1, \dots, \frac{\alpha}{p} + m_p \right). \quad (4.27)$$

The interpretation of the counts m_j , however, differs between DART and DAVT. In DART, m_j denotes the total number of splitting rules involving predictor x_j across the ensemble. Since a predictor may be selected multiple times within a single tree, its count reflects both how broadly it is used across the forest and how frequently it is reused within individual learners. In AddiVortes, variable inclusion is instead determined by membership of the active dimension set D_l . A predictor can therefore appear at most once within any individual tessellation. Consequently, the count m_j is redefined as the number of distinct tessellations in which predictor x_j is active:

$$m_j = \sum_{l=1}^m \mathbb{I}(j \in D_l), \quad 0 \leq m_j \leq m. \quad (4.28)$$

DAVT therefore retains the same conjugate Dirichlet updating mechanism as DART, but the structural evidence contributing to each Dirichlet parameter is defined differently. Rather than accumulating repeated splitting events, the posterior probability assigned to a predictor increases according to the number of distinct tessellations in which it is activated. All remaining aspects of the Dirichlet sparsity framework are retained from DART. In particular, the adaptive updating of the concentration parameter α are unchanged and follows the procedure described in Section 4.5.3. The DAVT extension therefore modifies only the structural counts entering the Dirichlet update, while preserving the remaining prior specification and adaptive sparsity mechanism. Variable selection can subsequently be performed using the MPVIP as suggested in Linero (2018) and detailed in Section 4.5.3 without any further modifications.

4.7 Simulation Study

4.7.1 Introduction and Primary Objectives

This simulation study provides an empirical evaluation of variable selection accuracy across the BART and AddiVortes techniques developed in this chapter. The core objective is to determine whether substituting the tree-based engine of BART with the spatial Voronoi tessellations of the AddiVortes engine improves upon the variable selection capabilities of established BART variable selection methods. Luo et al. (2021) provides a comprehensive overview of the BART variable selection literature alongside a simulation methodology. We replicate their methodology to ensure a direct comparison with the AddiVortes extension. We theorise that, due to the improved flexibility of tessellation partitioning, this simple switching of engine will provide improvement in selection accuracy across settings where BART’s restriction to axis-orthogonal splits limits the ability to capture complex signals. The objective of this study is strictly limited to evaluating variable selection accuracy. We explicitly omit results of additional inferences such as regression predictive performance or computational efficiency.

4.7.2 Data Generation and Ground Truth

To thoroughly test variable selection capabilities, we simulate N independent datasets across two distinct data generating processes:

Configuration C.C.1: Independent Uniform Predictors (Friedman)

The first configuration is a high dimensional adaptation of the standard Friedman (1991) nonlinear equation. For an overview of this function refer to Appendix C.2. For each dataset we generate $n = 250$ observations with dimensions scaling across $p \in \{500, 1000\}$. The predictors $x_1, \dots, x_p$ are sampled independently from a standard Uniform distribution $\mathcal{U}(0, 1)$. The continuous response y is drawn from a Normal distribution $\mathcal{N}(f_0(\mathbf{x}), \sigma^2)$ where the true functional form is defined as

$$f_0(\mathbf{x}) = 10 \sin(\pi x_1 x_2) + 20(x_3 - 0.5)^2 + 10x_4 + 5x_5. \quad (4.29)$$

The noise variance scales across $\sigma^2 \in \{1.0, 5.0\}$. In this design only the first five predictors $x_1, \dots, x_5$ possess true functional influence over the response while the remaining $p - 5$ components act as independent stochastic noise. This configuration establishes a baseline for assessing a model’s ability to recover the active predictors across varying settings of p and σ^2 .

Configuration C.C.2: Correlated Predictors with Pure Interactions

Following the correlated interaction setting of Zhu et al. (2015) this configuration assesses susceptibility to correlation bias when irrelevant predictors are correlated with variables that contribute to the true response. For each dataset we generate $n = 500$ observations with dimensions scaling across $p \in \{100, 200\}$. The predictors $x_1, \dots, x_p$ are sampled from a multivariate Normal distribution $\mathcal{N}_p(0, \Sigma)$ where Σ follows an autoregressive correlation structure

$$\Sigma_{jk} = 0.3^{|j-k|} \quad j, k = 1, \dots, p. \quad (4.30)$$

The continuous response y is drawn from a Normal distribution $\mathcal{N}(f_0(\mathbf{x}), \sigma^2)$ where the functional form is defined as

$$f_0(\mathbf{x}) = 2x_1x_4 + 2x_7x_{10}. \quad (4.31)$$

The noise standard deviation is fixed at $\sigma = 1$. Consequently the true active predictors are x_1, x_4, x_7 , and x_{10} . Each contributes to the response exclusively through a pairwise interaction. The absence of marginal main effects makes this configuration particularly challenging for variable selection while the correlation structure introduces the possibility of falsely selecting irrelevant predictors correlated with the true active predictors.

Evaluation Metrics

To evaluate variable selection performance, we compute True Positives (TP), False Positives (FP), and False Negatives (FN) to derive three primary metrics:

- **Precision:** The proportion of selected variables that are truly active, defined as $\frac{TP}{TP+FP}$.
- **Recall:** The proportion of truly active variables successfully identified by the model, defined as $\frac{TP}{TP+FN}$.
- **F_1 Score:** The harmonic mean of precision and recall, calculated as $2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$.

A highly effective model will demonstrate high recall by capturing all true underlying signals, alongside strict precision by aggressively filtering out irrelevant noise. The F_1 score serves as the primary benchmark for balanced selection.

4.7.3 Model Configurations and Selection Criteria

We simulate $N = 50$ independent datasets for each experimental configuration (C.C.1 and C.C.2). For every generated dataset, we fit the comprehensive suite of variable selection algorithms introduced in this chapter. Each algorithm includes both a standard BART and an AddiVortes counterpart. To facilitate a direct and equitable comparison, we ensure hyperparameters are as similar as possible across these pairs, deviating only in engine-exclusive parameters. Full hyperparameters specification is detailed in Section 4.7.3. The fitting procedure and the mechanism for extracting the predicted set of active predictors operates as follows for each framework:

- **Permutation Methods (VIP and MI):** For both the AddiVortes and BART engines, the model is fitted 10 times on the observed data to establish the observed VIP and MI scores. The model is then fit 50 times on data with a randomly permuted response vector to construct a variable-specific null distribution of splits. A predictor enters the predicted active set only if its observed score strictly exceeds the predefined Local, Global, or Max threshold derived from this null distribution.
- **Spike-and-Slab (ABC):** We evaluate both engines using the ABC approach at two distinct ensemble sizes ($m = 10$ and $m = 20$). Variables are formally selected into the active set if their MPVIPs meet or exceed the threshold levels of 0.25 and 0.50.
- **Dirichlet Prior (DART and DAVT):** We employ a sparsity-inducing Dirichlet prior on both engines, setting the ensemble size to $m \in \{20, 200\}$. The models are executed 10 times, and the resulting MPVIPs for each variable are averaged. Final variable selection is executed via the Median Probability Model (MPM), isolating any variable with a mean MPVIP ≥ 0.50 .
- **Baseline Models** We incorporate two standard baselines for comparison. First, a Random Forest ensemble of 500 trees serves as a comparative baseline for the AddiVortes and BART permutation methods. The model is fitted 10 times on the observed data to establish a mean out of bag permutation importance score for each predictor. The model is then fitted 50 times on data with a randomly permuted response vector to construct a null distribution. Predictors are selected if their observed importance strictly exceeds the 95th percentile of their specific empirical null distribution. Second, a 10-fold cross

validated Lasso isolates variables that retain non-zero coefficients under the one-standard-error penalty rule (see Hastie et al. (2009)).

Framework	Engines	Hyperparameters	Metric	Threshold
Permutation	BART, AddiVortes	$m = 20, 1000$ burn in, 1000 posterior, 10 observed runs, 50 permutation runs	VIP and MI scores	Local, Max, Global
ABC Spike and Slab	BART, AddiVortes	$m \in \{10, 20\}$, 200 burn in, 1 posterior draw	MPVIP	$\epsilon \in \{0.25, 0.50\}$
Dirichlet Prior	BART, AddiVortes	$m \in \{20, 200\}$, 1000 burn in, 1000 posterior, 10 runs averaged	MPVIP	≥ 0.50 (MPM)
Random Forest	Standard Trees	500 trees, 10 observed runs, 50 permutation runs	Increase in out of bag MSE	Local
LASSO	Linear Model	10 fold cross validation	Non zero coefficients	—

Table 4.2: Standardised model configurations and selection rules grouped by methodological framework.

Immediately after a model finalises its variable selection for a given dataset, the predicted active set is evaluated against the ground truth active predictors defined by the data generating equation. We record the True Positives, False Positives, and False Negatives to calculate the Precision, Recall, Miss Rate, and F_1 score for that specific run. This sequence (data generation, model fitting, threshold application, and metric extraction) is executed sequentially across all 50 datasets. The iteration scores are then aggregated to report the mean performance and standard deviation for each method within the specific experimental setting.

4.7.4 Analysis of Results: Friedman (C.C.1):

We now present the results of this study on the C.C.1 data configuration across settings $p \in \{500, 1000\}$ and $\sigma^2 \in \{1, 5\}$. To highlight the effects of increasing dimensionality and noise, we detail the transition from the easiest setting to the hardest setting and present the remaining cases in Appendix C alongside full precision-recall breakdowns.

VIP and MI (Permutation)

When evaluating the VIP metric, AddiVortes maintains only a slight edge over BART, with both exhibiting a shared sensitivity to increased dimensionality. Across both engines, the MI metric consistently outperforms VIP. This is expected since the added consideration of the MH acceptance ratios naturally carries more information regarding variable importance than just raw inclusion frequencies alone. In addition, the MI metric proves to be robust to the increase in dimensionality and noise, with MI-Global and MI-Local changing only marginally between the two data settings presented, whereas VIP-based methods see a clear deterioration. For instance, Addi-MI-Global marginally decreases from 0.98 to 0.90 between the settings. Conversely Addi-VIP-Global sees a complete deterioration from 0.82 to 0.51.

ABC (Spike and Slab):

Performance between BART and AddiVortes is similar in the ABC framework, with neither model showing a clear advantage. This lack of differentiation is likely since the individual ABC models are made to be lightweight by only being run for a small number of MCMC iterations. As a result, it is possible that AddiVortes was not run for enough MCMC iterations to fully develop its theoretical advantages over the standard BART engine. Consequently, both engines end up capturing the exact same “low hanging fruit” rather than exploring deeper structural improvements in the posterior space.

DART and DAVT (Dirichlet Prior)

The Dirichlet prior based methods stand out as the overall best performers across all C.C.1 settings. They maintain near perfect F_1 scores with $m = 200$ even at the highest noise and dimensionality levels. The Dirichlet prior proves consistently more effective at $m = 200$ than at $m = 20$. This directly contrasts with standard BART and AddiVortes frameworks which require reducing m to improve variable selection capabilities. This distinction is highly significant as reducing the ensemble

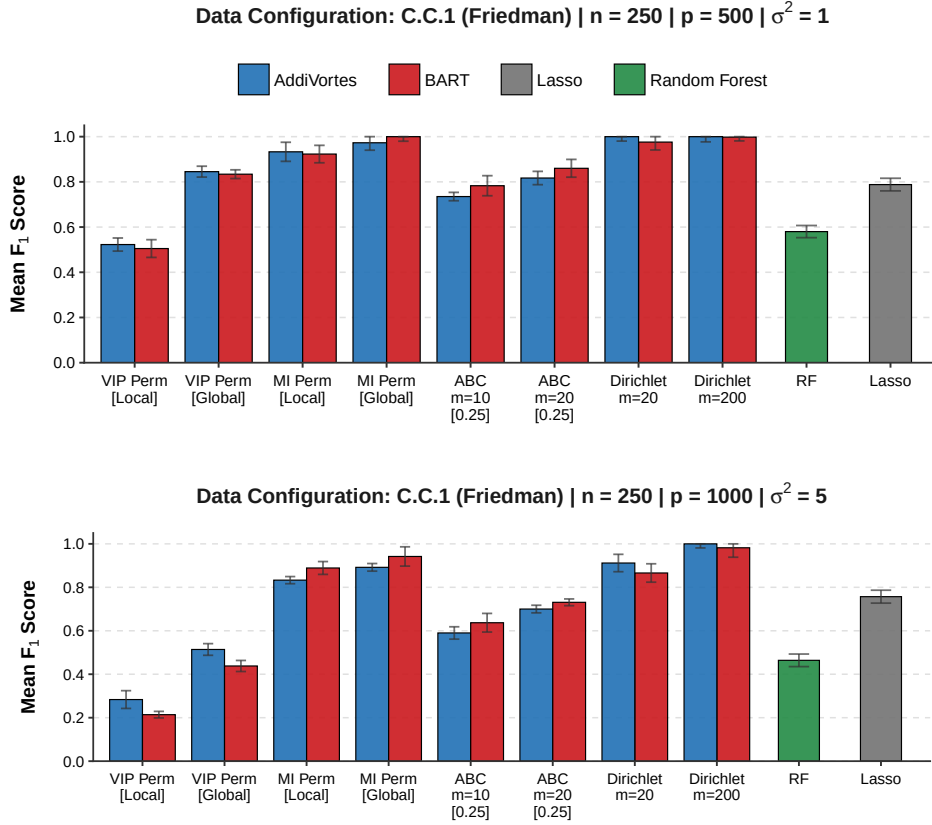


Figure 4.13: Mean F_1 scores for the C.C.1 Friedman scenario across the easiest ($p = 500, \sigma^2 = 1$) and hardest ($p = 1000, \sigma^2 = 5$) configurations.

size severely degrades regression accuracy as demonstrated by Linero (2018). Consequently, DART and DAVT outperform dedicated variable selection techniques whilst potentially preserving the regression predictive power of a large ensemble (see Section 4.7.6 for further discussion on this).

Baseline Models (Lasso and Random Forest)

The cross validated Lasso provides a steady baseline with F_1 scores resilient to changes in dimensionality and noise, remaining fixed around 0.75. Random Forest fails to exceed the performance of any BART or AddiVortes configuration across all settings. This behavior is expected as Chapter 1 documented the development of BART as a means to overcome the limitations of Random Forest in settings such as C.C.1.

4.7.5 Analysis of Results: Correlated Predictors (C.C.2):

We now consider the case of correlated predictors in the C.C.2 data configuration across settings $p \in \{100, 200\}$ and $\sigma^2 = 1$. This setting is particularly challenging due to the use of only interaction effects combined with correlations between variables. The sample size is fixed at $n = 500$, as we find that sample sizes below this threshold yield poor performance across all methods in this difficult setting. Furthermore, our results suggest that simulating higher noise levels would cause all methods to deteriorate completely, preventing any meaningful comparison.

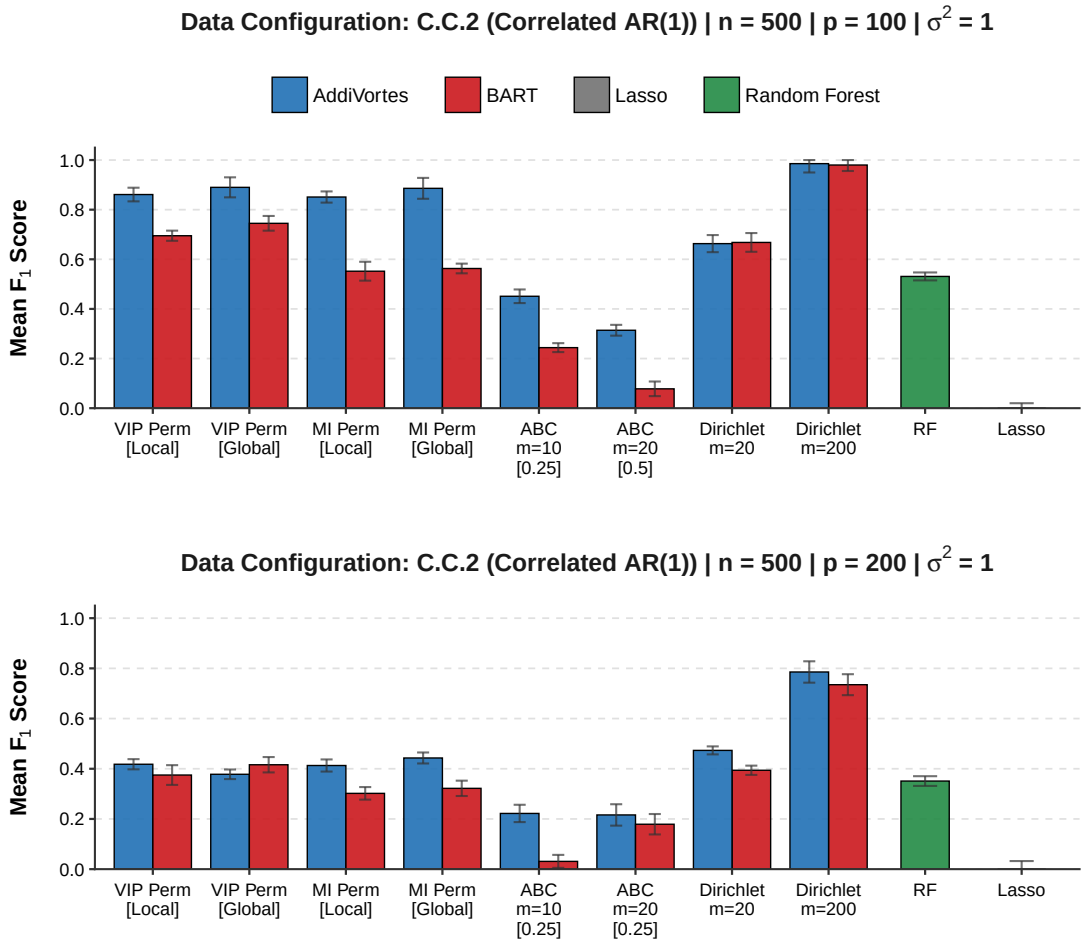


Figure 4.14: Mean F_1 scores for the C.C.2 correlated AR(1) scenario across the evaluated dimensions.

In contrast to C.C.1 where performance between BART and AddiVortes was comparable, C.C.2 reveals a clear preference for the AddiVortes engine across all frameworks. In C.C.1, we saw that the MI metric was comparable for both engines, whereas here we see that the MI values for AddiVortes are much higher than their BART counterpart. For Instance the first setting shows MI-Global having a dis-

crepancy of ≈ 0.3 between the engines. We theorise that this improvement stems directly from the flexibility of non-axis orthogonal splits. While the rigid splits of BART struggle to partition correlated feature spaces, AddiVortes slices efficiently through the correlated AR(1) structure to isolate active variables.

The Dirichlet prior methods maintain their dominance, though their behaviour shifts regarding ensemble size. While C.C.1 showed consistent performance across $m = 20$ and $m = 200$, the difficult correlated signal in C.C.2 requires an ensemble size of $m = 200$ to capture the underlying structure effectively. The DAVT model achieves an impressive F_1 score of 0.87 even in the more challenging configuration of $p = 200$ where all other methods fell to ≤ 0.5 . This result is exceptionally strong as the Dirichlet prior models not only outperform dedicated variable selection techniques but also preserve the full ensemble size necessary for regression.

While ABC performed reasonably well in the independent covariate setting of C.C.1, it experiences a complete breakdown under the correlated structures of C.C.2. The lightweight models used in the ABC framework are far too weak to capture the complex, correlated relationships among predictors. Consequently, both BART and AddiVortes struggle significantly, failing to isolate the true active set.

The Lasso model suffers complete deterioration: this failure is expected as Lasso is designed around linearity and fails fundamentally when confronted with pure interaction effects and severe multicollinearity. The Random Forest model performs marginally worse than BART, suffering from the same limitation of axis-aligned splits incapable of capturing the correlated structure.

4.7.6 Conclusion of Simulation Results

The aim of this simulation study was to assess if switching the engine of established BART variable selection methods outlined in Luo et al. (2021) to instead use an AddiVortes engine could improve upon their selection abilities. Our results presented in this section establish that the AddiVortes engine possess strong variable selection capabilities, and in complex data settings such as correlated predictors or high noise environments AddiVortes proves to be more effective than BART counterparts. In particular, we have provided evidence that in the case of correlated predictors, AddiVortes offers a significant improvement upon all BART variable selection frameworks and theorise that this arises from the flexibility of tessellations in navigating the AR(1) correlation structure.

In particular, this study highlights the success of Linero (2018)’s DART algorithm as a highly robust variable selection method. This result is supported by the

findings of Luo et al. (2021), who similarly identify DART as a dominant approach across a range of data configurations. To further illustrate the capabilities of the DART model, consider that all other methods evaluated in this chapter function purely as variable selection tools. For example, the ABC procedure entirely lacks a framework for regression. Furthermore, relying on a reduced ensemble size of $m = 20$ to make the VIP and MI metrics effective inherently sacrifices regression accuracy. While this lack of regression capability is acceptable when the sole objective is variable selection, DART and DAVT provide a clear advantage by outperforming dedicated selection methods while simultaneously maintaining robust regression performance.

To demonstrate that DART retains regression predictive power compared to BART ensembles constrained to $m = 20$, we present a final experiment. We generate high-dimensional data under the Friedman C.C.1 configuration with $n = 250$ observations, $p = 500$ predictors, and $\sigma^2 = 1$. We then fit two models with varying ensemble sizes. First, we fit a BART model with $m = 20$ to mirror the variable selection setup utilised for the permutation methods in this chapter. Next, we fit a DART model using a full ensemble of $m = 200$. The resulting regression performance of these models is shown in Figure 4.15.

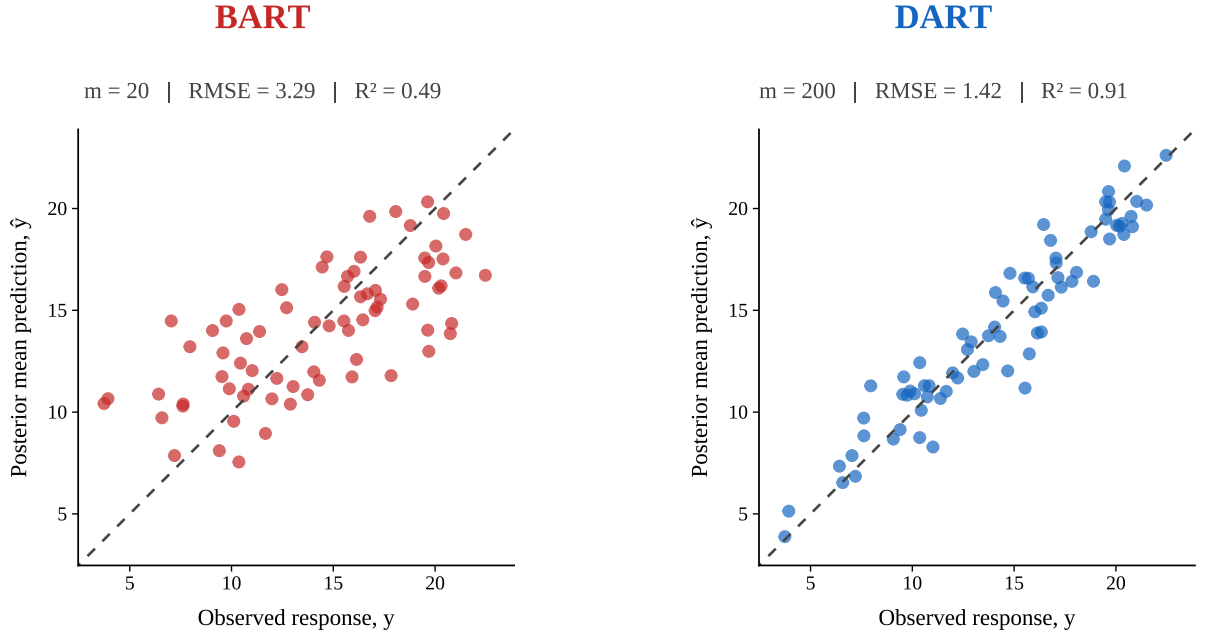


Figure 4.15: Regression predictive accuracy for BART ($m = 20$) and DART ($m = 200$) on the high-dimensional C.C.1 data structure. DART demonstrates the distinct advantage of maintaining full regression estimation capabilities within a single framework. The results presented are consistent across MCMC runs and datasets.

These results confirm the deterioration in regression ability that BART faces when m is decreased. In particular, this experiment shows that the RMSE of BART with $m = 20$ is double that of DART when fit to the same data. This deterioration is acceptable assuming the primary goal is variable selection. DART and DAVT remove the need for practitioners to choose between these two goals by addressing both objectives within a single modelling framework. The sparse priors encourage the models to concentrate splitting probability on a small subset of active predictors, while the larger ensemble retains the flexibility required to accurately estimate a complex regression function.

Chapter 5

Conclusion

BART models a response surface as a sum of decision trees, allowing complex non-linear relationships to be captured through axis-aligned partitions. AddiVortes provides an alternative partitioning framework by replacing the axis-aligned tree architecture with flexible Voronoi tessellations. The variable selection literature for BART is well established and can broadly be divided into two approaches. The first summarises posterior information from an unconstrained BART model to assess variable importance, typically through measures based on variable inclusion frequencies. The second incorporates variable selection into the prior by restricting which predictors are available during model construction. In particular, we reviewed an external spike-and-slab approach and an internal Dirichlet prior construction. Recent work by Stone and Gosling (2025a) demonstrated that AddiVortes can provide improved regression and predictive performance relative to BART, motivating the central question of whether these benefits extend to variable selection.

5.1 Innovations of this Thesis

This thesis provides two primary methodological contributions. First we fully specified the mathematical derivations for all Metropolis-Hastings acceptance ratios within the AddiVortes framework. These complete derivations were previously absent from the available literature. Secondly, we developed a formal variable selection methodology for AddiVortes. The original AddiVortes publication suggested using inclusion frequency counts to measure variable importance but did not define the threshold required to select a final subset of variables. We resolved this gap by adapting four distinct variable selection techniques to AddiVortes. These include Variable Inclusion Proportions (VIP), Metropolis Information (MI), an Approximate Bayesian Computation approach, and the use of Dirichlet priors. We presented a

simulated study evaluating these methods by extending the simulation study of Luo et al. (2021) as a benchmark against our AddiVortes methods. The results show that replacing the underlying partitioning mechanism with AddiVortes can improve variable selection performance, particularly in settings with correlated predictors where axis-aligned partitions may provide a less effective representation of the response surface.

5.2 Suggestions of Further Research

The simulation configurations evaluated in this thesis strictly utilised continuous predictors and continuous responses. Future research should investigate alternative variable cardinalities to evaluate potential structural biases. Recall from Section 4.2.2 that tree based methods such as BART and Random Forests suffer from cardinality bias. When using these models, variable importance metrics based on covariate usage frequencies naturally favour variables with a larger number of potential splitting rules. AddiVortes possesses a distinct structural advantage in this context because a single variable can define at most one dimension within a given tessellation. This strict geometric constraint theoretically prevents redundant splitting on high cardinality predictors. We deferred this investigation because the current software implementation of AddiVortes handles nominal covariates exclusively through one hot encoding. This preprocessing step encodes a single categorical predictor across multiple independent binary dummy variables. This fragmentation complicates the aggregation of variable importance metrics, preventing accurate evaluation of the true active set.

Future work should also explore extensive hyperparameter tuning. Aside from altering the ensemble size, m , we maintained standard default hyperparameter settings across all AddiVortes models. If the objective shifts from baseline comparison with BART to absolute performance maximisation, researchers should investigate which prior hyperparameters are optimal for variable selection. For instance, recall from Section 3.2.3 that the parameter ω dictates the expected number of active dimensions within each tessellation. Decreasing ω reduces the available space for variables to enter the model. This restriction forces covariates to compete more aggressively for inclusion and exerts a regularising effect similar to lowering the ensemble size m . Exploring the optimal balance between ω and the ensemble size m represents a promising avenue for maximising variable selection accuracy.

Subsequent research could also evaluate the impact of employing alternative distance metrics. Throughout this paper we have constructed tessellations exclusively

using euclidean distance. The AddiVortes architecture, however, can accommodate alternative spatial measures such as Manhattan or Mahalanobis distances. Modifying the underlying distance metric fundamentally alters the geometric profile and orientation of the spatial partitions. Investigating how these alternative tessellation structures influence signal recovery and variable selection performance represents an interesting direction for future study.

The Metropolis Information (MI) metric evaluates variable importance through MH acceptance probabilities. Section 4.3 established that this approach measures the posterior support for each proposed inclusion and naturally down-weights redundant deep partitions, helping to mitigate cardinality bias. However, the current BART MI formulation considers only GROW proposals. While these proposals measure the support for introducing a predictor into a terminal node, other MCMC moves also contain information about variable importance. A SWAP proposal, for example, compares competing splitting predictors and therefore provides information about the relative importance of the variables involved. A PRUNE proposal provides complementary information by measuring the support for removing an existing split. This is useful for identifying variables that contribute little to the fitted model, which is as relevant to variable selection as identifying variables with strong support for inclusion. Restricting MI to GROW proposals therefore discards potentially useful information contained in the remaining MCMC moves.

A natural extension would be to construct a unified importance measure that aggregates information from GROW, SWAP, and PRUNE acceptance probabilities. The main difficulty is determining how these contributions should be weighted. The moves are not proposed with equal frequency, so a simple average could allow the more common proposals to dominate the resulting measure. A principled weighting scheme would therefore be required to combine the different sources of information without allowing proposal frequency to determine their relative importance. Such a measure could make fuller use of the information already generated by the BART MCMC sampler and provide a more comprehensive basis for variable selection.

Appendix A

BART: Supplementary Material

A.1 A Simple BART Prediction Illustration

Consider the following decision trees shown in Figure A.1 as two individual trees within a BART ensemble with $m = 2$:

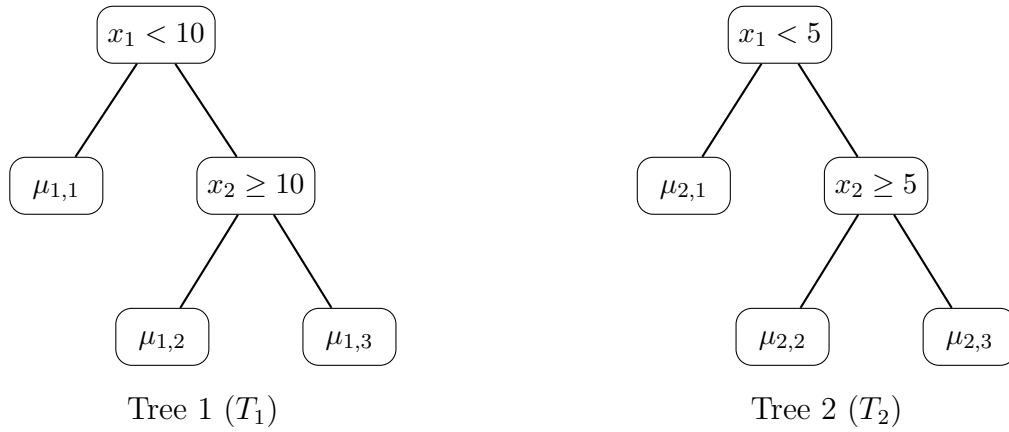


Figure A.1: Two simple regression trees within a BART ensemble with $m = 2$

BART then takes the two assigned leaf values from each of the trees and adds them together. For instance, consider $x_1 = 7$ and $x_2 = 10$. Tree 1 and Tree 2 will return the following:

$$g(x_1 = 7, x_2 = 10; (T_1, M_1)) = \mu_{1,1} \quad (\text{A.1})$$

$$g(x_1 = 7, x_2 = 10; (T_2, M_2)) = \mu_{2,2} \quad (\text{A.2})$$

BART then returns the sum of the individual trees:

$$\begin{aligned}
g(x; (T_1, M_1), (T_2, M_2)) &= g(x; (T_1, M_1)) + g(x; (T_2, M_2)) \\
&= \mu_{1,1} + \mu_{2,2}
\end{aligned}
\tag{A.3}$$

A.2 An Overview of Gibbs Sampling

Suppose the aim is to generate samples from a joint posterior distribution $\pi(\theta \mid y)$, where $\theta = (\theta_1, \dots, \theta_d) \in \Theta$. Direct sampling from this distribution may be difficult when the parameter space is high dimensional. Gibbs sampling avoids this by updating each component of θ in turn from its full conditional distribution. For the i th component, the full conditional distribution is

$$\pi(\theta_i \mid \theta_{-i}, y),$$

where θ_{-i} denotes all components of θ other than θ_i . Since terms that do not depend on θ_i are constant with respect to this conditional distribution, it can be written as

$$\pi(\theta_i \mid \theta_{-i}, y) \propto \pi(\theta_i, \theta_{-i}, y). \tag{A.4}$$

Starting from an initial value $\theta^{(0)} = (\theta_1^{(0)}, \dots, \theta_d^{(0)})$, the sampler updates the components sequentially. At iteration t , the updates are

$$\theta_1^{(t)} \sim \pi(\theta_1 \mid \theta_2^{(t-1)}, \dots, \theta_d^{(t-1)}, y), \tag{A.5}$$

$$\theta_2^{(t)} \sim \pi(\theta_2 \mid \theta_1^{(t)}, \theta_3^{(t-1)}, \dots, \theta_d^{(t-1)}, y), \tag{A.6}$$

$$\vdots$$

$$\theta_i^{(t)} \sim \pi(\theta_i \mid \theta_1^{(t)}, \dots, \theta_{i-1}^{(t)}, \theta_{i+1}^{(t-1)}, \dots, \theta_d^{(t-1)}, y), \tag{A.7}$$

$$\vdots$$

$$\theta_d^{(t)} \sim \pi(\theta_d \mid \theta_1^{(t)}, \dots, \theta_{d-1}^{(t)}, y). \tag{A.8}$$

Thus, each update uses the most recently available values of the remaining parameters. Under standard conditions ensuring that the resulting Markov chain is ergodic, the distribution of $\theta^{(t)}$ converges to the target posterior $\pi(\theta \mid y)$. After an appropriate burn-in period, the resulting MCMC draws can therefore be used to estimate posterior quantities, for example through averages of the sampled values.

A.3 An Overview of The Metropolis Hastings Algorithm

The Metropolis Hastings (MH) algorithm is a Markov Chain Monte Carlo (MCMC) method used to generate a sequence of dependent samples from a probability distribution that is difficult to sample from directly. The method was originally developed by Metropolis et al. (1953) for statistical mechanics and was later generalised by Hastings (1970) for broader applications in statistical inference.

Suppose the objective is to draw samples from a target posterior distribution $\pi(\theta \mid y)$. At each iteration t , given the current state $\theta^{(t)}$, a candidate state θ^* is generated from a proposal distribution $q(\theta^* \mid \theta^{(t)})$. The proposed state is then accepted with probability

$$\alpha(\theta^{(t)}, \theta^*) = \min \left(1, \frac{\pi(\theta^* \mid y) q(\theta^{(t)} \mid \theta^*)}{\pi(\theta^{(t)} \mid y) q(\theta^* \mid \theta^{(t)})} \right). \quad (\text{A.9})$$

If the candidate is accepted, $\theta^{(t+1)} = \theta^*$; otherwise, the chain remains at its current state, so that $\theta^{(t+1)} = \theta^{(t)}$. The normalising constant of the posterior cancels from the acceptance ratio, so MH only requires the unnormalised posterior.

The acceptance probability is constructed to satisfy detailed balance,

$$\pi(\theta \mid y) K(\theta, \theta') = \pi(\theta' \mid y) K(\theta', \theta), \quad (\text{A.10})$$

where K is the transition kernel. This ensures that $\pi(\theta \mid y)$ is an invariant distribution of the Markov chain. With a suitable proposal distribution and sufficient iterations, the resulting samples can therefore be used to approximate the target posterior.

A.4 Full Derivation of the Terminal Node Parameter Posterior

By Bayes' theorem, the conditional posterior of the terminal node parameter μ_η is proportional to the product of the likelihood of the residuals allocated to that node and the prior on the node parameter:

$$\pi(\mu_\eta \mid T_j, R_j, \sigma^2) \propto \pi(R_{j,\eta} \mid \mu_\eta, \sigma^2) \pi(\mu_\eta \mid \sigma_\mu^2).$$

Substituting the densities for the independent Gaussian likelihood and the Gaus-

sian prior:

$$\pi(\mu_\eta \mid T_j, R_j, \sigma^2) \propto \left(\prod_{i=1}^{n_\eta} \frac{1}{\sqrt{2\pi\sigma^2}} \exp \left(-\frac{(R_{\eta,i} - \mu_\eta)^2}{2\sigma^2} \right) \right) \frac{1}{\sqrt{2\pi\sigma_\mu^2}} \exp \left(-\frac{\mu_\eta^2}{2\sigma_\mu^2} \right).$$

We can drop the normalising constants $(2\pi\sigma^2)^{-n_\eta/2}$ and $(2\pi\sigma_\mu^2)^{-1/2}$ because they do not depend on μ_η . We then combine the product over n_η independent observations into a summation within a single exponential:

$$\pi(\mu_\eta \mid T_j, R_j, \sigma^2) \propto \exp \left(-\frac{1}{2\sigma^2} \sum_{i=1}^{n_\eta} (R_{\eta,i} - \mu_\eta)^2 \right) \exp \left(-\frac{\mu_\eta^2}{2\sigma_\mu^2} \right).$$

Next, we combine the two exponentials and expand the squared difference $(R_{\eta,i} - \mu_\eta)^2$:

$$\pi(\mu_\eta \mid T_j, R_j, \sigma^2) \propto \exp \left(-\frac{1}{2} \left[\frac{1}{\sigma^2} \sum_{i=1}^{n_\eta} (R_{\eta,i}^2 - 2\mu_\eta R_{\eta,i} + \mu_\eta^2) + \frac{\mu_\eta^2}{\sigma_\mu^2} \right] \right).$$

We distribute the summation across the three terms. Notice that the term $\sum_{i=1}^{n_\eta} R_{\eta,i}^2$ does not contain μ_η ; therefore, $\exp \left(-\sum R_{\eta,i}^2 / 2\sigma^2 \right)$ acts as a constant multiplier and can be absorbed into the proportionality constant. Additionally, summing μ_η^2 over n_η observations simply yields $n_\eta \mu_\eta^2$:

$$\pi(\mu_\eta \mid T_j, R_j, \sigma^2) \propto \exp \left(-\frac{1}{2} \left[-\frac{2\mu_\eta}{\sigma^2} \sum_{i=1}^{n_\eta} R_{\eta,i} + \frac{n_\eta \mu_\eta^2}{\sigma^2} + \frac{\mu_\eta^2}{\sigma_\mu^2} \right] \right).$$

We then group the terms by μ_η^2 and μ_η :

$$\pi(\mu_\eta \mid T_j, R_j, \sigma^2) \propto \exp \left(-\frac{1}{2} \left[\mu_\eta^2 \left(\frac{n_\eta}{\sigma^2} + \frac{1}{\sigma_\mu^2} \right) - 2\mu_\eta \left(\frac{\sum_{i=1}^{n_\eta} R_{\eta,i}}{\sigma^2} \right) \right] \right).$$

To complete the square, we want to manipulate the expression inside the bracket into the standard quadratic form $A\mu_\eta^2 - 2B\mu_\eta$. Let us define:

$$A = \frac{n_\eta}{\sigma^2} + \frac{1}{\sigma_\mu^2},$$

$$B = \frac{\sum_{i=1}^{n_\eta} R_{\eta,i}}{\sigma^2}.$$

Factoring out A , the exponential becomes:

$$\exp \left(-\frac{1}{2} [A\mu_\eta^2 - 2B\mu_\eta] \right) = \exp \left(-\frac{A}{2} \left[\mu_\eta^2 - 2\frac{B}{A}\mu_\eta \right] \right).$$

Adding and subtracting $(B/A)^2$ inside the bracket completes the square:

$$\exp \left(-\frac{A}{2} \left[\left(\mu_\eta - \frac{B}{A} \right)^2 - \left(\frac{B}{A} \right)^2 \right] \right).$$

The remaining constant term $\exp(A(B/A)^2/2)$ is independent of μ_η and is again absorbed into the proportionality constant. This leaves the exact kernel of a Gaussian distribution:

$$\pi(\mu_\eta \mid T_j, R_j, \sigma^2) \propto \exp \left(-\frac{A}{2} \left(\mu_\eta - \frac{B}{A} \right)^2 \right).$$

This is a Gaussian density $\mathcal{N}(\mu_\eta \mid \text{mean, variance})$ where the variance is $1/A$ and the mean is B/A .

To finalise the derivation, we reintroduce the sufficient statistics defined in the main text. Let the sum of residuals be $S_\eta = \sum_{i=1}^{n_\eta} R_{\eta,i}$, and the marginal variance factor be $V_\eta = \sigma^2 + n_\eta \sigma_\mu^2$.

First, we simplify the precision term A by finding a common denominator:

$$A = \frac{n_\eta}{\sigma^2} + \frac{1}{\sigma_\mu^2} = \frac{n_\eta \sigma_\mu^2 + \sigma^2}{\sigma^2 \sigma_\mu^2}.$$

Recognising that the numerator is exactly V_η , the posterior variance is simply the reciprocal:

$$\text{Variance} = \frac{1}{A} = \frac{\sigma^2 \sigma_\mu^2}{V_\eta}.$$

Next, we rewrite B using the sufficient statistic S_η :

$$B = \frac{S_\eta}{\sigma^2}.$$

We then calculate the posterior mean B/A :

$$\text{Mean} = \frac{B}{A} = B \times \frac{1}{A} = \frac{S_\eta}{\sigma^2} \times \frac{\sigma^2 \sigma_\mu^2}{V_\eta} = \frac{S_\eta \sigma_\mu^2}{V_\eta}.$$

Substituting the explicit mean and variance back into the Gaussian distribution yields the final exact closed form for the terminal node posterior draw:

$$\mu_\eta \mid T_j, R_j, \sigma^2 \sim \mathcal{N} \left(\frac{\sigma_\mu^2 S_\eta}{V_\eta}, \frac{\sigma^2 \sigma_\mu^2}{V_\eta} \right). \quad (\text{A.11})$$

A.5 BART Machinery: A Computational Demonstration

The purpose of this section is to demonstrate the operational mechanics of the BART algorithm. To illustrate how the theoretical posterior distributions and transdimensional integrals reduce to standard arithmetic operations, we execute a manual calculation of a single Gibbs cycle. The example is specifically constructed using a simple step function and a two tree ensemble. This minimal specification isolates the core computational principles of the model without unnecessary algebraic overhead. By tracing the exact numerical calculations sequentially, this demonstration explicitly exhibits two fundamental behaviours of the algorithm: the Bayesian backfitting process passing residual vectors between trees, and the regularisation mechanism of the Metropolis Hastings ratio rejecting structural modifications that model unstructured noise.

Consider a minimal dataset containing $n = 4$ observations. Suppose the dataset contains $p = 10$ candidate predictors, but the true signal depends exclusively on a single continuous predictor x_1 , forming a perfect step function:

$$x_1 = \begin{bmatrix} 1 \\ 2 \\ 8 \\ 9 \end{bmatrix}, \quad y = \begin{bmatrix} 2.0 \\ 2.0 \\ 8.0 \\ 8.0 \end{bmatrix}. \quad (\text{A.12})$$

For this demonstration, we initialise an ensemble of $m = 2$ trees. At iteration $t = 0$, both T_1 and T_2 begin as single root nodes predicting a value of 0. To ensure traceability, we fix the hyperparameters as follows: the current residual variance is $\sigma^2 = 1.0$, the prior leaf variance is $\sigma_\mu^2 = 4.0$, and the tree structure regularisation parameters are $\alpha = 0.5$ and $\beta = 2.0$. The proposal probabilities for the structural moves are fixed at $\mathbb{P}(\text{GROW}) = 0.25$ and $\mathbb{P}(\text{PRUNE}) = 0.25$.

Iteration 1: Updating T_1

Step 1a: Partial Residuals. The backfitting algorithm begins by isolating the residual variation unexplained by the rest of the ensemble. The partial residual

vector R_1 is computed by subtracting the predictions of T_2 from the response y :

$$R_1 = y - g(x; T_2) = \begin{bmatrix} 2.0 \\ 2.0 \\ 8.0 \\ 8.0 \end{bmatrix} - \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 2.0 \\ 2.0 \\ 8.0 \\ 8.0 \end{bmatrix}. \quad (\text{A.13})$$

Step 1b: Metropolis Hastings update. Suppose the algorithm proposes a GROW move to the root node η . It selects predictor x_1 and proposes a split at the midpoint threshold $x_1 < 5$. This candidate topology T_1^* partitions the data into a left child η_L containing observations $\{1, 2\}$, and a right child η_R containing observations $\{8, 9\}$. We calculate the sufficient statistics for the parent and child nodes:

$$\begin{aligned} S_\eta &= 2.0 + 2.0 + 8.0 + 8.0 = 20.0, & V_\eta &= 1.0 + 4(4.0) = 17.0, \\ S_{\eta_L} &= 2.0 + 2.0 = 4.0, & V_{\eta_L} &= 1.0 + 2(4.0) = 9.0, \\ S_{\eta_R} &= 8.0 + 8.0 = 16.0, & V_{\eta_R} &= 1.0 + 2(4.0) = 9.0. \end{aligned}$$

Substituting these statistics into the GROW likelihood ratio derived in Equation 2.45 yields:

$$\begin{aligned} \frac{\pi(R_1 \mid T_1^*, \sigma^2)}{\pi(R_1 \mid T_1, \sigma^2)} &= \sqrt{\frac{1.0 \times 17.0}{9.0 \times 9.0}} \exp\left(\frac{4.0}{2.0} \left[\frac{16.0}{9.0} + \frac{256.0}{9.0} - \frac{400.0}{17.0}\right]\right) \\ &\approx 0.458 \times \exp(2.0 [30.222 - 23.529]) \\ &\approx 0.458 \times 655450 \approx 300196. \end{aligned} \quad (\text{A.14})$$

The massive likelihood ratio reflects the structural perfection of the proposed split. Next, we compute the tree structure ratio using Equation 2.50. Because $p = 10$ and there are 3 possible internal split points for x_1 in this dataset, $p_{\text{adj}}(\eta) = 10$ and $n_{k,\text{adj}}(\eta) = 3$. With $d_\eta = 0$, the ratio evaluates to:

$$\frac{\pi(T_1^*)}{\pi(T_1)} = 0.5 \left(\frac{(1 - 0.5/4)^2}{(1 - 0.5) \times 10 \times 3} \right) = \frac{0.5 \times 0.7656}{15.0} \approx 0.0255. \quad (\text{A.15})$$

Finally, the transition ratio derived in Equation 2.36 evaluates to $(0.25/0.25) \times (1 \times 10 \times 3/1) = 30.0$. Multiplying these three components generates the Metropolis Hastings acceptance probability:

$$\alpha(T_1, T_1^*) = \min(1, 300196 \times 0.0255 \times 30.0) = \min(1, 229649) = 1. \quad (\text{A.16})$$

The proposal is strictly accepted, and the topology of T_1 is updated to include the split.

Step 1c: Leaf Update. With the new structure confirmed, the algorithm draws new terminal leaf parameters using the Gaussian posterior derived in Equation 2.53:

$$\mu_{1,\eta_L} \sim \mathcal{N}\left(\frac{4.0 \times 4.0}{9.0}, \frac{1.0 \times 4.0}{9.0}\right) \equiv \mathcal{N}(1.78, 0.44), \quad (\text{A.17})$$

$$\mu_{1,\eta_R} \sim \mathcal{N}\left(\frac{4.0 \times 16.0}{9.0}, \frac{1.0 \times 4.0}{9.0}\right) \equiv \mathcal{N}(7.11, 0.44). \quad (\text{A.18})$$

We simulate independent draws from these distributions, returning $\mu_{1,\eta_L} = 1.8$ and $\mu_{1,\eta_R} = 7.2$. The update for T_1 is now complete, resulting in the structure displayed in Figure A.2.

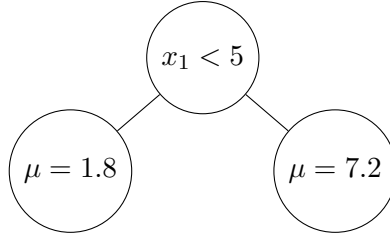


Figure A.2: The accepted structure and terminal parameters for T_1 after the first sequential update.

Iteration 1: Updating T_2 and Backfitting

Step 2a: Partial Residuals. The backfitting mechanic is explicitly demonstrated when updating the second tree. T_2 does not model the raw response y ; it models the residual vector left unexplained by T_1 . Subtracting the newly updated predictions of T_1 yields:

$$R_2 = y - g(x; T_1) = \begin{bmatrix} 2.0 \\ 2.0 \\ 8.0 \\ 8.0 \end{bmatrix} - \begin{bmatrix} 1.8 \\ 1.8 \\ 7.2 \\ 7.2 \end{bmatrix} = \begin{bmatrix} 0.2 \\ 0.2 \\ 0.8 \\ 0.8 \end{bmatrix}. \quad (\text{A.19})$$

Because T_1 successfully captured the primary step function, the target vector R_2 now contains only residual noise.

Step 2b: Metropolis Hastings update. Suppose the random generator again proposes a GROW move at $x_1 < 5$ for T_2 . The variance factors V_η remain identical to step 1, but the sum of residuals S_η collapses because the signal has been absorbed by T_1 . The new sufficient statistics are $S_\eta = 2.0$, $S_{\eta_L} = 0.4$, and $S_{\eta_R} = 1.6$. Substituting

these into the likelihood ratio yields:

$$\begin{aligned}
\frac{\pi(R_2 \mid T_2^*, \sigma^2)}{\pi(R_2 \mid T_2, \sigma^2)} &= 0.458 \times \exp \left(\frac{4.0}{2.0} \left[\frac{0.16}{9.0} + \frac{2.56}{9.0} - \frac{4.0}{17.0} \right] \right) \\
&\approx 0.458 \times \exp(2.0 [0.017 + 0.284 - 0.235]) \\
&\approx 0.458 \times \exp(0.132) \approx 0.522.
\end{aligned} \tag{A.20}$$

The structural improvement is negligible, resulting in a likelihood ratio strictly less than 1. The tree structure ratio and transition ratio remain 0.0255 and 30.0, respectively. The final acceptance probability collapses to:

$$\alpha(T_2, T_2^*) = \min(1, 0.522 \times 0.0255 \times 30.0) = 0.039. \tag{A.21}$$

A uniform random variable $u \sim \mathcal{U}(0, 1)$ is drawn. Supposing $u = 0.45$, the condition $u \leq \alpha$ fails. The proposal is strictly rejected, and T_2 remains a single root node. This illustrates the regularisation mechanic: BART actively resists growing trees to fit unstructured noise.

Step 2c: Leaf Update. Because T_2 remains a root node, we draw a single parameter μ_2 from the root posterior, using the statistics $S_\eta = 2.0$ and $V_\eta = 17.0$:

$$\mu_2 \sim \mathcal{N} \left(\frac{4.0 \times 2.0}{17.0}, \frac{1.0 \times 4.0}{17.0} \right) \equiv \mathcal{N}(0.47, 0.23). \tag{A.22}$$

Assuming a simulated draw returns $\mu_2 = 0.3$, the ensemble update is complete. The global ensemble prediction is exactly $g(x; T_1) + g(x; T_2)$.

Iteration 1: Variance Update

The iteration concludes by updating the global variance parameter σ^2 . A final residual vector ε is computed across the entire ensemble by subtracting all tree predictions from the response data: $\varepsilon = y - g(x; T_1) - g(x; T_2)$. A new value for σ^2 is then drawn from the scaled inverse chi squared posterior distribution conditioned on $\sum_{i=1}^n \varepsilon_i^2$. The algorithm then loops back to T_1 , passing the latest residuals forward, continuously refining the joint posterior approximation.

Appendix B

AddiVortes: Supplementary Material

B.1 illustrations of moves

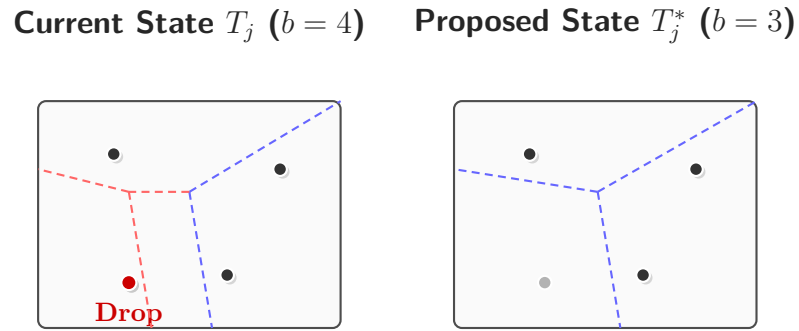


Figure B.1: **Remove Centre (RC)**. The inverse of the AC move. A randomly selected centre is deleted. The spatial territory previously occupied by the deleted centre's cell is immediately absorbed by its nearest geometric neighbors.

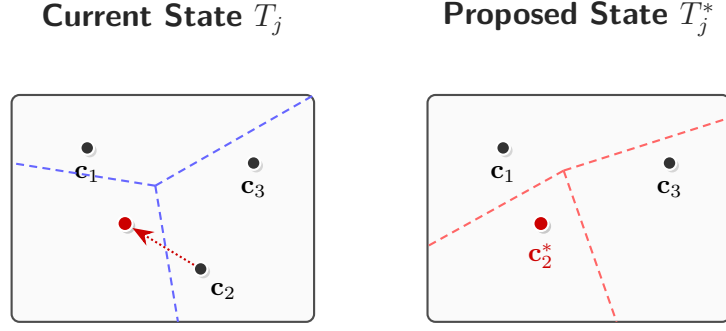


Figure B.2: **Change Centre (CC)**. Dimensionality and complexity (b) remain constant. A selected centre is moved to a newly sampled spatial coordinate drawn from the Gaussian spatial prior. Because the move is structurally and spatially symmetric, the final evaluation relies completely on the data likelihood.

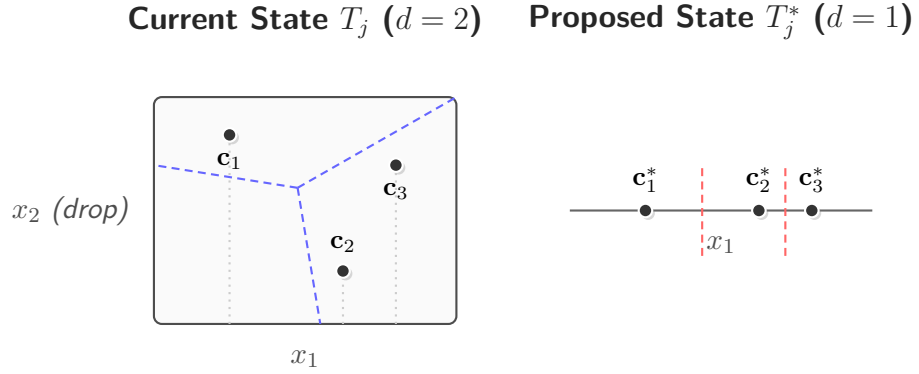


Figure B.3: **Remove Dimension (RD)**. The exact structural inverse of the AD move. Removing a dimension permanently discards the coordinates of all generating centres along that axis. The spatial topology collapses into a lower-dimensional subspace.

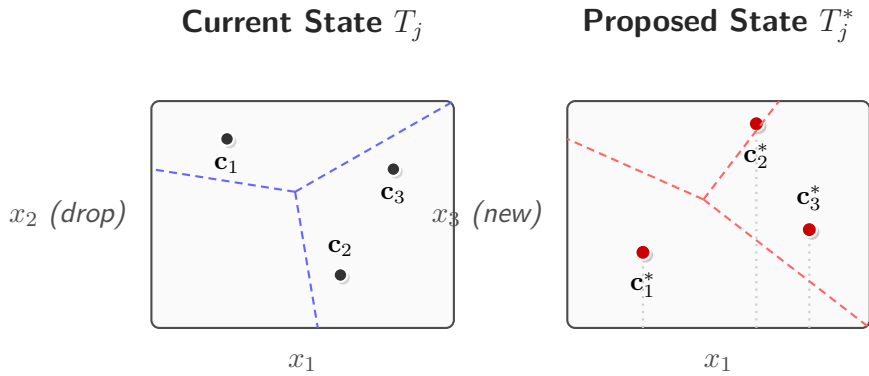


Figure B.4: **Swap Dimension (SD)**. Dimensionality d remains constant. The unslected dimension x_1 coordinates act as geometric anchors, while the x_2 coordinates are discarded and replaced by new coordinates drawn from a zero-mean normal distribution along x_3 . Because the prior and transition probabilities are structurally symmetric, the acceptance of the move is driven solely by the improvement to the data likelihood.

B.2 AddiVortes MH acceptance ratio derivations

B.2.1 Deriving The Remove Dimension (RD) Acceptance Ratio

Let T_j be the current tessellation with d active dimensions. We propose T_j^* by deleting a dimension k^* , chosen uniformly at random from the d active covariates. This reduces the dimensionality to $d - 1$ and effectively collapses the coordinate space of the generating centres along dimension k^* . Because the RD move is the exact structural inverse of the AD move, its constituent prior and transition ratios are easily derived by inverting the logic of the AD proposal:

$$\frac{\pi(T_j^*)}{\pi(T_j)} = \frac{d-1}{d} \left(\frac{p-\Omega}{\Omega} \right) \times \left(\prod_{l=1}^b \phi(c_{lk^*}; \sigma_c^2) \right)^{-1} \quad (\text{B.1})$$

$$\frac{q(T_j | T_j^*)}{q(T_j^* | T_j)} = \frac{p_{AD}}{p_{RD}} \times \frac{d}{p-d+1} \times \prod_{l=1}^b \phi(c_{lk^*}; \sigma_c^2) \quad (\text{B.2})$$

Multiplying the prior ratio by the transition ratio causes the spatial coordinate densities and the d terms to cancel out perfectly. Merging this result with the universal likelihood ratio yields the final acceptance probability for the Remove Dimension proposal:

$$\alpha_{RD}(T_j \rightarrow T_j^*) := \min \left\{ 1, \frac{\pi(R_j | T_j^*, \sigma^2)}{\pi(R_j | T_j, \sigma^2)} \times \frac{p_{AD}}{p_{RD}} \left(\frac{d-1}{p-d+1} \right) \left(\frac{p-\Omega}{\Omega} \right) \right\} \quad (\text{B.3})$$

B.2.2 Deriving The Swap Dimension (SD) Acceptance Ratio

Let T_j be the current tessellation with d active dimensions. We propose T_j^* by swapping an active dimension for an inactive one. The algorithm selects one dimension k_{out} uniformly at random from the d active covariates to remove, and one dimension k_{in} uniformly at random from the $p-d$ inactive covariates to add. The total dimensionality remains d . Consequently, the coordinates for the b generating centres along k_{out} are discarded, and new coordinates are drawn along k_{in} .

Because the number of dimensions does not change, the priors on the dimension count and the dimension subset are identical for both T_j and T_j^* , cancelling perfectly to in te final ratio. The prior ratio is driven entirely by the spatial coordinates of the new versus the old axis:

$$\frac{\pi(T_j^*)}{\pi(T_j)} = \frac{\prod_{l=1}^b \phi(c_{lk_{in}}; \sigma_c^2)}{\prod_{l=1}^b \phi(c_{lk_{out}}; \sigma_c^2)} \quad (\text{B.4})$$

For the transition ratio, both the forward and reverse moves share the same underlying probabilities: selecting the Swap Dimension move (p_{SD}), selecting one of the d active covariates to drop ($1/d$), and selecting one of the $p-d$ inactive covariates to add ($1/(p-d)$). The only asymmetry lies in generating the new coordinates for k_{in} during the forward move, and regenerating the original coordinates for k_{out} during the reverse move:

$$\frac{q(T_j | T_j^*)}{q(T_j^* | T_j)} = \frac{p_{SD} \times \frac{1}{d} \times \frac{1}{p-d} \times \prod_{l=1}^b \phi(c_{lk_{out}}; \sigma_c^2)}{p_{SD} \times \frac{1}{d} \times \frac{1}{p-d} \times \prod_{l=1}^b \phi(c_{lk_{in}}; \sigma_c^2)} = \frac{\prod_{l=1}^b \phi(c_{lk_{out}}; \sigma_c^2)}{\prod_{l=1}^b \phi(c_{lk_{in}}; \sigma_c^2)} \quad (\text{B.5})$$

When multiplying the prior ratio by the transition ratio, the spatial coordinate densities cancel each other out completely. The entire structural ratio collapses to exactly 1.

Therefore, the acceptance probability for the Swap Dimension proposal relies strictly on the universal marginal likelihood ratio:

$$\alpha_{SD}(T_j \rightarrow T_j^*) := \min \left\{ 1, \frac{\pi(R_j | T_j^*, \sigma^2)}{\pi(R_j | T_j, \sigma^2)} \right\} \quad (\text{B.6})$$

B.2.3 Deriving The Remove Centre (RC) Acceptance Ratio

Let T_j be the current tessellation with d active dimensions and b generating centres. We propose T_j^* by deleting a centre $\mathbf{c}^*$, chosen uniformly at random from the b active centres. This reduces the total number of centres to $b-1$ and effectively deletes its spatial coordinates. Because the RC move is the exact structural inverse of the AC move, its constituent prior and transition ratios are easily derived by inverting the logic of the AC proposal:

$$\frac{\pi(T_j^*)}{\pi(T_j)} = \frac{b-1}{\lambda_c \cdot b} \times \left(\prod_{m=1}^d \phi(c_m^*; \sigma_c^2) \right)^{-1} \quad (\text{B.7})$$

$$\frac{q(T_j | T_j^*)}{q(T_j^* | T_j)} = \frac{p_{AC}}{p_{RC}} \times b \times \prod_{m=1}^d \phi(c_m^*; \sigma_c^2) \quad (\text{B.8})$$

Multiplying the prior ratio by the transition ratio causes the spatial coordinate densities and the b denominator to cancel out perfectly. Merging this collapsed structural ratio with the universal likelihood ratio yields the final acceptance probability

for the Remove Centre proposal:

$$\alpha_{RC}(T_j \rightarrow T_j^*) := \min \left\{ 1, \frac{\pi(R_j \mid T_j^*, \sigma^2)}{\pi(R_j \mid T_j, \sigma^2)} \times \frac{p_{AC}}{p_{RC}} \left(\frac{b-1}{\lambda_c} \right) \right\} \quad (\text{B.9})$$

B.2.4 Deriving The Change Centre (CC) Acceptance Ratio

Let T_j be the current tessellation with d active dimensions and b generating centres. We propose T_j^* by moving an existing centre to a new location. The algorithm selects one centre uniformly at random from the b active centres, discards its current coordinates $\mathbf{c}_{old}^*$, and draws new coordinates $\mathbf{c}_{new}^*$ along all d dimensions. The total number of centres remains b .

Because the number of centres and dimensions does not change, the priors on the centre count, dimension count, and dimension subset are identical for both T_j and T_j^* , cancelling perfectly to in the final ratio. The prior ratio is driven entirely by the spatial coordinates of the new versus the old location:

$$\frac{\pi(T_j^*)}{\pi(T_j)} = \frac{\prod_{m=1}^d \phi(c_{new,m}^*; \sigma_c^2)}{\prod_{m=1}^d \phi(c_{old,m}^*; \sigma_c^2)} \quad (\text{B.10})$$

For the transition ratio, both the forward and reverse moves share the same underlying probabilities: selecting the Change Centre move (p_{CC}) and selecting one of the b active centres to modify ($1/b$). The only asymmetry lies in generating the new coordinates for $\mathbf{c}_{new}^*$ during the forward move, and regenerating the original coordinates for $\mathbf{c}_{old}^*$ during the reverse move:

$$\frac{q(T_j \mid T_j^*)}{q(T_j^* \mid T_j)} = \frac{p_{CC} \times \frac{1}{b} \times \prod_{m=1}^d \phi(c_{old,m}^*; \sigma_c^2)}{p_{CC} \times \frac{1}{b} \times \prod_{m=1}^d \phi(c_{new,m}^*; \sigma_c^2)} = \frac{\prod_{m=1}^d \phi(c_{old,m}^*; \sigma_c^2)}{\prod_{m=1}^d \phi(c_{new,m}^*; \sigma_c^2)} \quad (\text{B.11})$$

When multiplying the prior ratio by the transition ratio, the spatial coordinate densities cancel each other out completely. The entire structural ratio collapses to exactly 1.

Therefore, the acceptance probability for the Change Centre proposal relies strictly on the universal marginal likelihood ratio:

$$\alpha_{CC}(T_j \rightarrow T_j^*) := \min \left\{ 1, \frac{\pi(R_j \mid T_j^*, \sigma^2)}{\pi(R_j \mid T_j, \sigma^2)} \right\} \quad (\text{B.12})$$

Appendix C

Variable Selection: Supplementary Material

C.1 A worked example for BART VIP

Table C.1: Demonstration of BART Variable Inclusion Proportion (VIP) Calculation

Panel A: Raw Splitting Node Counts (c_{ki}) & Row Sums

Iteration	X_1	X_2	X_3	X_4	X_5	Total Splits ($\sum c_{ki}$)
1995	20	15	2	3	0	40
1996	22	14	1	2	1	40
1997	19	18	3	1	1	42
1998	25	12	0	2	0	39
1999	21	16	2	2	2	43
2000	23	15	1	1	1	41

Panel B: Within-Iteration Proportions & Column Means (VIP)

Iteration	X_1	X_2	X_3	X_4	X_5
1995	0.500	0.375	0.050	0.075	0.000
1996	0.550	0.350	0.025	0.050	0.025
1997	0.452	0.429	0.071	0.024	0.024
1998	0.641	0.308	0.000	0.051	0.000
1999	0.488	0.372	0.047	0.047	0.047
2000	0.561	0.366	0.024	0.024	0.024
Mean (VIP)	0.532	0.367	0.036	0.045	0.020

C.2 Empirical Benchmark: The Friedman Dataset

To evaluate the variable selection capabilities of the methodologies developed within this chapter, we extensively utilise the synthetic dataset introduced by Friedman (1991) in the development of Multivariate Adaptive Regression Splines (MARS). Commonly referred to as the “Friedman data”, this formulation has established itself as a standard benchmark within the statistical learning literature for evaluating non-parametric regression and dimension reduction algorithms. The continuous response variable is generated according to the following underlying function:

$$y_i = 10 \sin(\pi x_{i1} x_{i2}) + 20(x_{i3} - 0.5)^2 + 10x_{i4} + 5x_{i5} + \varepsilon_i \quad (\text{C.1})$$

where the covariate vectors $x_i = (x_{i1}, \dots, x_{ip})^\top$ are drawn independently from a standard uniform distribution, $x_{ij} \sim \mathcal{U}(0, 1)$, and the residual error term follows a zero-mean Gaussian distribution, $\varepsilon_i \sim \mathcal{N}(0, \sigma^2)$.

This data generating process is highly effective for benchmarking because the true underlying signal depends strictly on the first five covariates $(x_{i1}, \dots, x_{i5})$. These active predictors exhibit a structurally complex composition consisting of non-linear multivariate interactions, $\sin(\pi x_{i1} x_{i2})$, quadratic effects, $(x_{i3} - 0.5)^2$, and strictly linear additive trends. To explicitly evaluate variable selection performance, the dataset is typically generated by appending $p - 5$ additional noise variables $(x_{i6}, \dots, x_{ip})$ that possess exactly zero structural relationship with the response y_i . Because the ground truth of the active set $\mathcal{S}^*$ is perfectly defined, this simulation framework serves as a rigorous baseline and is consistently utilised throughout the BARTS literature Chipman et al. (2010); Bleich et al. (2014b); Linero (2018).

C.3 BART VIP as Gini Coefficient

The general Gini importance, denoted IMP_{Gini} , for a predictor x_j in a tree-based ensemble is defined by the weighted sum of impurity decreases across all internal nodes where x_j is selected for partitioning. Following Louppe (2014), this importance metric is expressed as:

$$\text{IMP}_{\text{Gini}}(x_j) = \frac{1}{M} \sum_{m=1}^M \sum_{\eta \in \phi_m} \mathbf{I}(j_\eta = j) [w(\eta) \Delta i(\eta)], \quad (\text{C.2})$$

where:

- M defines the total number of trees within the ensemble.

- ϕ_m represents the set of all internal splitting nodes within the m^{th} tree.
- $I(j_\eta = j)$ is an indicator function equating to 1 if predictor x_j is utilised at node η , and 0 otherwise.
- $w(\eta)$ denotes the proportional weight assigned to node η .
- $\Delta i(\eta)$ quantifies the empirical decrease in node impurity achieved by the split at η .

As outlined by Luo et al. (2021), the BART VIP metric is algebraically equivalent to a specialised case of Gini importance. To derive the BART VIP as specified in Definition 3.1, we apply three strict structural constraints to the generalised Gini framework:

1. **Posterior Averaging:** The expectation is taken over K posterior samples generated via MCMC, rather than M independent trees.
2. **Unit Impurity Decrease:** The reduction in impurity is treated as a uniform constant across all splits, such that $\Delta i(\eta) = 1$. Unlike standard tree-building algorithms that weight importance by the strict magnitude of variance reduction, the standard BART VIP is an unweighted frequency metric. By recording only the discrete presence of a split, the formulation reduces to a pure counting measure.
3. **Inverse Frequency Weighting:** The node weight is defined strictly as the inverse of the total number of internal splitting rules present within the k^{th} posterior sample: $w(\eta) = \frac{1}{c_k}$.

Substituting these explicit constraints into the generalised Gini formula yields:

$$v_j^{\text{VIP}} = \frac{1}{K} \sum_{k=1}^K \sum_{\eta \in \phi_k} I(j_\eta = j) \left[\frac{1}{c_k} \cdot 1 \right]. \quad (\text{C.3})$$

Because the weighting term $\frac{1}{c_k}$ is constant across all nodes η within a specific posterior sample k , it factors out of the inner summation:

$$v_j^{\text{VIP}} = \frac{1}{K} \sum_{k=1}^K \frac{1}{c_k} \left(\sum_{\eta \in \phi_k} I(j_\eta = j) \right). \quad (\text{C.4})$$

The inner summation, $\sum_{\eta \in \phi_k} I(j_\eta = j)$, explicitly counts the absolute frequency that predictor x_j serves as a splitting variable in sample k . Following our established notation, this count is defined as c_{kj} , resulting in the final expression:

$$v_j^{\text{VIP}} = \frac{1}{K} \sum_{k=1}^K \frac{c_{kj}}{c_k}. \quad (\text{C.5})$$

This derivation establishes the structural equivalence between the BART VIP metric and an unweighted, frequency-based Gini coefficient. In turn, this motivates an investigation into whether it also inherits the cardinality bias associated with Gini importance.

C.4 A Worked Example of Metropolis Importance

To demonstrate the corrective mechanism of the Metropolis Importance (MI) metric, we return to the single tree configuration presented in Figure C.1. In this tree, the binary predictor x_1 splits once at the root node ($d_\eta = 0$), while the continuous predictor x_2 splits six times at deeper levels: twice at depth $d_\eta = 1$, and four times at depth $d_\eta = 2$.

As established in Section 4.3.1, the transition ratio heavily penalises proposals at greater depths. Let us assume standard BART prior parameters ($\alpha = 0.95, \beta = 2.0$), which yield monotonically decreasing Metropolis acceptance probabilities (π_{BIRTH}) as tree depth increases. For this illustration, the decision nodes are annotated with their corresponding MCMC proposal mechanism and acceptance ratio. Note that for this example we assume all variables entered the tree via BIRTH to help illustrate the calculation. In reality this would not be the case and some variables could enter via SWAP.

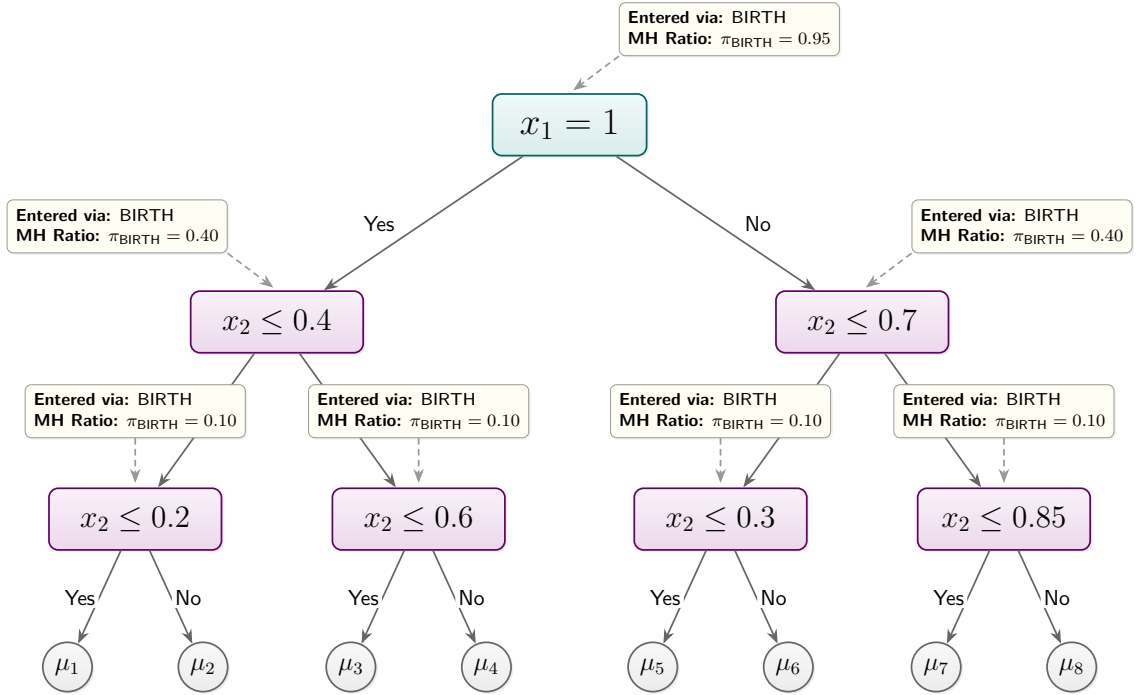


Figure C.1: A depth-4 decision tree annotated with the MCMC modification mechanisms and their corresponding Metropolis acceptance probabilities (π_{GROW}). The transition prior enforces a reduction in π_{BIRTH} values on splits occurring at greater depths

First we will reconsider the standard BART VIP calculation. We find the raw inclusion counts (c_j) and the standard Variable Inclusion Proportion (v_j^{VIP}) for each predictor:

$$\begin{aligned} c_1 = 1 &\implies v_1^{\text{VIP}} = \frac{1}{7} \approx 14.3\% \\ c_2 = 6 &\implies v_2^{\text{VIP}} = \frac{6}{7} \approx 85.7\% \end{aligned}$$

Next, we evaluate the unnormalized Metropolis Importance ($\tilde{u}_j$), which computes the average acceptance probability across all instances where the predictor was selected:

$$\begin{aligned} \tilde{u}_1 &= \frac{0.95}{1} = 0.95 \\ \tilde{u}_2 &= \frac{(2 \times 0.40) + (4 \times 0.10)}{6} = \frac{1.20}{6} = 0.20 \end{aligned}$$

Normalizing these averages yields the final Metropolis Importance metric (v_j^{MI}):

$$\begin{aligned} v_1^{\text{MI}} &= \frac{0.95}{0.95 + 0.20} = \frac{0.95}{1.15} \approx 82.6\% \\ v_2^{\text{MI}} &= \frac{0.20}{0.95 + 0.20} = \frac{0.20}{1.15} \approx 17.4\% \end{aligned}$$

The standard VIP framework inherently favours x_2 by treating all six deep splits as equal to the single root split. By transitioning to the MI framework, the redundant splits are penalised and therefore the MI metric for x_2 is reduced. This structural penalty neutralises the cardinality bias, correctly elevating the binary predictor x_1 as the dominant functional component of the tree. Table C.2 summarises this result.

Table C.2: A side-by-side comparison of the standard Variable Inclusion Proportion (VIP) and the Metropolis Importance (MI) metric.

Predictor	Standard VIP Framework		Metropolis Importance Framework (MI)		
	Splits (c_j)	VIP (v_j^{VIP})	Sum π_{BIRTH}	Avg π_{BIRTH} ($\tilde{u}_j$)	MI (v_j^{MI})
$\mathbf{x}_1$ (Binary)	1	14.3%	0.95	0.95	82.6%
$\mathbf{x}_2$ (Continuous)	6	85.7%	1.20	0.20	17.4%

C.5 Worked Example: Permutation-Based Thresholding

To explicitly demonstrate how these three thresholding strategies operate in practice, we present a simplified toy example. Suppose we have a dataset with three predictors (X_1, X_2, X_3). We fit our observed data to a BART model and extract the true observed VIP vector, $\mathbf{v}$. We then execute $S = 5$ permutations of the response vector $\mathbf{y}$, fitting a new BART model each time, to generate our null distribution matrix. For this example, we set our significance level to $\alpha = 0.20$. Because $S = 5$, the $(1 - \alpha)$ quantile corresponds to the 4th value in ascending order (the 80th percentile) in any sorted list of permutation results.

Step 1: Generating the Null Distribution

Table C.3 displays the observed VIPs alongside the matrix of permuted VIPs generated under the null hypothesis.

Predictor	Observed	Permuted VIPs ($v_{s,k}^*$)				
	VIP (v_k)	$s = 1$	$s = 2$	$s = 3$	$s = 4$	$s = 5$
X_1 (Strong)	0.450	0.35	0.40	0.31	0.33	0.30
X_2 (Weak)	0.365	0.32	0.31	0.37	0.34	0.35
X_3 (Noise)	0.185	0.33	0.29	0.32	0.33	0.35

Table C.3: Observed VIPs vs. Permuted VIPs for $S = 5$ iterations

Note on the Global Maximum: For the Global Maximum threshold, we must extract the highest VIP score achieved across all predictors within each specific permutation ($v_{\max,s}^*$). Looking at the columns above, these maximums are $\max(s_1) = 0.35$, $\max(s_2) = 0.40$, $\max(s_3) = 0.37$, $\max(s_4) = 0.34$, and $\max(s_5) = 0.35$.

Step 2: Calculating the Thresholds

We now calculate the three specific cut-off values as defined in Section 4.4.3.

- **Local (q_k^*):** The 80th percentile of each predictor’s specific row. For X_1 , the sorted null VIPs are 0.30, 0.31, 0.33, **0.35**, 0.40. The threshold is 0.35.
- **Global Max (g^*):** The 80th percentile of the “Permutation Max” values we extracted above. The sorted maxima are 0.34, 0.35, 0.35, **0.37**, 0.40. The threshold is 0.37.
- **Global SE ($m_k + C^*s_k$):** We calculate the mean (m_k) and standard deviation

(s_k) for each row. Suppose that we find that a constant multiplier of $C^* = 1.05$ satisfies the simultaneous $\alpha = 0.20$ coverage across all predictors.

	Local	Global Max	Global SE		
Predictor	q_k^*	g^*	Mean (m_k)	SD (s_k)	$SE_k = m_k + C^* s_k$
X_1 Threshold	0.350	0.370	0.338	0.038	0.377
X_2 Threshold	0.350	0.370	0.338	0.024	0.363
X_3 Threshold	0.330	0.370	0.324	0.022	0.347

Table C.4: The calculated rejection thresholds for the three methodologies.

Step 3: Final Variable Selection

Finally, we compare the original observed VIP vector ($\mathbf{v}$) against the calculated thresholds. A predictor is included if its observed VIP is strictly greater than the threshold.

	Observed	Local	Global Max	Global SE
Predictor	v_k	$(v_k > q_k^*)$	$(v_k > g^*)$	$(v_k > SE_k^*)$
X_1 (Strong)	0.450	Select	Select	Select
X_2 (Weak)	0.365	Select	Reject	Select
X_3 (Noise)	0.185	Reject	Reject	Reject

Table C.5: Final variable selection decisions for each of the three considered thresholds. Observations which pass their respective thresholds are marked as Select and observations which fall below are marked as Reject. Thresholds are found in Table C.4

As Table C.5 illustrates, X_1 possesses a strong enough signal to survive all thresholds, while the X_3 is universally rejected. The behaviour of X_2 , however, perfectly encapsulates the trade-off inherent in these methods. The Global Maximum threshold conditions on the single largest noise value generated anywhere in the dataset which acts as a strict, highly conservative filter. It successfully guards against Type I errors (false positives) but commits a Type II error (false negative) by discarding X_2 . Conversely, the Global SE threshold adapts to the tighter variance of X_2 's specific null distribution, successfully recovering the weak signal while maintaining global coverage constraints required to reject X_3 .

C.6 Additional Simulation Results

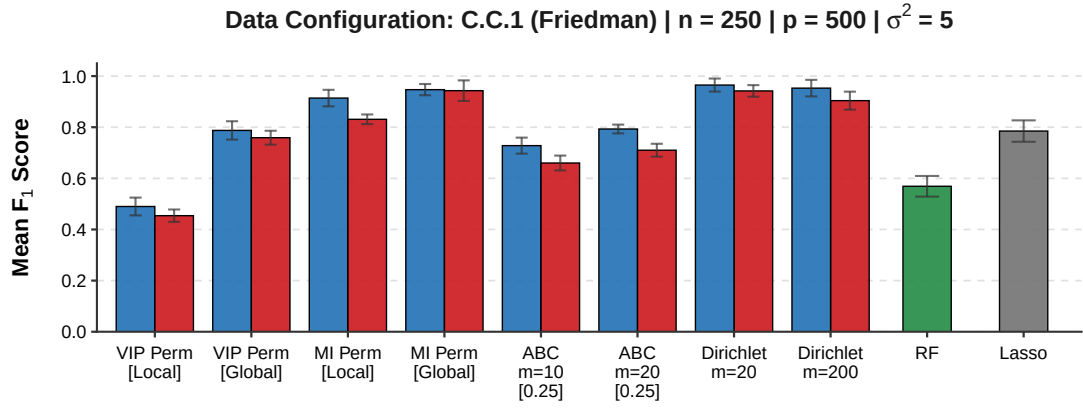


Figure C.2: Mean F_1 scores for the C.C.1 Friedman scenario across the $(p = 500, \sigma^2 = 5)$ configuration

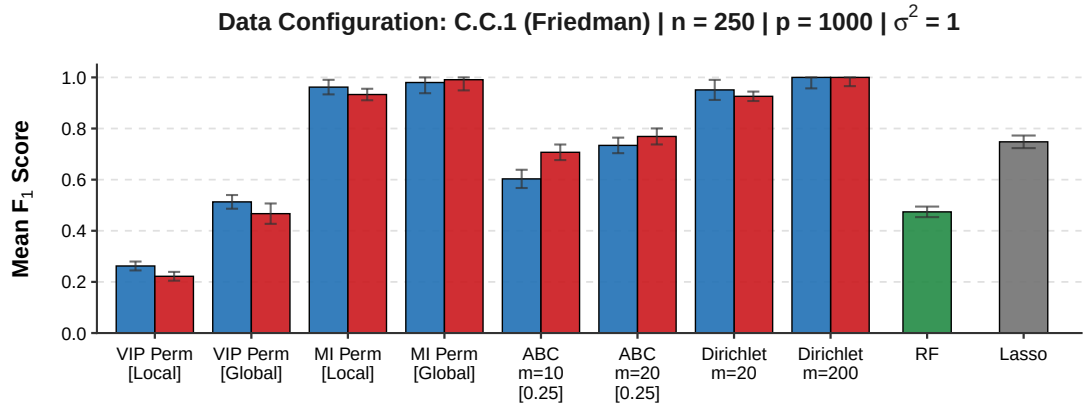


Figure C.3: Mean F_1 scores for the C.C.1 Friedman scenario across the $(p = 1000, \sigma^2 = 1)$ configuration

Table C.6: Simulation Results for C.C.1 ($n = 250$, $p = 500$, $\sigma^2 = 1$)

Method	Threshold	Precision	Recall	F_1 Score
Addi_ABC_m10	0.25	0.940 (0.240)	0.624 (0.234)	0.735 (0.238)
Addi_ABC_m10	0.5	0.880 (0.328)	0.372 (0.194)	0.510 (0.236)
Addi_ABC_m20	0.25	0.980 (0.141)	0.728 (0.228)	0.817 (0.202)
Addi_ABC_m20	0.5	0.920 (0.274)	0.520 (0.249)	0.642 (0.264)
Addi_MI_Perm	Global	0.950 (0.081)	1.000 (0.000)	0.973 (0.044)
Addi_MI_Perm	Local	0.884 (0.138)	1.000 (0.000)	0.933 (0.082)
Addi_MI_Perm	Max	1.000 (0.000)	0.840 (0.158)	0.906 (0.097)
Addi_Perm	Global	0.955 (0.083)	1.000 (0.000)	0.975 (0.047)
Addi_Perm	Local	0.918 (0.111)	1.000 (0.000)	0.953 (0.065)
Addi_Perm	Max	1.000 (0.000)	0.788 (0.159)	0.873 (0.103)
BART_ABC_m10	0.25	0.980 (0.141)	0.664 (0.169)	0.783 (0.159)
BART_ABC_m10	0.5	0.940 (0.240)	0.444 (0.211)	0.582 (0.230)
BART_ABC_m20	0.25	1.000 (0.000)	0.772 (0.162)	0.860 (0.131)
BART_ABC_m20	0.5	0.940 (0.240)	0.560 (0.229)	0.684 (0.238)
BART_MI_Perm	Global	1.000 (0.000)	1.000 (0.000)	1.000 (0.000)
BART_MI_Perm	Local	0.864 (0.127)	1.000 (0.000)	0.923 (0.073)
BART_MI_Perm	Max	0.800 (0.422)	0.640 (0.398)	0.696 (0.394)
BART_Perm	Global	0.726 (0.144)	1.000 (0.000)	0.834 (0.097)
BART_Perm	Local	0.342 (0.078)	1.000 (0.000)	0.505 (0.083)
BART_Perm	Max	1.000 (0.000)	0.952 (0.086)	0.973 (0.048)
DART_m20	MPM_0.50	1.000 (0.000)	0.956 (0.084)	0.976 (0.046)
DART_m200	MPM_0.50	0.997 (0.024)	1.000 (0.000)	0.998 (0.013)
Lasso	lambda.1se	0.808 (0.217)	0.800 (0.000)	0.788 (0.125)
RF_Perm	Global	0.649 (0.142)	0.892 (0.108)	0.742 (0.107)
RF_Perm	Local	0.424 (0.091)	0.936 (0.094)	0.580 (0.094)
RF_Perm	Max	1.000 (0.000)	0.740 (0.101)	0.847 (0.069)

Table C.7: Simulation Results for C.C.1 ($n = 250$, $p = 1000$, $\sigma^2 = 5$)

Method	Threshold	Precision	Recall	F_1 Score
Addi_ABC_m10	0.25	0.960 (0.198)	0.444 (0.186)	0.590 (0.201)
Addi_ABC_m10	0.5	0.840 (0.370)	0.192 (0.114)	0.307 (0.160)
Addi_ABC_m20	0.25	1.000 (0.000)	0.556 (0.158)	0.700 (0.145)
Addi_ABC_m20	0.5	0.980 (0.141)	0.292 (0.129)	0.437 (0.151)
Addi_MI_Perm	Global	0.895 (0.115)	0.900 (0.105)	0.892 (0.084)
Addi_MI_Perm	Local	0.783 (0.168)	0.900 (0.105)	0.833 (0.132)
Addi_MI_Perm	Max	1.000 (0.000)	0.360 (0.207)	0.502 (0.202)
Addi_Perm	Global	0.916 (0.118)	0.940 (0.093)	0.922 (0.083)
Addi_Perm	Local	0.830 (0.136)	0.944 (0.091)	0.877 (0.095)
Addi_Perm	Max	0.920 (0.274)	0.328 (0.197)	0.463 (0.220)
BART_ABC_m10	0.25	0.980 (0.141)	0.484 (0.152)	0.637 (0.159)
BART_ABC_m10	0.5	0.900 (0.303)	0.232 (0.117)	0.362 (0.160)
BART_ABC_m20	0.25	0.980 (0.141)	0.596 (0.164)	0.731 (0.161)
BART_ABC_m20	0.5	0.940 (0.240)	0.352 (0.169)	0.497 (0.198)
BART_MI_Perm	Global	0.922 (0.146)	0.980 (0.063)	0.942 (0.093)
BART_MI_Perm	Local	0.818 (0.177)	1.000 (0.000)	0.889 (0.123)
BART_MI_Perm	Max	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)
BART_Perm	Global	0.283 (0.046)	0.992 (0.040)	0.438 (0.056)
BART_Perm	Local	0.120 (0.016)	1.000 (0.000)	0.214 (0.025)
BART_Perm	Max	1.000 (0.000)	0.512 (0.108)	0.670 (0.100)
DART_m20	MPM_0.50	1.000 (0.000)	0.772 (0.121)	0.866 (0.078)
DART_m200	MPM_0.50	0.986 (0.048)	0.980 (0.061)	0.982 (0.045)
Lasso	lambda.1se	0.769 (0.262)	0.804 (0.028)	0.757 (0.167)
RF_Perm	Global	0.545 (0.129)	0.888 (0.115)	0.665 (0.101)
RF_Perm	Local	0.313 (0.081)	0.936 (0.094)	0.464 (0.089)
RF_Perm	Max	1.000 (0.000)	0.644 (0.093)	0.780 (0.067)

Table C.8: Simulation Results for C.C.2 ($n = 500$, $p = 100$, $\sigma^2 = 1$)

Method	Threshold	Precision	Recall	F_1 Score
Addi_ABC_m10	0.25	0.369 (0.282)	0.715 (0.385)	0.451 (0.274)
Addi_ABC_m10	0.5	0.060 (0.240)	0.015 (0.060)	0.024 (0.096)
Addi_ABC_m20	0.25	0.113 (0.207)	0.850 (0.343)	0.150 (0.188)
Addi_ABC_m20	0.5	0.447 (0.473)	0.265 (0.313)	0.314 (0.344)
Addi_MI_Perm	Global	0.921 (0.114)	0.875 (0.172)	0.886 (0.117)
Addi_MI_Perm	Local	0.798 (0.175)	0.925 (0.143)	0.851 (0.148)
Addi_MI_Perm	Max	0.800 (0.410)	0.412 (0.272)	0.529 (0.307)
Addi_Perm	Global	0.903 (0.142)	0.900 (0.160)	0.890 (0.124)
Addi_Perm	Local	0.803 (0.159)	0.945 (0.116)	0.861 (0.124)
Addi_Perm	Max	0.800 (0.404)	0.395 (0.272)	0.509 (0.304)
BART_ABC_m10	0.25	0.421 (0.448)	0.185 (0.213)	0.244 (0.263)
BART_ABC_m10	0.5	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)
BART_ABC_m20	0.25	0.376 (0.294)	0.640 (0.411)	0.438 (0.290)
BART_ABC_m20	0.5	0.160 (0.370)	0.055 (0.145)	0.078 (0.193)
BART_MI_Perm	Global	0.618 (0.329)	0.537 (0.284)	0.563 (0.285)
BART_MI_Perm	Local	0.498 (0.195)	0.650 (0.262)	0.552 (0.203)
BART_MI_Perm	Max	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)
BART_Perm	Global	0.774 (0.199)	0.750 (0.237)	0.745 (0.197)
BART_Perm	Local	0.590 (0.165)	0.875 (0.162)	0.695 (0.151)
BART_Perm	Max	0.300 (0.463)	0.120 (0.204)	0.166 (0.270)
DART_m20	MPM_0.50	0.881 (0.266)	0.565 (0.252)	0.668 (0.242)
DART_m200	MPM_0.50	0.964 (0.078)	1.000 (0.000)	0.980 (0.043)
Lasso	lambda.1se	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)
RF_Perm	Global	0.543 (0.253)	0.490 (0.252)	0.491 (0.214)
RF_Perm	Local	0.486 (0.174)	0.635 (0.238)	0.531 (0.166)
RF_Perm	Max	0.240 (0.431)	0.070 (0.134)	0.107 (0.198)

Table C.9: Simulation Results for C.C.2 ($n = 500$, $p = 200$, $\sigma^2 = 1$)

Method	Threshold	Precision	Recall	F_1 Score
Addi_ABC_m10	0.25	0.411 (0.468)	0.165 (0.206)	0.222 (0.259)
Addi_ABC_m10	0.5	0.020 (0.141)	0.005 (0.035)	0.008 (0.057)
Addi_ABC_m20	0.25	0.164 (0.194)	0.560 (0.345)	0.216 (0.145)
Addi_ABC_m20	0.5	0.040 (0.198)	0.010 (0.049)	0.016 (0.079)
Addi_MI_Perm	Global	0.592 (0.300)	0.375 (0.207)	0.443 (0.220)
Addi_MI_Perm	Local	0.426 (0.190)	0.425 (0.200)	0.413 (0.173)
Addi_MI_Perm	Max	0.200 (0.410)	0.050 (0.103)	0.080 (0.164)
Addi_Perm	Global	0.509 (0.353)	0.325 (0.254)	0.378 (0.262)
Addi_Perm	Local	0.456 (0.252)	0.410 (0.230)	0.418 (0.218)
Addi_Perm	Max	0.060 (0.240)	0.015 (0.060)	0.024 (0.096)
BART_ABC_m10	0.25	0.055 (0.222)	0.025 (0.116)	0.031 (0.130)
BART_ABC_m10	0.5	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)
BART_ABC_m20	0.25	0.270 (0.369)	0.145 (0.203)	0.179 (0.238)
BART_ABC_m20	0.5	0.040 (0.198)	0.010 (0.049)	0.016 (0.079)
BART_MI_Perm	Global	0.360 (0.245)	0.325 (0.216)	0.322 (0.193)
BART_MI_Perm	Local	0.245 (0.110)	0.412 (0.219)	0.302 (0.141)
BART_MI_Perm	Max	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)
BART_Perm	Global	0.373 (0.177)	0.510 (0.226)	0.416 (0.170)
BART_Perm	Local	0.256 (0.075)	0.720 (0.218)	0.375 (0.106)
BART_Perm	Max	0.060 (0.240)	0.020 (0.085)	0.029 (0.121)
DART_m20	MPM_0.50	0.564 (0.459)	0.320 (0.286)	0.394 (0.330)
DART_m200	MPM_0.50	0.730 (0.283)	0.805 (0.273)	0.735 (0.249)
Lasso	lambda.1se	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)
RF_Perm	Global	0.371 (0.255)	0.345 (0.247)	0.340 (0.218)
RF_Perm	Local	0.295 (0.158)	0.470 (0.229)	0.351 (0.160)
RF_Perm	Max	0.120 (0.328)	0.035 (0.101)	0.053 (0.150)

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